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(12) **United States Patent**
Minshull et al.(10) **Patent No.: US 12,404,504 B2**(45) **Date of Patent: Sep. 2, 2025**(54) **TRANSPPOSITION OF NUCLEIC ACIDS INTO EUKARYOTIC GENOMES WITH A TRANSPOSASE FROM HELIOTHIS**2020/0318135 A1 10/2020 Minshull et al.
2021/0292773 A1 9/2021 Minshull et al.
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CPC **C12N 15/11** (2013.01); **C12N 5/0602** (2013.01); **C12N 9/1241** (2013.01); **C12Y 207/07** (2013.01)(58) **Field of Classification Search**None
See application file for complete search history.(56) **References Cited**

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The present invention provides polynucleotide vectors for high expression of heterologous genes. Some vectors further comprise novel transposons and transposases that further improve expression. Further disclosed are vectors that can be used in a gene transfer system for stably introducing nucleic acids into the DNA of a cell. The gene transfer systems can be used in methods, for example, gene expression, bioprocessing, gene therapy, insertional mutagenesis, or gene discovery.

17 Claims, 1 Drawing Sheet**Specification includes a Sequence Listing.**

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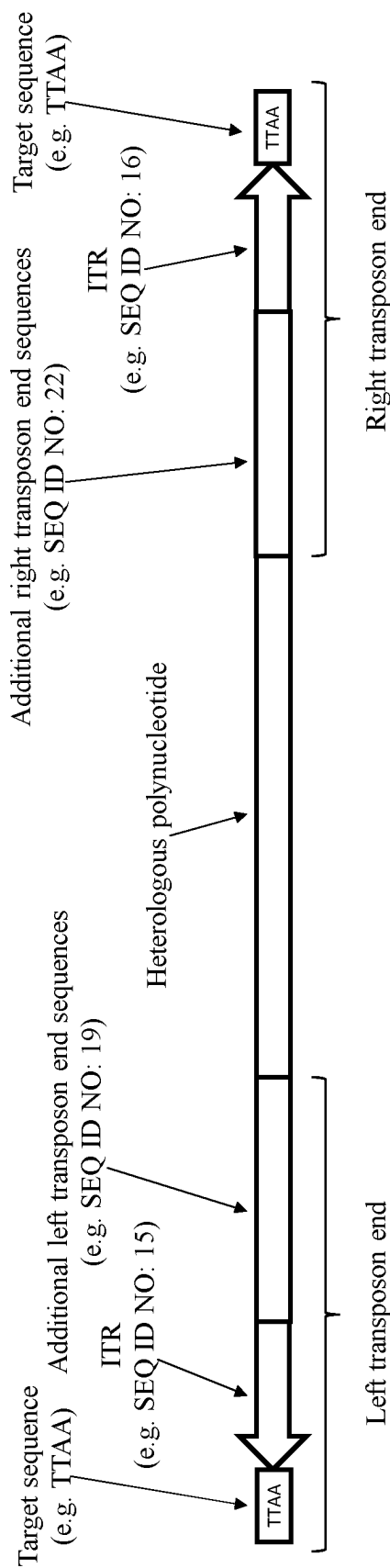
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TRANSPOSITION OF NUCLEIC ACIDS INTO EUKARYOTIC GENOMES WITH A TRANSPOSASE FROM HELIOTHIS

CROSS REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. application Ser. No. 16/842,707 filed Apr. 7, 2020, which claims priority to U.S. Provisional Application No. 62/831,103 filed Apr. 8, 2019, U.S. Provisional Application No. 62/873,346 filed Jul. 12, 2019 and U.S. Provisional Application No. 62/978,862 filed Feb. 20, 2020, each incorporated by reference in its entirety for all purposes.

REFERENCE TO A SEQUENCE LISTING

The application refers to sequences disclosed in a txt file named 557951SEQLIST.TXT, of 3,651,164 bytes, created Jun. 2, 2021, incorporated by reference.

1. FIELD OF THE INVENTION

The field of the present invention relates to configurations of DNA vectors for making stable modifications of the genomes of target cells, and the use of non-natural transposons and transposases.

2. BACKGROUND OF THE INVENTION

The expression levels of genes encoded on a polynucleotide integrated into the genome of a cell depend on the configuration of sequence elements within the polynucleotide. The efficiency of integration and thus the number of copies of the polynucleotide that are integrated into each genome, and the genomic loci where integration occurs also influence the expression levels of genes encoded on the polynucleotide. The efficiency with which a polynucleotide may be integrated into the genome of a target cell can often be increased by placing the polynucleotide into a transposon.

Transposons comprise two ends that are recognized by a transposase. The transposase acts on the transposon to remove it from one DNA molecule and integrate it into another. The DNA between the two transposon ends is transposed by the transposase along with the transposon ends. Heterologous DNA flanked by a pair of transposon ends, such that it is recognized and transposed by a transposase is referred to herein as a synthetic transposon. Introduction of a synthetic transposon and a corresponding transposase into the nucleus of a eukaryotic cell may result in transposition of the transposon into the genome of the cell. These outcomes are useful because they increase transformation efficiencies and because they can increase expression levels from integrated heterologous DNA. There is thus a need in the art for hyperactive transposases and transposons.

Transposition by a piggy Bac-like transposase is perfectly reversible. The transposon is initially integrated at an integration target sequence in a recipient DNA molecule, during which the target sequence becomes duplicated at each end of the transposon inverted terminal repeats (ITRs). Subsequent transposition removes the transposon and restores the recipient DNA to its former sequence, with the target sequence duplication and the transposon removed. However, this is not sufficient to remove a transposon from a genome into which it has been integrated, as it is highly likely that the

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transposon will be excised from the first integration target sequence but transposed into a second integration target sequence in the genome. Transposases that are deficient for the integration (or transposition) function, on the other hand, can excise the transposon from the first target sequence, but will be unable to integrate into a second target sequence. Integration-deficient transposases are thus useful for reversing the genomic integration of a transposon.

One application for transposases is for the engineering of eukaryotic genomes. Such engineering may require the integration of more than one different polynucleotide into the genome. These integrations may be simultaneous or sequential. When transposition into a genome of a first transposon comprising a first heterologous polynucleotide by a first transposase is followed by transposition into the same genome of a second transposon comprising a second heterologous polynucleotide by a second transposase, it is advantageous that the second transposase not recognize and transpose the first transposon. This is because the location of a polynucleotide sequence within the genome influences the expressibility of genes encoded on said polynucleotide, so transposition of the first transposon to a different chromosomal location by the second transposase could change the expression properties of any genes encoded on the first heterologous polynucleotide. There is therefore a need for a set of transposons and their corresponding transposases in which the transposases within the set recognize and transpose only their corresponding transposons, but not any other transposons in the set.

Since its discovery in 1983, the piggy Bac transposon and transposase from the looper moth *Trichoplusia ni* has been widely used for inserting heterologous DNA into the genomes of target cells from many different organisms. The piggyBac system is a particularly valuable transposase system because of: “its activity in a wide range of organisms, its ability to integrate multiple large transgenes with high efficiency, the ability to add domains to the transposase without loss of activity, and excision from the genome without leaving a footprint mutation” (Doherty et al., Hum. Gene Ther. 23, 311-320 (2012), at p. 312, LHC, ¶ 12).

The value and versatility of the piggyBac system has inspired significant efforts to identify other active piggyBac-like transposons (commonly referred to as piggyBac-like elements, or PLEs) but these have been largely unsuccessful. “Since piggyBac is one of the most popular transposons used for transgenesis, searching for new active PLEs has attracted lots of attention. However, only a few active PLEs have been reported to date.” (Luo et al., BMC Molecular Biology 15, 28 (2014) world wide web biomedcentral.com/1471-2199/15/28. p. 4 of 12, RHC, ¶ 1 “Discussion”).

Although there are large numbers of homologs of piggy Bac transposons and transposases in sequence databases, few active ones have been identified because the vast majority are inactivated by their hosts to avoid activity deleterious to the hosts as illustrated by the following excerpts: “Related piggy Bac transposable elements have been found in plants, fungi and animals, including humans [125], although they are probably inactive due to mutation.” (Munoz-Lopez & Garcia-Perez, Current Genomics 11, 115-128 (2010) at p. 120, RHC, ¶ 1). “It is believed that transposons invade a genome and subsequently spread throughout it during evolution. The “selfish” mobility of transposons is harmful to the host; hence, they are eliminated or inactivated by the host through natural selection. Even harmless transposons lose the activity eventually because of the absence of conservative selection for them. Thus, in general, transposons have a short life span in a host

and they subsequently become fossils in the genome.” (Hikosaka et al., Mol. Biol. Evol. 24, 2648-3656 (2007) at p. 2648, LHC, ¶ 1 “Introduction”). “Frequent movement of transposable elements in a genome is harmful (Belancio et al., 2008; Deininger & Batzer, 1999; Le Rouzic & Capy, 2006; Oliver & Greene, 2009). As a result, most transposable elements are inactivated shortly after they invade a new host.” (Luo et al., Insect Science 18, 652-662 (2011) at p. 660, LHC, ¶ 1).

Three classes of piggyBac-like elements have been found: (1) those that are very similar to the original piggy Bac from the looper moth (typically >95% identical at the nucleotide level), (2) those that are moderately related (typically 30-50% identical at the amino acid level), and (3) those that are very distantly related (Wu et al., Insect Science 15, 521-528 (2008) at p. 521, RHC, ¶ 2).

PiggyBac-like transposases highly related to the looper moth transposase have been described by several groups. They are extremely highly conserved. Very similar transposase sequences to the original piggyBac (95-98% nucleotide identity) have been reported in three different strains of the fruit fly *Bactrocera dorsalis* (Handler & McCombs, Insect Molecular Biology 9, 605-612, (2000)). Comparably conserved piggyBac sequences have been found in other *Bactrocera* species (Bonizzoni et al., Insect Molecular Biology 16, 645-650 (2007)). Two species of noctuid moth (*Helicoverpa zea* and *Helicoverpa armigera*) and other strains of the looper moth *Trichoplusia ni* had genomic copies of the piggy Bac transposase with 93-100% nucleotide identity to the original piggyBac sequence (Zimowska & Handler, Insect Biochemistry and Molecular Biology, 36, 421-428 (2006)). Zimowska & Handler also found multiple copies of much more significantly mutated (and truncated) versions of the piggy Bac transposase in both *Helicoverpa* species, as well as a homolog in the armyworm *Spodoptera frugiperda*. None of these groups attempted to measure any activity for these transposases. Wu et. al (2008), supra, reported isolating a transposase from *Macdunnoughia crassisigna* with 99.5% sequence identity with the looper moth piggy Bac. They also demonstrated that this transposon and transposase are active, by showing that they could measure both excision and transposition. Their Discussion summarized previous results as follows: “Other reportedly closely related IFP2 class sequences were in various *Bactrocera* species, *T. ni* genome, *Helicoverpa armigera*, and *H. zea* (Handler & McCombs, 2000; Zimowska & Handler, 2006; Bonizzoni et al., 2007). These sequences were partial fragments of piggy Bac-like elements, and most of them were truncated or inactivated by accumulating random mutations.” (Wu et. al., Insect Science 15, 521-528 (2008) at p. 526, LHC, ¶ 3.)

It has proved very difficult to identify active piggy Bac-like transposases that are moderately related to the looper moth enzyme simply by looking at sequence. The presence of features that are known to be necessary: a full-length open reading frame, catalytic aspartate residues and intact ITRs, has not proven to be predictive of activity. “A large diversity of PLEs in eukaryotes has been documented in a computational analysis of genomic sequence data [citations omitted]. However, few elements were isolated with an intact structure consistent with function, and only the original IFP2 piggyBac has been developed into a vector for routine transgenesis.” (Wu et al., Genetica 139, 149-154 (2011), at p. 152, RHC, ¶ 2). Wu et al.’s group from Nanjing University (the “Nanjing group”) published several papers over a 6-year period, each identifying moderately related piggyBac homologs. Although the Nanjing group showed in 2008 that

they could measure both excision and transposition of the *Macdunnoughia crassisigna* transposon by its corresponding transposase, and in each subsequent paper they express the desire to identify novel active piggy Bac-like transposases, they only show excision activity and that only for one transposase from *Aphis gossypii*. They conclude that the usefulness of this transposase “remains to be explored with further experiments” (Luo et. al. 2011, p. 660, LHC ¶ 2 “Discussion”). However, none of the other papers published by the Nanjing group in which piggy Bac-like sequences were identified from a variety of other insects, show that any activity was found. Three papers identifying other putative active piggyBac-like transposases were published by a group at Kansas State University. None of these papers reports any activity data. Wang et al., Insect Molecular Biology 15, 435-443 (2006) found multiple copies of piggyBac-like sequences in the genome of the tobacco budworm *Heliothis virescens*. Many of these had obvious mutations or deletions that led the authors not to consider them to be candidate active transposases. Wang et. al., Insect Biochemistry and Molecular Biology 38, 490-498 (2008) reported more than 30 piggyBac-like sequences in the genome of the red flour beetle *Tribolium castaneum*. They concluded “All the TcPLEs identified here, except TcPLE1, were apparently defective due to the presence of multiple stop codons and/or indels in the putative transposase encoding regions.” Even for TcPLE1 there was “no evidence supporting recent or current mobilization events” (p. 492, section 3.1, ¶ 2&3). Wang et al. (2010) used PCR to identify piggyBac-like sequences from the pink bollworm *Pectinophora gossypiella*. Again, they found many obviously defective copies, as well as one transposase with characteristics the authors believe to be consistent with activity (page 179, RHC, ¶ 2). But no follow up report indicating transposase activity can be found. Other groups have also attempted to identify active piggyBac-like transposases. These reports conclude with statements that the piggy Bac-like elements identified are undergoing testing for activity, but there are no subsequent reports of success. For example, Sarkar et. al. (2003) conclude their Discussion by re-stating the value of novel active piggyBac-like transposons, and describing their ongoing efforts to identify one: “The mobility of the original *T. ni* piggyBac element in various insects suggests that piggy Bac family transposons might prove to be useful genetic tools in organisms other than insects. We are currently isolating an intact piggy Bac element from *An. gambiae* (AgaPB1) to test its mobility in various organisms.” (Mol. Gen. Genomics 270, 173-180 at p. 179, LHC, 11). There appear to be no further published reports of this putative active transposase. Xu et al. analyzed the silkworm genome looking for piggyBac-like sequences (Xu et al., Mol Gen Genomics 276, 31-40 (2006)). They found 98 piggyBac-like sequences and performed various computational analyses of putative transposase sequence and ITR sequences. They conclude: “We have isolated several intact piggyBac-like elements from *B. mori* and are currently testing their activity and the feasibility of using them as transformation vectors.” (p 38, RHC, ¶ 3). There appear to be no further published reports of these putative active transposases.

Four published papers discussing the third class of distantly related piggy Bac-like transposases. The first three of these demonstrate only the excision part of the reaction and acknowledge that this is different from full transposition. Hikosaka et. al., Mol Biol Evol 24, 2648-2656 (2007) reported that “In the present study, we demonstrated that the Xtr-Urib2 Tase has excision activity toward the target

transposon, although there is no evidence for the integration of the excised target into the genome thus far.” (page 2654, RHC, ¶ 2). Luo et. al., *Insect Science* 18, 652-662 (2011) reported “These results demonstrated the activity of the Ago-PLE1.1 transposase in mediating the first step of the cut and-paste movement of the element” (page 658, LHC, ¶ 1). Daimon et. al., *Genome* 53, 585-593 (2010) discussed the transposase systems yabusabe-1 and yabusabe-W. Although Daimon et al. reported detecting an excision event by PCR, they also report screening approximately 100,000 recovered plasmids for the excision of yabusame-1 and yabusame-W without identifying a single recovered plasmid from which the elements had excised. By contrast Daimon reports the transposition frequency of wildtype piggy Bac enzyme as around 0.3-1.4. Thus, it appears from Daimon et al. that the excision frequency of yabusabe-1 or -W is less than 0.001% (1:100,000). This is at least 2-3 orders of magnitude less than can be achieved with a wild-type piggy Bac enzyme and even less than available genetically engineered variants of piggy Bac transposase, which achieve ten-fold higher transposition than wildtype. The implied transposition frequency for yabusame-1 from Daimon et al. is also two orders of magnitude lower than random integration frequency in mammalian cells (which is of the order of 0.1%). Thus, Daimon et al. show that yabusame-1 was essentially inactive and would not be useful as a genetic engineering tool. Such a view likely underlies Daimon et al.’s own conclusion: “Although we could detect the excision event in the highly sensitive PCR-based assay, our data indicate that both elements have lost their excision activity almost entirely.” This also suggests that the PCR-based excision assay used to show activity of Uribo2 and Ago-PLE1.1 is not predictive of transposition activity that will be useful for inserting heterologous DNA into the genome of a target cell. The only report of a fully active piggy Bac-like transposase (competent for both excision and integration) of the third category of distantly related transposases to the original piggy Bac transposase from *Trichoplusia Ni* is one from the bat *Myotis lucifugus* (Mitra et. al., *Proc. Natl. Acad. Sci.* 110, 234-239 (2013)). These authors used a yeast system to demonstrate both excision and transposition activities for the bat transposase. All of the work described here shows that it has been extremely difficult to identify fully active piggy-Bac-like transposases, even though there are a large number of candidate sequences. There is therefore a need for new piggyBac-like transposons and their corresponding transposases.

3. SUMMARY OF THE INVENTION

Heterologous gene expression from polynucleotide constructs that stably integrate into a target cell genome can be improved by placing the expression polynucleotide between a pair of transposon ends: sequence elements that are recognized and transposed by transposases. DNA sequences inserted between a pair of transposon ends can be excised by a transposase from one DNA molecule and inserted into a second DNA molecule. Three novel piggy Bac-like transposon-transposase systems are disclosed that are not derived from the looper moth *Trichoplusia ni*. They are derived from; the tobacco budworm moth *Heliothis virescens* (the *Heliothis* transposase); the cutworm *Agrotis ipsilon* (the *Agrotis* transposase) and the bollworm *Helicoverpa armigera* (the *Helicoverpa* transposase). Each of these comprise sequences that function as transposon ends and that can be used in conjunction with a corresponding transposase that recognizes and acts on those transposon ends, as gene

transfer systems for stably introducing nucleic acids into the DNA of a cell. The gene transfer systems of the invention can be used in methods including but not limited to genomic engineering of eukaryotic cells, heterologous gene expression, gene therapy, cell therapy, insertional mutagenesis, or gene discovery.

Transposition may be effected using a polynucleotide comprising an open reading frame encoding a *Heliothis* transposase, the amino acid sequence of which is at least 90% identical to SEQ ID NO: 39, operably linked to a heterologous promoter. The heterologous promoter may be active in a eukaryotic cell. The heterologous promoter may be active in a mammalian cell. mRNA may be prepared from a polynucleotide comprising an open reading frame encoding a *Heliothis* transposase, the amino acid sequence of which is at least 90% identical to SEQ ID NO: 39, operably linked to a heterologous promoter that is active in an in vitro transcription reaction. The transposase may comprise a mutation as shown in columns C and D in Table 1, relative to the sequence of SEQ ID NO: 39. The transposase may comprise a mutation at an amino acid position selected from 41, 43, 81, 83, 85, 125, 126, 131, 136, 140, 149, 151, 152, 153, 155, 162, 169, 212, 238, 239, 241, 264, 268, 280, 297, 299, 300, 305, 312, 316, 322, 357, 360, 396, 397, 421, 430, 447, 449, 450, 476, 485, 492, 495, 507, 512, 585, 589, 595, and 603, relative to the sequence of SEQ ID NO: 39. The transposase may comprise a mutation selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39, the transposase optionally including at least 2, 3, 4, or 5 selected from the group. The amino acid sequence of the transposase may be selected from SEQ ID NO: 131-490. The transposase can excise or transpose a transposon from SEQ ID NO: 66. The excision activity or transposition activity of the transposase is at least 2-fold higher than the activity of SEQ ID NO: 39, optionally 2 to 10 fold higher. Codons of the open reading frame of the transposase may be selected for mammalian cell expression. An isolated mRNA may encode a polypeptide, the amino acid sequence of which is at least 90% identical with SEQ ID NO: 39, and wherein the mRNA sequence comprises at least 10 synonymous codon differences relative to SEQ ID NO: 1181 at corresponding positions between the mRNA and SEQ ID NO: 1181, optionally wherein codons in the mRNA at the corresponding positions are selected for mammalian cell expression. The open reading frame encoding the transposase may further encode a heterologous nuclear localization sequence fused to the transposase. The open reading frame encoding the transposase may further encode a heterologous DNA binding domain (for example derived from a Casr Cas system, or a zinc finger protein, or a TALE protein) fused to the transposase. A non-naturally occurring polynucleotide may encode a polypeptide, the sequence of which is at least 90% identical to SEQ ID NO: 39.

A *Heliothis* transposon comprises SEQ ID NO: 15 and SEQ ID NO: 16 flanking a heterologous polynucleotide. The transposon may further comprise a sequence at least 90% identical to SEQ ID NO: 19 on one side of the heterologous polynucleotide and a sequence at least 90% identical to SEQ ID NO: 22 on the other. The heterologous polynucleotide

may comprise a heterologous promoter that is active in eukaryotic cells. The promoter may be operably linked to at least one or more of: i) an open reading frame; ii) a nucleic acid encoding a selectable marker; iii) a nucleic acid encoding a counter-selectable marker; iv) a nucleic acid encoding a regulatory protein; v) a nucleic acid encoding an inhibitory RNA. The heterologous promoter may comprise a sequence selected from SEQ ID NOs: 723-812. The heterologous polynucleotide may comprise a heterologous enhancer that is active in eukaryotic cells. The heterologous enhancer may be selected from SEQ ID NOs: 702-722. The heterologous polynucleotide may comprise a heterologous intron that is spliceable in eukaryotic cells. The nucleotide sequence of the heterologous intron may be selected from SEQ ID NO: 813-873. The heterologous polynucleotide may comprise an insulator sequence. The nucleic acid sequence of the insulator may be selected from SEQ ID NO: 684-690. The heterologous polynucleotide may comprise two open reading frames, each operably linked to a separate promoter. The heterologous polynucleotide may comprise a sequence selected from SEQ ID NOs: 997-1180. The heterologous polynucleotide may comprise or encode a selectable marker. The selectable marker may be selected from a glutamine synthetase enzyme, a dihydrofolate reductase enzyme, a puromycin acetyltransferase enzyme, a blasticidin acetyltransferase enzyme, a hygromycin B phosphotransferase enzyme, an aminoglycoside 3'-phosphotransferase enzyme and a fluorescent protein. A eukaryotic cell whose genome comprises SEQ ID NO: 15 and SEQ ID NO: 16 flanking a heterologous polynucleotide is an embodiment of the invention. The cell may be an animal cell, a mammalian cell, a rodent cell or a human cell.

A transposon may be integrated into the genome of a eukaryotic cell by (a) introducing into the cell a transposon comprising SEQ ID NO: 15 and SEQ ID NO: 16 flanking a heterologous polynucleotide, (b) introducing into the cell a transposase, the sequence of which is at least 90% identical with SEQ ID NO: 39 wherein the transposase transposes the transposon to produce a genome comprising SEQ ID NO: 15 and SEQ ID NO: 16 flanking the heterologous polynucleotide. The transposase may be introduced as a polynucleotide encoding the transposase, the polynucleotide may be an mRNA molecule or a DNA molecule. The transposase may be introduced as a protein. The heterologous polynucleotide may also encode a selectable marker, and the method may further comprise selecting a cell comprising the selectable marker. The cell may be an animal cell, a mammalian cell, a rodent cell or a human cell. The human cell may be a human immune cell, for example a B-cell or a T-cell. The heterologous polynucleotide may encode a chimeric antigen receptor. A polypeptide may be expressed from the transposon integrated into the genome of the eukaryotic cell. The polypeptide may be purified. The purified polypeptide may be incorporated into a pharmaceutical composition.

4. BRIEF DESCRIPTION OF THE FIGURES

FIG. 1. Structure of a *Heliothis* transposon. A *Heliothis* transposon comprises a left transposon end and a right transposon end flanking a heterologous polynucleotide. The left transposon end comprises (i) a left target sequence, which is often 5'-TTAA-3', although a number of other target sequences are used at lower frequency (Li et al., 2013. Proc. Natl. Acad. Sci. vol. 110, no. 6, E478-487); (ii) a left ITR (e.g. SEQ ID NO: 15) and (iii) (optionally) additional left transposon end sequences (e.g. SEQ ID NO: 19). The right transposon end comprises (i) (optionally) additional right

transposon end sequences (e.g. SEQ ID NO: 22); (ii) a right ITR (e.g. SEQ ID NO: 16) which is a perfect or imperfect repeat of the left ITR, but in inverted orientation relative to the left ITR and (iii) a right target sequence which is typically the same as the left target sequence.

5. DETAILED DESCRIPTION OF THE INVENTION

5.1 Definitions

Use of the singular forms "a," "an," and "the" include plural references unless the context clearly dictates otherwise. Thus, for example, reference to "a polynucleotide" includes a plurality of polynucleotides, reference to "a substrate" includes a plurality of such substrates, reference to "a variant" includes a plurality of variants, and the like.

Terms such as "connected," "attached," "linked," and "conjugated" are used interchangeably herein and encompass direct as well as indirect connection, attachment, linkage or conjugation unless the context clearly dictates otherwise. Where a range of values is recited, it is to be understood that each intervening integer value, and each fraction thereof, between the recited upper and lower limits of that range is also specifically disclosed, along with each subrange between such values. The upper and lower limits of any range can independently be included in or excluded from the range, and each range where either, neither or both limits are included is also encompassed within the invention. Where a value being discussed has inherent limits, for example where a component can be present at a concentration of from 0 to 100%, or where the pH of an aqueous solution can range from 1 to 14, those inherent limits are specifically disclosed. Where a value is explicitly recited, it is to be understood that values which are about the same quantity or amount as the recited value are also within the scope of the invention. Where a combination is disclosed, each sub combination of the elements of that combination is also specifically disclosed and is within the scope of the invention. Conversely, where different elements or groups of elements are individually disclosed, combinations thereof are also disclosed. Where any element of an invention is disclosed as having a plurality of alternatives, examples of that invention in which each alternative is excluded singly or in any combination with the other alternatives are also hereby disclosed; more than one element of an invention can have such exclusions, and all combinations of elements having such exclusions are hereby disclosed.

Unless defined otherwise herein, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Singleton, et al., *Dictionary of Microbiology and Molecular Biology*, 2nd Ed., John Wiley and Sons, New York (1994), and Hale & Marham, *The Harper Collins Dictionary of Biology*, Harper Perennial, NY, 1991, provide one of skill with a general dictionary of many of the terms used in this invention. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, the preferred methods and materials are described. Unless otherwise indicated, nucleic acids are written left to right in 5' to 3' orientation; amino acid sequences are written left to right in amino to carboxy orientation, respectively. The terms defined immediately below are more fully defined by reference to the specification as a whole.

The term "codon usage" or "codon bias" refers to the relative frequencies with which different synonymous

codons are used to encode an amino acid within an open reading frame. A nucleic acid sequence having codon preferences for a particular target cell has a balance of synonymous codon choices that result in efficient translation in that cell type. This balance is often not calculable from observed genomic codon frequencies, but must be empirically determined, for example as described in U.S. Pat. Nos. 7,561,972 and 7,561,973 and 8,401,798 and in Welch et. al. (2009) "Design Parameters to Control Synthetic Gene Expression in *Escherichia coli*". PLOS ONE 4 (9): e7002. <https://doi.org/10.1371/journal.pone.0007002>. A nucleic acid originally isolated from one cell type to be introduced into a target cell of another type can undergo selection of codon preferences for the target site cell such that at least 1 and sometimes, 5, 20, 15, 20, 50, 100 or more choices among synonymous codons differ between the nucleic acid introduced into the target cell from the original nucleic acid.

The "configuration" of a polynucleotide means the functional sequence elements within the polynucleotide, and the order and direction of those elements.

The terms "corresponding transposon" and "corresponding transposase" are used to indicate an activity relationship between a transposase and a transposon. A transposase transposes its corresponding transposon. Many transposases may correspond with a single transposon, and many transposons may correspond with a single transposase.

The term "counter-selectable marker" means a polynucleotide sequence that confers a selective disadvantage on a host cell. Examples of counter-selectable markers include *sacB*, *rpsL*, *tetAR*, *pheS*, *thyA*, *gata-1*, *ccdB*, *kid* and *barnase* (Bernard, 1995, *Journal/Gene*, 162:159-160; Bernard et al., 1994, *Journal/Gene*, 148:71-74; Gabant et al., 1997, *Journal/Biotechniques*, 23:938-941; Gabant et al., 1998, *Journal/Gene*, 207:87-92; Gabant et al., 2000, *Journal/Biotechniques*, 28:784-788; Galvao and de Lorenzo, 2005, *Journal/Appl Environ Microbiol*, 71:883-892; Hartzog et al., 2005, *Journal/Yeast*, 22:789-798; Knipfer et al., 1997, *Journal/Plasmid*, 37:129-140; Reytrat et al., 1998, *Journal/Infect Immun*, 66:4011-4017; Soderholm et al., 2001, *Journal/Biotechniques*, 31:306-310, 312; Tamura et al., 2005, *Journal/Appl Environ Microbiol*, 71:587-590; Yazynin et al., 1999, *Journal/FEBS Lett*, 452:351-354). Counter-selectable markers often confer their selective disadvantage in specific contexts. For example, they may confer sensitivity to compounds that can be added to the environment of the host cell, or they may kill a host with one genotype but not kill a host with a different genotype. Conditions which do not confer a selective disadvantage on a cell carrying a counter-selectable marker are described as "permissive". Conditions which do confer a selective disadvantage on a cell carrying a counter-selectable marker are described as "restrictive".

The term "coupling element" or "translational coupling element" means a DNA sequence that allows the expression of a first polypeptide to be linked to the expression of a second polypeptide. Internal ribosome entry site elements (IRES elements) and cis-acting hydrolase elements (CHYSEL elements) are examples of coupling elements.

The terms "DNA sequence", "RNA sequence" or "polynucleotide sequence" mean a contiguous nucleic acid sequence. The sequence can be an oligonucleotide of 2 to 20 nucleotides in length to a full length genomic sequence of thousands or hundreds of thousands of base pairs.

The term "expression construct" means any polynucleotide designed to transcribe an RNA. For example, a construct that contains at least one promoter which is or may be operably linked to a downstream gene, coding region, or polynucleotide sequence (for example, a cDNA or genomic

DNA fragment that encodes a polypeptide or protein, or an RNA effector molecule, for example, an antisense RNA, triplex-forming RNA, ribozyme, an artificially selected high affinity RNA ligand (aptamer), a double-stranded RNA, for example, an RNA molecule comprising a stem-loop or hairpin dsRNA, or a bi-finger or multi-finger dsRNA or a microRNA, or any RNA). An "expression vector" is a polynucleotide comprising a promoter which can be operably linked to a second polynucleotide. Transfection or transformation of the expression construct into a recipient cell allows the cell to express an RNA effector molecule, polypeptide, or protein encoded by the expression construct. An expression construct may be a genetically engineered plasmid, virus, recombinant virus, or an artificial chromosome derived from, for example, a bacteriophage, adenovirus, adeno-associated virus, retrovirus, lentivirus, poxvirus, or herpesvirus. Such expression vectors can include sequences from bacteria, viruses or phages. Such vectors include chromosomal, episomal and virus-derived vectors, for example, vectors derived from bacterial plasmids, bacteriophages, yeast episomes, yeast chromosomal elements, and viruses, vectors derived from combinations thereof, such as those derived from plasmid and bacteriophage genetic elements, cosmids and phagemids. An expression construct can be replicated in a living cell, or it can be made synthetically. For purposes of this application, the terms "expression construct", "expression vector", "vector", and "plasmid" are used interchangeably to demonstrate the application of the invention in a general, illustrative sense, and are not intended to limit the invention to a particular type of expression construct.

The term "expression polypeptide" means a polypeptide encoded by a gene on an expression construct.

The term "expression system" means any in vivo or in vitro biological system that is used to produce one or more gene product encoded by a polynucleotide.

A "gene" refers to a transcriptional unit including a promoter and sequence to be expressed from it as an RNA or protein. The sequence to be expressed can be genomic or cDNA among other possibilities. Other elements, such as introns, and other regulatory sequences may or may not be present.

A "gene transfer system" comprises a vector or gene transfer vector, or a polynucleotide comprising the gene to be transferred which is cloned into a vector (a "gene transfer polynucleotide" or "gene transfer construct"). A gene transfer system may also comprise other features to facilitate the process of gene transfer. For example, a gene transfer system may comprise a vector and a lipid or viral packaging mix for enabling a first polynucleotide to enter a cell, or it may comprise a polynucleotide that includes a transposon and a second polynucleotide sequence encoding a corresponding transposase to enhance productive genomic integration of the transposon. The transposases and transposons of a gene transfer system may be on the same nucleic acid molecule or on different nucleic acid molecules. The transposase of a gene transfer system may be provided as a polynucleotide or as a polypeptide.

Two elements are "heterologous" to one another if not naturally associated. For example, a nucleic acid sequence encoding a protein linked to a heterologous promoter means a promoter other than that which naturally drives expression of the protein. A heterologous nucleic acid flanked by transposon ends or ITRs means a heterologous nucleic acid not naturally flanked by those transposon ends or ITRs, such as a nucleic acid encoding a polypeptide other than a transposase, including an antibody heavy or light chain. A

nucleic acid is heterologous to a cell if not naturally found in the cell or if naturally found in the cell but in a different location (e.g., episomal or different genomic location) than the location described.

The term “host” means any prokaryotic or eukaryotic organism that can be a recipient of a nucleic acid. A “host,” as the term is used herein, includes prokaryotic or eukaryotic organisms that can be genetically engineered. For examples of such hosts, see Maniatis et al., *Molecular Cloning*, A Laboratory Manual, Cold Spring Harbor Laboratory, Cold Spring Harbor, N.Y. (1982). As used herein, the terms “host,” “host cell,” “host system” and “expression host” can be used interchangeably.

A “hyperactive” transposase is a transposase that is more active than the naturally occurring transposase from which it is derived. “Hyperactive” transposases are thus not naturally occurring sequences.

“Integration defective” or “transposition defective” means a transposase that can excise its corresponding transposon, but that integrates the excised transposon at a lower frequency into the host genome than a corresponding naturally occurring transposase.

An “IRES” or “internal ribosome entry site” means a specialized sequence that directly promotes ribosome binding, independent of a cap structure.

An “isolated” polypeptide or polynucleotide means a polypeptide or polynucleotide that has been either removed from its natural environment, produced using recombinant techniques, or chemically or enzymatically synthesized. Polypeptides or polynucleotides of this invention may be purified, that is, essentially free from any other polypeptide or polynucleotide and associated cellular products or other impurities.

The terms “nucleoside” and “nucleotide” include those moieties which contain not only the known purine and pyrimidine bases, but also other heterocyclic bases which have been modified. Such modifications include methylated purines or pyrimidines, acylated purines or pyrimidines, or other heterocycles. Modified nucleosides or nucleotides can also include modifications on the sugar moiety, for example, where one or more of the hydroxyl groups are replaced with halogen, aliphatic groups, or is functionalized as ethers, amines, or the like. The term “nucleotidic unit” is intended to encompass nucleosides and nucleotides.

An “Open Reading Frame” or “ORF” means a portion of a polynucleotide that, when translated into amino acids, contains no stop codons. The genetic code reads DNA sequences in groups of three base pairs, which means that a double-stranded DNA molecule can read in any of six possible reading frames—three in the forward direction and three in the reverse. An ORF typically also includes an initiation codon at which translation may start.

The term “operably linked” refers to functional linkage between two sequences such that one sequence modifies the behavior of the other. For example, a first polynucleotide comprising a nucleic acid expression control sequence (such as a promoter, IRES sequence, enhancer or array of transcription factor binding sites) and a second polynucleotide are operably linked if the first polynucleotide affects transcription and/or translation of the second polynucleotide. Similarly, a first amino acid sequence comprising a secretion signal or a subcellular localization signal and a second amino acid sequence are operably linked if the first amino acid sequence causes the second amino acid sequence to be secreted or localized to a subcellular location.

The term “orthogonal” refers to a lack of interaction between two systems. A first transposon and its correspond-

ing first transposase and a second transposon and its corresponding second transposase are orthogonal if the first transposase does not excise or transpose the second transposon and the second transposase does not excise or transpose the first transposon.

The term “overhang” or “DNA overhang” means the single-stranded portion at the end of a double-stranded DNA molecule. Complementary overhangs are those which will base-pair with each other.

A “piggy Bac-like transposase” means a transposase with at least 20% sequence identity as identified using the TBLASTN algorithm to the piggy Bac transposase from *Trichoplusia ni* (SEQ ID NO: 45), and as more fully described in Sakar, A. et. al., (2003). *Mol. Gen. Genomics* 270:173-180. “Molecular evolutionary analysis of the widespread piggy Bac transposon family and related ‘domesticated’ species”, and further characterized by a DDE-like DDD motif, with aspartate residues at positions corresponding to D268, D346, and D447 of *Trichoplusia ni* piggyBac transposase on maximal alignment. Piggy Bac-like transposases are also characterized by their ability to excise their transposons precisely with a high frequency. A “piggyBac-like transposon” means a transposon having transposon ends which are the same or at least 80% and preferably at least 90, 95, 96, 97, 98 or 99% or 100% identical to the transposon ends of a naturally occurring transposon that encodes a piggy Bac-like transposase. A piggy Bac-like transposon includes an inverted terminal repeat (ITR) sequence of approximately 12-16 bases at each end, and is flanked on each side by a 4 base sequence corresponding to the integration target sequence which is duplicated on transposon integration (the Target Site Duplication or Target Sequence Duplication or TSD). PiggyBac-like transposons and transposases occur naturally in a wide range of organisms including *Argyrogramma agnate* (GU477713), *Anopheles gambiae* (XP_312615; XP_320414; XP_310729), *Aphis gossypii* (GU329918), *Acyrtosiphon pisum* (XP_001948139), *Agrotis ipsilon* (GU477714), *Bombyx mori* (BAD11135), *Ciona intestinalis* (XP_002123602), *Chilo suppressalis* (JX294476), *Drosophila melanogaster* (AAL39784), *Daphnia pulicaria* (AAM76342), *Helicoverpa armigera* (ABS18391), *Homo sapiens* (NP_689808), *Heliothis virescens* (ABD76335), *Macdunnoughia crassisigna* (EU287451), *Macaca fascicularis* (ABI79012), *Mus musculus* (NP_741958), *Pectinophora gossypiella* (GU270322), *Rattus norvegicus* (XP_220453), *Tribolium castaneum* (XP_001814566) and *Trichoplusia ni* (AAA87375) and *Xenopus tropicalis* (BAF82026), although transposition activity has been described for almost none of these.

The terms “polynucleotide,” “oligonucleotide,” “nucleic acid” and “nucleic acid molecule” are used interchangeably to refer to a polymeric form of nucleotides of any length, and may comprise ribonucleotides, deoxyribonucleotides, analogs thereof, or mixtures thereof. This term refers only to the primary structure of the molecule. Thus, the term includes triple-, double- and single-stranded deoxyribonucleic acid (“DNA”), as well as triple-, double- and single-stranded ribonucleic acid (“RNA”). It also includes modified, for example by alkylation, and/or by capping, and unmodified forms of the polynucleotide. More particularly, the terms “polynucleotide,” “oligonucleotide,” “nucleic acid” and “nucleic acid molecule” include polydeoxyribonucleotides (containing 2-deoxy-D-ribose), polyribonucleotides (containing D-ribose), including tRNA, IRNA, hRNA, siRNA and mRNA, whether spliced or unspliced, any other type of polynucleotide which is an N- or C-glycoside of a purine or pyrimidine base, and other polymers containing non-nucleo-

tidic backbones, for example, polyamide (for example, peptide nucleic acids ("PNAs")) and polymorpholino (commercially available from the Anti-Virals, Inc., Corvallis, Oreg., as Neugene) polymers, and other synthetic sequence-specific nucleic acid polymers providing that the polymers contain nucleobases in a configuration which allows for base pairing and base stacking, such as is found in DNA and RNA. There is no intended distinction in length between the terms "polynucleotide," "oligonucleotide," "nucleic acid" and "nucleic acid molecule," and these terms are used interchangeably herein. These terms refer only to the primary structure of the molecule. Thus, these terms include, for example, 3'-deoxy-2', 5'-DNA, oligodeoxyribonucleotide N3' P5' phosphoramidates, 2'-O-alkyl-substituted RNA, double- and single-stranded DNA, as well as double- and single-stranded RNA, and hybrids thereof including for example hybrids between DNA and RNA or between PNAs and DNA or RNA, and also include known types of modifications, for example, labels, alkylation, "caps," substitution of one or more of the nucleotides with an analog, internucleotide modifications such as, for example, those with uncharged linkages (for example, methyl phosphonates, phosphotriesters, phosphoramidates, carbamates, or the like) with negatively charged linkages (for example, phosphorothioates, phosphorodithioates, or the like), and with positively charged linkages (for example, aminoalkylphosphoramidates, aminoalkylphosphotriesters), those containing pendant moieties, such as, for example, proteins (including enzymes (for example, nucleases), toxins, antibodies, signal peptides, poly-L-lysine, or the like), those with intercalators (for example, acridine, psoralen, or the like), those containing chelates (of, for example, metals, radioactive metals, boron, oxidative metals, or the like), those containing alkylators, those with modified linkages (for example, alpha anomeric nucleic acids, or the like), as well as unmodified forms of the polynucleotide or oligonucleotide.

A "promoter" means a nucleic acid sequence sufficient to direct transcription of an operably linked nucleic acid molecule. A promoter can be used with or without other transcription control elements (for example, enhancers) that are sufficient to render promoter-dependent gene expression controllable in a cell type-specific, tissue-specific, or temporal-specific manner, or that are inducible by external signals or agents; such elements may be within the 3' region of a gene or within an intron. Desirably, a promoter is operably linked to a nucleic acid sequence, for example, a cDNA or a gene sequence, or an effector RNA coding sequence, in such a way as to enable expression of the nucleic acid sequence, or a promoter is provided in an expression cassette into which a selected nucleic acid sequence to be transcribed can be conveniently inserted. A regulatory element such as promoter active in a mammalian cells means a regulatory element configurable to result in a level of expression of at least 1 transcript per cell in a mammalian cell into which the regulatory element has been introduced.

The term "selectable marker" means a polynucleotide segment or expression product thereof that allows one to select for or against a molecule or a cell that contains it, often under particular conditions. These markers can encode an activity, such as, but not limited to, production of RNA, peptide, or protein, or can provide a binding site for RNA, peptides, proteins, inorganic and organic compounds or compositions. Examples of selectable markers include but are not limited to: (1) DNA segments that encode products which provide resistance against otherwise toxic compounds

(e.g., antibiotics); (2) DNA segments that encode products which are otherwise lacking in the recipient cell (e.g., tRNA genes, auxotrophic markers); (3) DNA segments that encode products which suppress the activity of a gene product; (4) DNA segments that encode products which can be readily identified (e.g., phenotypic markers such as beta-galactosidase, green fluorescent protein (GFP), and cell surface proteins); (5) DNA segments that bind products which are otherwise detrimental to cell survival and/or function; (6) DNA segments that otherwise inhibit the activity of any of the DNA segments described in Nos. 1-5 above (e.g., antisense oligonucleotides); (7) DNA segments that bind products that modify a substrate (e.g. restriction endonucleases); (8) DNA segments that can be used to isolate a desired molecule (e.g. specific protein binding sites); (9) DNA segments that encode a specific nucleotide sequence which can be otherwise non-functional (e.g., for PCR amplification of subpopulations of molecules); and/or (10) DNA segments, which when absent, directly or indirectly confer sensitivity to particular compounds.

Sequence identity can be determined by aligning sequences using algorithms, such as BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package Release 7.0, Genetics Computer Group, 575 Science Dr., Madison, Wis.), using default gap parameters, or by inspection, and the best alignment (i.e., resulting in the highest percentage of sequence similarity over a comparison window). Percentage of sequence identity is calculated by comparing two optimally aligned sequences over a window of comparison, determining the number of positions at which the identical residues occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of matched and mismatched positions not counting gaps in the window of comparison (i.e., the window size), and multiplying the result by 100 to yield the percentage of sequence identity. Unless otherwise indicated the window of comparison between two sequences is defined by the entire length of the shorter of the two sequences.

A "target nucleic acid" is a nucleic acid into which a transposon is to be inserted. Such a target can be part of a chromosome, episome or vector.

An "integration target sequence" or "target sequence" or "target site" for a transposase is a site or sequence in a target DNA molecule into which a transposon can be inserted by a transposase. The piggyBac transposase from *Trichoplusia ni* inserts its transposon predominantly into the target sequence 5'-TTAA-3'. Other useable target sequences for piggyBac transposons are 5'-CTAA-3', 5'-TTAG-3', 5'-ATAA-3', 5'-TCAA-3', 5'-AGTT-3', 5'-ATTA-3', 5'-GTTA-3', 5'-TTGA-3', 5'-TTTA-3', 5'-TTAC-3', 5'-ACTA-3', 5'-AGGG-3', 5'-CTAG-3', 5'-GTAA-3', 5'-AGGT-3', 5'-ATCA-3', 5'-CTCC-3', 5'-TAAA-3', 5'-TCTC-3', 5'-TGAA-3', 5'-AAAT-3', 5'-AATC-3', 5'-ACAA-3', 5'-ACAT-3', 5'-ACTC-3', 5'-AGTG-3', 5'-ATAG-3', 5'-CAAA-3', 5'-CACA-3', 5'-CATA-3', 5'-CCAG-3', 5'-CCCA-3', 5'-CGTA-3', 5'-CTGA-3', 5'-GTCC-3', 5'-TAAG-3', 5'-TCTA-3', 5'-TGAG-3', 5'-TGTT-3', 5'-TTCA-3', 5'-TTCT-3' and 5'-TTTT-3' (Li et al., 2013. Proc. Natl. Acad. Sci vol. 110, no. 6, E478-487) and 5'-TTAT. PiggyBac-like transposases transpose their transposons using a cut-and-paste mechanism, which results in duplication of their 4 base pair target sequence on insertion into a DNA molecule. The target sequence is thus found on each side of an integrated piggy Bac-like transposon.

The term “translation” refers to the process by which a polypeptide is synthesized by a ribosome ‘reading’ the sequence of a polynucleotide.

A ‘transposase’ is a polypeptide that catalyzes the excision of a corresponding transposon from a donor polynucleotide, for example a vector, and (providing the transposase is not integration-deficient) the subsequent integration of the transposon into a target nucleic acid. A “*Heliothis* transposase” means a transposase with at least 80, 90, 95, 96, 7, 98, 99 or 100% sequence identity to SEQ ID NO: 39, including hyperactive variants of SEQ ID NO: 39, that are able to transposase a corresponding transposon. An “*Agrotis* transposase” means a transposase with at least 80, 90, 95, 96, 97, 98, 99 or 100% sequence identity to SEQ ID NO: 40, including hyperactive variants of SEQ ID NO: 40, that are able to transposase a corresponding transposon. A “*Helicoverpa* transposase” means a transposase with at least 80, 90, 95, 96, 7, 98, 99 or 100% sequence identity to SEQ ID NO: 41, including hyperactive variants of SEQ ID NO: 41, that are able to transposase a corresponding transposon. A hyperactive transposase is a transposase that is more active than the naturally occurring transposase from which it is derived, for excision activity or transposition activity or both. A hyperactive transposase is preferably at least 1.5-fold more active, or at least 2-fold more active, or at least 5-fold more active, or at least 10-fold more active than the naturally occurring transposase from which it is derived, e.g., 2-5 fold or 2-10 fold. A transposase may or more not be fused to one or more additional domains such as a nuclear localization sequence or DNA binding protein.

The term “transposition” is used herein to mean the action of a transposase in excising a transposon from one polynucleotide and then integrating it, either into a different site in the same polynucleotide, or into a second polynucleotide.

The term “transposon” means a polynucleotide that can be excised from a first polynucleotide, for instance, a vector, and be integrated into a second position in the same polynucleotide, or into a second polynucleotide, for instance, the genomic or extrachromosomal DNA of a cell, by the action of a corresponding trans-acting transposase. A transposon comprises a first transposon end and a second transposon end, which are polynucleotide sequences recognized by and transposed by a transposase. A transposon usually further comprises a first polynucleotide sequence between the two transposon ends, such that the first polynucleotide sequence is transposed along with the two transposon ends by the action of the transposase. This first polynucleotide in natural transposons frequently comprises an open reading frame encoding a corresponding transposase that recognizes and transposes the transposon. Transposons of the present invention are “synthetic transposons” comprising a heterologous polynucleotide sequence which is transposable by virtue of its juxtaposition between two transposon ends. Synthetic transposons may or may not further comprise flanking polynucleotide sequence(s) outside the transposon ends, such as a sequence encoding a transposase, a vector sequence or sequence encoding a selectable marker.

The term “transposon end” means the cis-acting nucleotide sequences that are sufficient for recognition by and transposition by a corresponding transposase. Transposon ends of piggyBac-like transposons comprise perfect or imperfect repeats such that the respective repeats in the two transposon ends are reverse complements of each other. These are referred to as inverted terminal repeats (ITR) or terminal inverted repeats (TIR). A transposon end may or may not include additional sequence proximal to the ITR that promotes or augments transposition.

The term “vector” or “DNA vector” or “gene transfer vector” refers to a polynucleotide that is used to perform a “carrying” function for another polynucleotide. For example, vectors are often used to allow a polynucleotide to be propagated within a living cell, or to allow a polynucleotide to be packaged for delivery into a cell, or to allow a polynucleotide to be integrated into the genomic DNA of a cell. A vector may further comprise additional functional elements, for example it may comprise a transposon.

5.2 Description

5.2.1 Genomic Integration

Expression of a gene from a heterologous polynucleotide in a eukaryotic host cell can be improved if the heterologous polynucleotide is integrated into the genome of the host cell. Integration of a polynucleotide into the genome of a host cell also generally makes it stably heritable, by subjecting it to the same mechanisms that ensure the replication and division of genomic DNA. Such stable heritability is desirable for achieving good and consistent expression over long growth periods. This is particularly important for cell therapies in which cells are genetically modified and then placed into the body. It is also important for the manufacturing of biomolecules, particularly for therapeutic applications where the stability of the host and consistency of expression levels is also important for regulatory purposes. Cells with gene transfer vectors, including transposon-based gene transfer vectors, integrated into their genomes are thus an important embodiment of the invention.

Heterologous polynucleotides may be more efficiently integrated into a target genome if they are part of a transposon (i.e., positioned between transposon ITRs), for example so that they may be integrated by a transposase. A particular benefit of a transposon is that the entire polynucleotide between the transposon ITRs is integrated. A transposon comprising target sites flanking ITRs flanking a heterologous polynucleotide integrates at a target site in a genome to result in the genome containing the heterologous polynucleotide flanked by the ITRs, flanked by target sites. This is in contrast to random integration, where a polynucleotide introduced into a eukaryotic cell is often fragmented at random in the cell, and only parts of the polynucleotide become incorporated into the target genome, usually at a low frequency. The piggyBac transposon from the looper moth *Trichoplusia ni* has been shown to be transposed by its transposase in cells from many organisms (see e.g. Keith et al (2008) BMC Molecular Biology 9:72 “Analysis of the piggyBac transposase reveals a functional nuclear targeting signal in the 94 c-terminal residues”). Heterologous polynucleotides incorporated into piggyBac-like transposons may be integrated into eukaryotic cells including animal cells, fungal cells or plant cells. Preferred animal cells can be vertebrate or invertebrate. Preferred vertebrate cells include cells from mammals including rodents such as rats, mice, and hamsters; ungulates, such as cows, goats or sheep; and swine. Preferred vertebrate cells also include cells from human tissues and human stem cells. Target cells types include hepatocytes, neural cells, muscle cells, blood cells, embryonic stem cells, somatic stem cells, hematopoietic cells, embryos, zygotes, sperm cells (some of which are open to be manipulated in an in vitro setting) and immune cells including lymphocytes such as T cells, B cells and natural killer cells, T-helper cells, antigen-presenting cells, dendritic cells, neutrophils and macrophages. Preferred cells can be pluripotent cells (cells whose descendants can differentiate into several restricted cell types, such as hema-

topoietic stem cells or other stem cells) or totipotent cells (i.e., a cell whose descendants can become any cell type in an organism, e.g., embryonic stem cells). Preferred culture cells are Chinese hamster ovary (CHO) cells or Human embryonic kidney (HEK293) cells. Preferred fungal cells are yeast cells including *Saccharomyces cerevisiae* and *Pichia pastoris*. Preferred plant cells are algae, for example *Chlorella*, tobacco, maize and rice (Nishizawa-Yokoi et al (2014) Plant J. 77:454-63 "Precise marker excision system using an animal derived piggyBac transposon in plants").

Preferred gene transfer systems comprise a transposon in combination with a corresponding transposase protein that transposases the transposon, or a nucleic acid that encodes the corresponding transposase protein and is expressible in the target cell. A preferred gene transfer system comprises a synthetic *Heliothis* transposon and a corresponding *Heliothis* transposase. A preferred gene transfer system comprises a synthetic *Helicoverpa* transposon and a corresponding *Helicoverpa* transposase. A preferred gene transfer system comprises a synthetic *Agrotis* transposon and a corresponding *Agrotis* transposase.

A transposase protein can be introduced into a cell as a protein or as a nucleic acid encoding the transposase, for example as a ribonucleic acid, including mRNA or any polynucleotide recognized by the translational machinery of a cell; as DNA, e.g. as extrachromosomal DNA including episomal DNA; as plasmid DNA, or as viral nucleic acid. Furthermore, the nucleic acid encoding the transposase protein can be transfected into a cell as a nucleic acid vector such as a plasmid, or as a gene expression vector, including a viral vector. The nucleic acid can be circular or linear. mRNA encoding the transposase may be prepared using DNA in which a gene encoding the transposase is operably linked to a heterologous promoter, such as the bacterial T7 promoter, which is active in vitro. DNA encoding the transposase protein can be stably inserted into the genome of the cell or into a vector for constitutive or inducible expression. Where the transposase protein is transfected into the cell or inserted into the vector as DNA, the transposase encoding sequence is preferably operably linked to a heterologous promoter. There are a variety of promoters that could be used including constitutive promoters, cell-type specific promoters, organism-specific promoters, tissue-specific promoters, inducible promoters, and the like. Where DNA encoding the transposase is operably linked to a promoter and transfected into a target cell, the promoter should be operable in the target cell. For example if the target cell is a mammalian cell, the promoter should be operable in a mammalian cell; if the target cell is a yeast cell, the promoter should be operable in a yeast cell; if the target cell is an insect cell, the promoter should be operable in an insect cell; if the target cell is a human cell, the promoter should be operable in a human cell; if the target cell is a human immune cell, the promoter should be operable in a human immune cell. All DNA or RNA sequences encoding piggy Bac-like transposase proteins are expressly contemplated. Alternatively, the transposase may be introduced into the cell directly as protein, for example using cell-penetrating peptides (e.g. as described in Ramsey and Flynn (2015) Pharmacol. Ther. 154:78-86 "Cell-penetrating peptides transport therapeutics into cells"); using small molecules including salt plus propanebetaine (e.g. as described in Astolfo et al (2015) Cell 161:674-690); or electroporation (e.g. as described in Morgan and Day (1995) Methods in Molecular Biology 48:63-71 "The introduction of proteins into mammalian cells by electroporation").

It is possible to insert the transposon into DNA of a cell through non-homologous recombination through a variety of reproducible mechanisms, and even without the activity of a transposase. The transposons described herein can be used for gene transfer regardless of the mechanisms by which the genes are transferred.

5.2.5 Gene Transfer Systems

Gene transfer systems comprise a polynucleotide to be transferred to a host cell. Preferably the polynucleotide comprises a *Heliothis* transposon or a *Helicoverpa* transposon or an *Agrotis* transposon wherein the polynucleotide is to be integrated into the genome of a target cell.

When there are multiple components of a gene transfer system, for example the one or more polynucleotides comprising genes for expression in the target cell and optionally comprising transposon ends, and a transposase (which may be provided either as a protein or encoded by a nucleic acid), these components can be transfected into a cell at the same time, or sequentially. For example, a transposase protein or its encoding nucleic acid may be transfected into a cell prior to, simultaneously with or subsequent to transfection of a corresponding transposon. Additionally, administration of either component of the gene transfer system may occur repeatedly, for example, by administering at least two doses of this component.

Any of the transposase proteins described herein may be encoded by polynucleotides including RNA or DNA. Similarly, the nucleic acid encoding the transposase protein or the transposon of this invention can be transfected into the cell as a linear fragment or as a circularized fragment, either as a plasmid or as recombinant viral DNA.

A *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may be provided as a DNA molecule expressible in the target cell. The sequence encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase should be operably linked to heterologous sequences that enable expression of the transposase in the target cell. A sequence encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may be operably linked to a heterologous promoter that is active in the target cell. For example, if the target cell is a mammalian cell, then the promoter should be active in a mammalian cell. If the target is a vertebrate cell, the promoter should be active in a vertebrate cell. If the target cell is a plant cell, the promoter should be active in a plant cell. If the promoter is an insect cell, the promoter should be active in an insect cell. The sequence encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may also be operably linked to other sequence elements required for expression in the target cell, for example polyadenylation sequences, terminator sequences etc.

A *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may be provided as an mRNA expressible in the target cell. mRNA is preferably prepared in an in vitro transcription reaction. For in vitro transcription, a sequence encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase is operably linked to a promoter that is active in an in vitro transcription reaction. Exemplary promoters active in an in vitro transcription reaction include a T7 promoter (5'-TAATACGACTCAC-TATAG-3') which enables transcription by T7 RNA polymerase, a T3 promoter (5'-AATTAACCCCTACTAAAG-3') which enables transcription by T3 RNA polymerase and an SP6 promoter (5'-ATTAGGTGACACTATAG-3') which enables transcription by SP6 RNA polymerase. Variants of these promoters and other promoters that can be used for in

vitro transcription may also be operably linked to a sequence encoding an *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase.

If the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase is provided as a polynucleotide (either DNA or mRNA) encoding the transposase, then it is advantageous to improve the expressibility of the transposase in the target cell. It is therefore advantageous to use a sequence other than a naturally occurring sequence to encode the transposase, in other words, to use codon-preferences of the cell type in which expression is to be performed. For example, if the target cell is a mammalian cell, then the codons should be biased toward the preferences seen in a mammalian cell. If the target is a vertebrate cell, then the codons should be biased toward the preferences seen in the particular vertebrate cell. If the target cell is a plant cell, then the codons should be biased toward the preferences seen in a plant cell. If the promoter is an insect cell, then the codons should be biased toward the preferences seen in an insect cell.

Preferable RNA molecules include those with appropriate cap structures to enhance translation in a eukaryotic cell, polyadenylic acid and other 3' sequences that enhance mRNA stability in a eukaryotic cell and optionally substitutions to reduce toxicity effects on the cell, for example substitution of uridine with pseudouridine, and substitution of cytosine with 5-methyl cytosine. mRNA encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may be prepared such that it has a 5'-cap structure to improve expression in a target cell. Exemplary cap structures are a cap analog (G(5')ppp(5')G), an anti-reverse cap analog (3'-O-Me-m7G(5')ppp(5')G), a clean cap (m7G(5')ppp(5')(2'OMeA)pG), an mCap (m7G(5')ppp(5')G). mRNA encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may be prepared such that some bases are partially or fully substituted, for example uridine may be substituted with pseudo-uridine, cytosine may be substituted with 5-methyl-cytosine. Any combinations of these caps and substitutions may be made.

The components of the gene transfer system may be transfected into one or more cells by techniques such as particle bombardment, electroporation, microinjection, combining the components with lipid-containing vesicles, such as cationic lipid vesicles, DNA condensing reagents (example, calcium phosphate, polylysine or polyethyleneimine), and inserting the components (that is the nucleic acids thereof into a viral vector and contacting the viral vector with the cell. Where a viral vector is used, the viral vector can include any of a variety of viral vectors known in the art including viral vectors selected from the group consisting of a retroviral vector, an adenovirus vector or an adeno-associated viral vector. The gene transfer system may be formulated in a suitable manner as known in the art, or as a pharmaceutical composition or kit.

5.2.3 Sequence Elements in Gene Transfer Systems

Expression of genes from a gene transfer polynucleotide such as a piggyBac-like transposon, including a *Heliothis* transposon or a *Helicoverpa* transposon or an *Agrotis* transposon, integrated into a host cell genome is often strongly influenced by the chromatin environment into which it integrates. Polynucleotides that are integrated into euchromatin have higher levels of expression than those that are either integrated into heterochromatin, or which become silenced following their integration. Silencing of a heterologous polynucleotide may be reduced if it comprises a chromatin control element. It is thus advantageous for gene transfer polynucleotides (including any of the transposons

described herein) to comprise chromatin control elements such as sequences that prevent the spread of heterochromatin (insulators). Advantageous gene transfer polynucleotides including a *Heliothis* transposon or a *Helicoverpa* transposon or an *Agrotis* transposon comprise an insulator sequence that is at least 95% identical to a sequence selected from one of SEQ ID NOS: 684-690, they may also comprise ubiquitously acting chromatin opening elements (UCOE) or stabilizing and anti-repressor elements (STARs), to increase long-term stable expression from the integrated gene transfer polynucleotide. Advantageous gene transfer polynucleotides may further comprise a matrix attachment region for example a sequence that is at least 95% identical to a sequence selected from one of SEQ ID NOS: 691-701.

In some cases, it is advantageous for a gene transfer polynucleotide to comprise two insulators, one on each side of the heterologous polynucleotide that contains the sequence(s) to be expressed and within the transposon ITRs. The insulators may be the same, or they may be different. Particularly advantageous gene transfer polynucleotides comprise an insulator sequence that is at least 95% identical to a sequence selected from one of SEQ ID NO: 689 or SEQ ID NO: 690 and an insulator sequence that is at least 95% identical to a sequence selected from one of SEQ ID NOS: 684-688. Insulators also shield expression control elements from one another. For example, when a gene transfer polynucleotide comprises genes encoding two open reading frames, each operably linked to a different promoter, one promoter may reduce expression from the other in a phenomenon known as transcriptional interference. Interposing an insulator sequence that is at least 95% identical to a sequence selected from one of SEQ ID NOS: 684-690 between the two transcriptional units can reduce this interference, increasing expression from one or both promoters.

Preferred gene transfer vectors comprise expression elements capable of driving high levels of gene expression. In eukaryotic cells, gene expression is regulated by several different classes of elements, including enhancers, promoters, introns, RNA export elements, polyadenylation sequences and transcriptional terminators.

Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells comprise an enhancer operably linked to a heterologous gene. Advantageous gene transfer polynucleotides for the transfer of genes for expression into mammalian cells comprise an enhancer from immediate early genes 1, 2 or 3 of cytomegalovirus (CMV) from either human, primate or rodent cells (for example sequences at least 95% identical to any of SEQ ID NOS: 702-720), an enhancer from the adenoviral major late protein enhancer (for example sequences at least 95% identical to SEQ ID NO: 721), or an enhancer from SV40 (for example sequences at least 95% identical to SEQ ID NO: 722), operably linked to a heterologous gene.

Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells comprise a promoter operably linked to a heterologous gene. Advantageous gene transfer polynucleotides for the transfer of genes for expression into mammalian cells comprise an EF1a promoter from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster, (for example any of SEQ ID NOS: 723-744); a promoter from the immediate early genes 1, 2 or 3 of cytomegalovirus (CMV) from either human, primate or rodent cells (for example any of SEQ ID NOS: 745-755); a promoter for eukaryotic elongation factor 2 (EEF2) from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster, (for example any of SEQ ID NOS: 756-

766); a GAPDH promoter from any mammalian or yeast species (for example any of SEQ ID NOS: 777-793), an actin promoter from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster (for example any of SEQ ID NOS: 767-776); a PGK promoter from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster (for example any of SEQ ID NOS: 794-800), or a ubiquitin promoter (for example SEQ ID NO: 801-804), operably linked to a heterologous gene. The promoter may be operably linked to i) a heterologous open reading frame; ii) a nucleic acid encoding a selectable marker; iii) a nucleic acid encoding a counter-selectable marker; iii) a nucleic acid encoding a regulatory protein; iv) a nucleic acid encoding an inhibitory RNA.

Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells comprise an intron within a heterologous polynucleotide spliceable in a target cell. Advantageous gene transfer polynucleotides for the transfer of genes for expression into mammalian cells comprise an intron from immediate early genes 1, 2 or 3 of cytomegalovirus (CMV) from either human, primate or rodent cells (for example sequences at least 95% identical to any of SEQ ID NOS: 813-823), an intron from EF1a from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster, (for example sequences at least 95% identical to any of SEQ ID NOS: 833-845), an intron from EEF2 from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster, (for example sequences at least 95% identical to any of SEQ ID NOS: 865-872); an intron from actin from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster (for example sequences at least 95% identical to any of SEQ ID NOS: 846-859), a GAPDH intron from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster (for example sequences at least 95% identical to any of SEQ ID NOS: 860-862); an intron comprising the adenoviral major late protein enhancer for example sequences at least 95% identical to SEQ ID NO: 863-864) or a hybrid/synthetic intron (for example sequences at least 95% identical to any of SEQ ID NOS: 824-832) within a heterologous polynucleotide.

Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells comprise an enhancer and promoter, operably linked to a heterologous coding sequence. Such gene transfer polynucleotides may comprise combinations of enhancers and promoters in which an enhancer from one gene is combined with a promoter from a different gene, that is the enhancer is heterologous to the promoter. For example, for the transfer of genes for expression into mammalian cells, an immediate early CMV enhancer from rodent or human or primate (such as a sequence selected from SEQ ID NOS: 702-720) is advantageously followed by a promoter from an EF1a gene (such as a sequence selected from SEQ ID NOS: 723-744), or a promoter from a heterologous CMV gene (such as a sequence selected from SEQ ID NOS: 745-755), or a promoter from an EEF2 gene (such as a sequence selected from SEQ ID NOS: 756-766), or a promoter from an actin gene (such as a sequence selected from SEQ ID NOS: 767-776), or a promoter from a GAPDH gene (such as a sequence selected from SEQ ID NOS: 777-793) operably linked to a heterologous sequence.

Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells comprise an operably linked promoter and an intron, operably linked to a heterologous open reading frame. Such gene transfer

polynucleotides may comprise combinations of promoters and introns in which a promoter from one gene is combined with an intron from a different gene, that is the intron is heterologous to the promoter. For example, for the transfer of genes for expression into mammalian cells, an immediate early CMV promoter from rodent or human or primate (such as a sequence selected from SEQ ID NOS: 745-755) is advantageously followed by an intron from an EF1a gene (such as a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS: 833-845) or an intron from an EEF2 gene (such as a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS: 865-872), or an intron from an actin gene (such as a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS: 846-859) operably linked to a heterologous sequence.

Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells, comprise composite transcriptional initiation regulatory elements comprising promoters that are operably linked to enhancers and/or introns, and the composite transcriptional initiation regulatory element is operably linked to a heterologous sequence. Examples of advantageous composite transcriptional initiation regulatory elements that may be operably linked to a heterologous sequence in gene transfer polynucleotides for the transfer of genes for expression into mammalian cells are sequences selected from SEQ ID NOS: 874-966.

Expression of two open reading frames from a single polynucleotide can be accomplished by operably linking the expression of each open reading frame to a separate promoter, each of which may optionally be operably linked to enhancers and introns as described above. This is particularly useful when expressing two polypeptides that need to interact at specific molar ratios, such as chains of an antibody or chains of a bispecific antibody, or a receptor and its ligand. It is often advantageous to prevent transcriptional promoter interference by placing a genetic insulator between the two open reading frames, for example to the 3' of the polyadenylation sequence operably linked to the first open reading frame and to the 5' of the promoter operably linked to the second open reading frame encoding the second polypeptide. Transcriptional promoter interference may also be prevented by effectively terminating transcription of the first gene. In many eukaryotic cells the use of strong poly A signal sequences between two open reading frames will reduce transcriptional promote interference. Examples of poly A signal sequences that can be used to effectively terminate transcription are given as SEQ ID NOS: 967-996. Advantageous gene transfer polynucleotides comprise a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS: 967-996 operably linked to a heterologous open reading frame. Advantageous composite regulatory elements for the termination of transcription of a first gene and the initiation of transcription of a second gene include sequences given as SEQ ID NOS: 997-1180. Particularly advantageous gene transfer polynucleotides for the transfer of a first and a second open reading frame for co-expression into mammalian cells comprise a sequence at least 90% identical or at least 95% identical or at least 99% identical to or 100% identical to a sequence selected from SEQ ID NOS: 997-1180, separating two heterologous open reading frames.

5.2.4 Selection of Target Cells Comprising Gene Transfer Polynucleotides

A target cell whose genome comprises a stably integrated transfer polynucleotide may be identified, if the gene transfer polynucleotide comprises an open reading frame encod-

ing a selectable marker, by exposing the target cells to conditions that favor cells expressing the selectable marker ("selection conditions"). It is advantageous for a gene transfer polynucleotide to comprise an open reading frame encoding a selectable marker such as an enzyme that confers resistance to antibiotics such as neomycin (resistance conferred by an aminoglycoside 3'-phosphotransferase e.g. a sequence selected from SEQ ID NOs: 513-516), puromycin (resistance conferred by puromycin acetyltransferase e.g. a sequence selected from SEQ ID NOs: 519-521), blasticidin (resistance conferred by a blasticidin acetyltransferase and a blasticidin deaminase e.g. SEQ ID NO: 523), hygromycin B (resistance conferred by hygromycin B phosphotransferase e.g. a sequence selected from SEQ ID NOs: 517-518) and zeocin (resistance conferred by a binding protein encoded by the ble gene, for example SEQ ID NO: 510). Other selectable markers include those that are fluorescent (such as open reading frames encoding GFP, RFP etc.) and can therefore be selected for example using flow cytometry. Other selectable markers include open reading frames encoding transmembrane proteins that are able to bind to a second molecule (protein or small molecule) that can be fluorescently labelled so that the presence of the transmembrane protein can be selected for example using flow cytometry.

A gene transfer polynucleotide may comprise a selectable marker open reading frame encoding glutamine synthetase (GS, for example a sequence selected from SEQ ID NOs: 524-528) which allows selection via glutamine metabolism. Glutamine synthase is the enzyme responsible for the biosynthesis of glutamine from glutamate and ammonia, it is a crucial component of the only pathway for glutamine formation in a mammalian cell. In the absence of glutamine in the growth medium, the GS enzyme is essential for the survival of mammalian cells in culture. Some cell lines, for example mouse myeloma cells do not express sufficient GS enzyme to survive without added glutamine. In these cells a transfected GS open reading frame can function as a selectable marker by permitting growth in a glutamine-free medium. In other cell lines, for example Chinese hamster ovary (CHO) cells express sufficient GS enzyme to survive without exogenously added glutamine. These cell lines can be manipulated by genome editing techniques including CRISPR/Cas9 to reduce or eliminate the activity of the GS enzyme. In all of these cases, GS inhibitors such as methionine sulfoximine (MSX) can be used to inhibit a cell's endogenous GS activity. Selection protocols include introducing a gene transfer polynucleotide comprising sequences encoding a first polypeptide and a glutamine synthase selectable marker, and then treating the cell with inhibitors of glutamine synthase such as methionine sulfoximine. The higher the levels of methionine sulfoximine that are used, the higher the level of glutamine synthase expression is required to allow the cell to synthesize sufficient glutamine to survive. Some of these cells will also show an increased expression of the first polypeptide.

Preferably the GS open reading frame is operably linked to a weak promoter or other sequence elements that attenuate expression as described herein, such that high levels of expression can only occur if many copies of the gene transfer polynucleotide are present, or if they are integrated in a position in the genome where high levels of expression occur. In such cases it may be unnecessary to use the inhibitor methionine sulfoximine: simply synthesizing sufficient glutamine for cell survival may provide a sufficiently stringent selection if expression of the glutamine synthetase is attenuated.

A gene transfer polynucleotide may comprise a selectable marker open reading frame encoding dihydrofolate reductase (DHFR, for example a sequence selected from SEQ ID NOs: 511-512) which is required for catalyzing the reduction of 5,6-dihydrofolate (DHF) to 5,6,7,8-tetrahydrofolate (THF). Some cell lines do not express sufficient DHFR to survive without added hypoxanthine and thymidine (HT). In these cells a transfected DHFR open reading frame can function as a selectable marker by permitting growth in a hypoxanthine and thymidine-free medium. DHFR-deficient cell lines, for example Chinese hamster ovary (CHO) cells can be produced by genome editing techniques including CRISPR/Cas9 to reduce or eliminate the activity of the endogenous DHFR enzyme. DHFR confers resistance to methotrexate (MTX). DHFR can be inhibited by higher levels of methotrexate. Selection protocols include introducing a construct comprising sequences encoding a first polypeptide and a DHFR selectable marker into a cell with or without a functional endogenous DHFR gene, and then treating the cell with inhibitors of DHFR such as methotrexate. The higher the levels of methotrexate that are used, the higher the level of DHFR expression is required to allow the cell to synthesize sufficient DHFR to survive. Some of these cells will also show an increased expression of the first polypeptide. Preferably the DHFR open reading frame is operably linked to a weak promoter or other sequence elements that attenuate expression as described above, such that high levels of expression can only occur if many copies of the gene transfer polynucleotide are present, or if they are integrated in a position in the genome where high levels of expression occur.

High levels of expression may be obtained from genes encoded on gene transfer polynucleotides that are integrated at regions of the genome that are highly transcriptionally active, or that are integrated into the genome in multiple copies, or that are present extrachromosomally in multiple copies. It is often advantageous to operably link the open reading frame encoding the selectable marker to expression control elements that result in low levels of expression of the selectable polypeptide from the gene transfer polynucleotide and/or to use conditions that provide more stringent selection. Under these conditions, for the expression cell to produce sufficient levels of the selectable polypeptide encoded on the gene transfer polynucleotide to survive the selection conditions, the gene transfer polynucleotide can either be present in a favorable location in the cell's genome for high levels of expression, or a sufficiently high number of copies of the gene transfer polynucleotide can be present, such that these factors compensate for the low levels of expression achievable because of the expression control elements.

Genomic integration of transposons in which a selectable marker is operably linked to regulatory elements that only weakly express the marker usually requires that the transposon be inserted into the target genome by a transposase, see for example Section 6.1.4. By operably linking the selectable marker to elements that result in weak expression, cells are selected which either incorporate multiple copies of the transposon, or in which the transposon is integrated at a favorable genomic location for high expression. Using a gene transfer system that comprises a transposon and a corresponding transposase increases the likelihood that cells will be produced with multiple copies of the transposon, or in which the transposon is integrated at a favorable genomic location for high expression. Gene transfer systems comprising a transposon and a corresponding transposase are thus particularly advantageous when the transposon com-

prises a selectable marker operably linked to a weak promoter. A weak promoter in this context is one which does not drive enough transcription of an operably linked selectable marker to permit target cell survival under restrictive conditions if only a single copy of the promoter and selectable marker are integrated into the target cell genome.

A nucleic acid to be expressed as an RNA or protein and a selectable marker may be included on the same gene transfer polynucleotide, but operably linked to different promoters. In this case low expression levels of the selectable marker may be achieved by using a weakly active constitutive promoter such as the phosphoglycerokinase (PGK) promoter (such as a promoter selected from SEQ ID NOs: 794-800), the Herpes Simplex Virus thymidine kinase (HSV-TK) promoter (e.g. a sequence selected from SEQ ID NOs: 805-806), the MC1 promoter (for example SEQ ID NO: 807), a ubiquitin promoter (for example a sequence selected from SEQ ID NOs: 801-804). Other weakly active promoters may be deliberately constructed, for example a promoter attenuated by truncation, such as a truncated SV40 promoter (for example a sequence selected from SEQ ID NO: 808-809), or a promoter attenuated by insertion of a 5'UTR unfavorable for expression (for example a sequence selected from SEQ ID NOS: 811-812) between a promoter and the open reading frame encoding the selectable polypeptide. Particularly advantageous gene transfer polynucleotides comprise a promoter sequence selected from SEQ ID NOS: 794-810, operably linked to an open reading frame encoding a selectable marker.

Expression levels of a selectable marker may also be advantageously reduced by other mechanisms such as the insertion of the SV40 small t antigen intron after the open reading frame for the selectable marker. The SV40 small t intron accepts aberrant 5' splice sites, which can lead to deletions within the preceding open reading frame in a fraction of the spliced mRNAs, thereby reducing expression of the selectable marker. Particularly advantageous gene transfer polynucleotides comprise intron SEQ ID NO: 873, operably linked to an open reading frame encoding a selectable marker. For this mechanism of attenuation to be effective, it is preferable for the open reading frame encoding the selectable marker to comprise an intron donor within its coding region. DNA sequences SEQ ID NOs: 529-532 are exemplary nucleic acid sequences that encode glutamine synthetase sequences with SEQ ID NOs: 524-527 respectively. Each of these nucleic acid sequences comprises an intron donor, and which may be operably linked to the SV40 small t antigen intron by placing the intron into the 3' UTR of the glutamine synthetase open reading frame. Sequence SEQ ID NO: 522 is an exemplary nucleic acid sequence encoding puromycin acetyl transferase SEQ ID NO: 521 which comprises an intron donor, and which may be operably linked to the SV40 small t antigen intron by placing the intron into the 3' UTR of the puromycin open reading frame. Advantageous gene transfer polynucleotides comprise a sequence at least 90% identical or at least 95% identical or at least 99% identical to, or 100% identical to a sequence selected from one of SEQ ID NOs: 529-532 or 522, in combination with SEQ ID NO: 873.

Expression levels of a selectable marker may also be advantageously reduced by other mechanisms such as insertion of an inhibitory 5'-UTR within the transcript, for example SEQ ID NOs: 811-812. Particularly advantageous gene transfer polynucleotides comprise a promoter operably linked to an open reading frame encoding a selectable marker, wherein a sequence that is at least 90% identical or at least 95% identical or at least 99% identical to, or 100%

identical to SEQ ID NO: 811-812 is interposed between the promoter and the selectable marker.

Exemplary nucleic acid sequences comprising the glutamine synthetase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 550-619 or 681-683. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 550-619 or 681-683, upon integration into the genome of a target cell, expresses glutamine synthetase, thereby helping a cell to grow in the absence of added glutamine or in the presence of MSX. Regulatory elements in these sequences have been balanced to produce low levels of expression of glutamine synthetase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 550-619 or 681-683, and they may further comprise a left transposon end and a right transposon end.

Exemplary nucleic acid sequences comprising the blasticidin-S-transferase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 620-626. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 620-626, upon integration into the genome of a target cell, expresses blasticidin-S-transferase, thereby helping a cell to grow in the presence of added blasticidin. Regulatory elements in these sequences have been balanced to produce low levels of expression of blasticidin-S-transferase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 620-626, and they may further comprise a left transposon end and a right transposon end.

Exemplary nucleic acid sequences comprising the hygromycin B phosphotransferase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 627-628. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 627-628, upon integration into the genome of a target cell, will express hygromycin B phosphotransferase, thereby helping a cell to grow in the presence of added hygromycin. Regulatory elements in these sequences have been balanced to produce low levels of expression of hygromycin B phosphotransferase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 627-628, and they may further comprise a left transposon end and a right transposon end.

Exemplary nucleic acid sequences comprising the aminoglycoside 3'-phosphotransferase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 629-631. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 629-631, upon integration into the genome of a target cell, expresses aminoglycoside 3'-phosphotransferase, thereby helping a cell to grow in the presence of added

neomycin. Regulatory elements in these sequences have been balanced to produce low levels of expression of aminoglycoside 3'-phosphotransferase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 629-631, and they may further comprise a left transposon end and a right transposon end.

Exemplary nucleic acid sequences comprising the puromycin acetyltransferase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 632-651 or 659-683. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 632-651 or 659-683, upon integration into the genome of a target cell, expresses puromycin acetyltransferase, thereby helping a cell to grow in the presence of added puromycin. Regulatory elements in these sequences have been balanced to produce low levels of expression of puromycin acetyltransferase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 632-651 or 659-683, and they may further comprise a left transposon end and a right transposon end.

Exemplary nucleic acid sequences comprising the ble gene coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 652-656. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 652-656, upon integration into the genome of a target cell, expresses the ble gene, thereby helping a cell to grow in the presence of added zeocin. Regulatory elements in these sequences have been balanced to produce low levels of expression of ble gene product, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 652-656, and they may further comprise a left transposon end and a right transposon end.

Exemplary nucleic acid sequences comprising the dihydrofolate reductase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 533-549 or 657-680. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 533-549 or 657-680, upon integration into the genome of a target cell, expresses dihydrofolate reductase, thereby helping a cell to grow in the absence of added hypoxanthine and thymidine or in the presence of MTX. Regulatory elements in these sequences have been balanced to produce low levels of expression of dihydrofolate reductase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 533-549 or

657-680, and they may further comprise a left transposon end and a right transposon end.

The use of transposons and transposases in conjunction with weakly expressed selectable markers has several advantages over non-transposon constructs. One is that linkage between expression of the first polypeptide and the selectable marker is better for transposons, because a transposase integrates the entire sequence that lies between the two transposon ends into the genome. In contrast when heterologous DNA is introduced into the nucleus of a eukaryotic cell, for example a mammalian cell, it is gradually broken into random fragments which may either be integrated into the cell's genome, or degraded. Thus if a gene transfer polynucleotide comprising sequences that encode a first polypeptide and a selectable marker is introduced into a population of cells, some cells will integrate the sequences encoding the selectable marker but not those encoding the first polypeptide, and vice versa. Selection of cells expressing high levels of selectable marker is thus only somewhat correlated with cells that also express high levels of the first polypeptide. In contrast, because the transposase integrates all of the sequences between the transposon ends, cells expressing high levels of selectable marker are highly likely to also express high levels of the first polypeptide.

A second advantage of transposons and transposases is that they are much more efficient at integrating DNA sequences into the genome. A much higher fraction of the cell population is therefore likely to integrate one or more copies of the gene transfer polynucleotide into their genomes, so there will be a correspondingly higher likelihood of good stable expression of both the selectable marker and the first polypeptide.

A third advantage of piggy Bac-like transposons and transposases is that piggy Bac-like transposases are biased toward inserting their corresponding transposons into transcriptionally active chromatin. Each cell is therefore likely to integrate the gene transfer polynucleotide into a region of the genome from which genes are well expressed, so there will be a correspondingly higher likelihood of good stable expression of both the selectable marker and the first polypeptide.

5.2.5 Novel Piggybac-Like Transposases

Natural DNA transposons undergo a 'cut and paste' system of replication in which the transposon is excised from a first DNA molecule and inserted into a second DNA molecule. DNA transposons are characterized by inverted terminal repeats (ITRs) and are mobilized by an element-encoded transposase. The piggy Bac transposon/transposase system is particularly useful because of the precision with which the transposon is integrated and excised (see for example "Fraser, M. J. (2001) The TTAA-Specific Family of Transposable Elements: Identification, Functional Characterization, and Utility for Transformation of Insects. Insect Transgenesis: Methods and Applications. A. M. Handler and A. A. James. Boca Raton, Fla., CRC Press: 249-268"; and "US20070204356 A1: PiggyBac constructs in vertebrates" and references therein).

Many sequences with sequence similarity to the piggy Bac transposase from *Trichoplusia ni* have been found in the genomes of phylogenetically distinct species from fungi to mammals, but very few have been shown to possess transposase activity (see for example Wu M, et al (2011) *Genetica* 139:149-54. "Cloning and characterization of piggy Bac-like elements in lepidopteran insects", and references therein).

Two properties of transposases that are of particular interest for genomic modifications are their ability to inte-

grate a polynucleotide into a target genome, and their ability to precisely excise a polynucleotide from a target genome. Both of these properties can be measured with a suitable system.

A system for measuring the first step of transposition, which is excision of a transposon from a first polynucleotide, comprises the following components: (i) A first polynucleotide encoding a first selectable marker operably linked to sequences that cause it to be expressed in a selection host and (ii) A transposon comprising transposon ends recognized by a transposase. The transposon is present in, and interrupts the coding sequence of, the first selectable marker, such that the first selectable marker is not active. The transposon is placed in the first selectable marker such that precise excision of the first transposon causes the first selectable marker to be reconstituted. If an active transposase that can excise the first transposon is introduced into a host cell which comprises the first polynucleotide, the host cell will express the active first selectable marker. The activity of the transposase in excising the transposon can be measured as the frequency with which the host cells become able to grow under conditions that require the first selectable marker to be active.

If the transposon comprises a second selectable marker, operably linked to sequences that make the second selectable marker expressible in the selection host, transposition of the second selectable marker into the genome of the host cell will yield a genome comprising active first and second selectable markers. The activity of the transposase in transposing the transposon into a second genomic location can be measured as the frequency with which the host cells become able to grow under conditions that require the first and second selectable markers to be active. In contrast, if the first selectable marker is present, but the second is not, then this indicates that the transposon was excised from the first polynucleotide, but was not subsequently transposed into a second polynucleotide. The selectable markers may, for example, be open reading frames encoding an antibiotic resistance protein, or an auxotrophic marker, or any other selectable marker.

We used such a system to test putative transposase/transposon combinations for activity, as described in Section 6.1. We used computational methods to search publicly available sequenced genomes for open reading frames with homology to known active piggy Bac-like transposases. We selected transposase sequences that appeared to possess the DDDE motif characteristic of active piggyBac-like transposases and searched the DNA sequences flanking these putative transposases for inverted repeat sequences adjacent to a 5'-TTAA-3' target sequence. Amongst those that we identified were putative transposons with intact transposases from: *Spodoptera litura* (Genbank accession number MTZO01002002.1, protein accession number XP_022823959) with an open reading frame encoding a putative transposase with SEQ ID NO: 46 flanked by a putative left end with SEQ ID NO: 91 and a putative right end with SEQ ID NO: 92; *Pieris rapae* (NCBI genomic reference sequence NW_019093607.1, Genbank protein accession number XP_022123753.1) with an open reading frame encoding a putative transposase with SEQ ID NO: 47 flanked by a putative left end with SEQ ID NO: 93 and a putative right end with SEQ ID NO: 94; *Myzus persicae* (NCBI genomic reference sequence NW_019100532.1, protein accession number XP_022166603) with an open reading frame encoding a putative transposase with SEQ ID NO: 48 flanked by a putative left end with SEQ ID NO: 95 and a putative right end with SEQ ID NO: 96; *Onthophagus*

taurus (NCBI genomic reference sequence NW_019280463, protein accession number XP_022900752) with an open reading frame encoding a putative transposase with SEQ ID NO: 49 flanked by a putative left end with SEQ ID NO: 97 and a putative right end with SEQ ID NO: 98; *Temnothorax curvispinosus* (NCBI genomic reference sequence NW_020220783.1, protein accession number XP_024881886) with an open reading frame encoding a putative transposase with SEQ ID NO: 50 flanked by a putative left end with SEQ ID NO: 99 and a putative right end with SEQ ID NO: 100; *Agrius planipenn* (NCBI genomic reference sequence NW_020442437.1, protein accession number XP_025836109) with an open reading frame encoding a putative transposase with SEQ ID NO: 51 flanked by a putative left end with SEQ ID NO: 101 and a putative right end with SEQ ID NO: 102; *Parasteatoda tepidariorum* (NCBI genomic reference sequence NW_018371884.1, protein accession number XP_015905033) with an open reading frame encoding a putative transposase with SEQ ID NO: 52 flanked by a putative left end with SEQ ID NO: 103 and a putative right end with SEQ ID NO: 104; *Pectinophora gossypiella* (Genbank accession number GU270322.1, protein ID ADB45159.1, also described in Wang et al, 2010. Insect Mol. Biol. 19, 177-184. "piggyBac-like elements in the pink bollworm, *Pectinophora gossypiella*") with an open reading frame encoding a putative transposase with SEQ ID NO: 53 flanked by a putative left end with SEQ ID NO: 105 and a putative right end with SEQ ID NO: 106; *Ctenophusia agnata* (NCBI accession number GU477713.1, protein accession number ADV17598.1, also described by Wu M, et al (2011) Genetica 139:149-54. "Cloning and characterization of piggyBac-like elements in lepidopteran insects") with an open reading frame encoding a putative transposase with SEQ ID NO: 54 flanked by a putative left end with SEQ ID NO: 107 and a putative right end with SEQ ID NO: 108; *Macrostomum lignano* (NCBI genomic reference sequence NIVC01003029.1, protein accession number PAA53757) with an open reading frame encoding a putative transposase with SEQ ID NO: 55 flanked by a putative left end with SEQ ID NO: 109 and a putative right end with SEQ ID NO: 110; *Orussus abietinus* (NCBI accession number XM_012421754, protein accession number XP_012277177) with an open reading frame encoding a putative transposase with SEQ ID NO: 56 flanked by a putative left end with SEQ ID NO: 111 and a putative right end with SEQ ID NO: 112; *Eufriesea mexicana* (NCBI genomic reference sequence NIVC01003029.1, protein accession number XP_017759329) with an open reading frame encoding a putative transposase with SEQ ID NO: 57 flanked by a putative left end with SEQ ID NO: 113 and a putative right end with SEQ ID NO: 114; *Spodoptera litura* (NCBI genomic reference sequence NC_036206.1, protein accession number XP_022824855) with an open reading frame encoding a putative transposase with SEQ ID NO: 58 flanked by a putative left end with SEQ ID NO: 115 and a putative right end with SEQ ID NO: 116; *Vanessa tameamea* (NCBI genomic reference sequence NW_020663261.1, protein accession number XP_026490968) with an open reading frame encoding a putative transposase with SEQ ID NO: 59 flanked by a putative left end with SEQ ID NO: 117 and a putative right end with SEQ ID NO: 118; *Blattella germanica* (NCBI genomic reference sequence PYGN01002011.1, protein accession number PSN31819) with an open reading frame encoding a putative transposase with SEQ ID NO: 60 flanked by a putative left end with SEQ ID NO: 119 and a putative right end with SEQ ID NO: 120;

Onthophagus taurus (NCBI genomic reference sequence NW_019281532.1, protein accession number XP_022910826) with an open reading frame encoding a putative transposase with SEQ ID NO: 61 flanked by a putative left end with SEQ ID NO: 121 and a putative right end with SEQ ID NO: 122; *Onthophagus taurus* (NCBI genomic reference sequence NW_019281689.1, protein accession number XP_022911139) with an open reading frame encoding a putative transposase with SEQ ID NO: 62 flanked by a putative left end with SEQ ID NO: 123 and a putative right end with SEQ ID NO: 124; *Onthophagus taurus* (NCBI genomic reference sequence NW_019286114.1, protein accession number XP_022913435) with an open reading frame encoding a putative transposase with SEQ ID NO: 63 flanked by a putative left end with SEQ ID NO: 125 and a putative right end with SEQ ID NO: 126; *Megachile rotundata* (NCBI genomic reference sequence NW_003797295, protein accession number XP_012145925) with an open reading frame encoding a putative transposase with SEQ ID NO: 64 flanked by a putative left end with SEQ ID NO: 127 and a putative right end with SEQ ID NO: 128; *Xiphophorus maculatus* (NCBI genomic reference sequence NC_036460.1, protein accession number XP_023207869) with an open reading frame encoding a putative transposase with SEQ ID NO: 65 flanked by a putative left end with SEQ ID NO: 129 and a putative right end with SEQ ID NO: 130; *Heliothis virescens* (NCBI accession number NWSH01000300.1, protein accession number PCG77621) with an open reading frame encoding a putative transposase with SEQ ID NO: 39 flanked by a putative left end with SEQ ID NO: 1 and a putative right end with SEQ ID NO: 3; *Agrotis ipsilon* (NCBI accession number GU477714.1, protein accession number ADV17599.1) with an open reading frame encoding a putative transposase with SEQ ID NO: 40 flanked by a putative left end with SEQ ID NO: 23 and a putative right end with SEQ ID NO: 24 and *Helicoverpa armigera* (NCBI accession number ABS18391.1, protein accession number ABS18391) with an open reading frame encoding a putative transposase with SEQ ID NO: 41 flanked by a putative left end with SEQ ID NO: 31 and a putative right end with SEQ ID NO: 32.

5.2.5.1 the *Heliothis* Transposase and its Corresponding Transposon

One active transposase and its corresponding transposon identified by transposition activity in yeast was a *Heliothis* transposase, as described in Section 6.1.2. A *Heliothis* transposase comprises a polypeptide sequence that is at least 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the sequence given by SEQ ID NO: 39, and which is capable of transposing the transposon from transposase reporter construct SEQ ID NO: 66, as described in Section 6.1.2. Exemplary *Heliothis* transposases include sequences given as SEQ ID NOs: 131-465.

A *Heliothis* transposase may be provided as a part of a gene transfer system as a protein, or as a polynucleotide encoding the *Heliothis* transposase, wherein the polynucleotide is expressible in the target cell. When provided as a polynucleotide, the *Heliothis* transposase may be provided as DNA or mRNA. If provided as DNA, the open reading frame encoding the *Heliothis* transposase is preferably operably linked to heterologous regulatory elements including a promoter that is active in the target cell such that the transposase is expressible in the target cell, for example a promoter that is active in a eukaryotic cell or a vertebrate cell or a mammalian cell. If provided as mRNA, the mRNA may be prepared in vitro from a DNA molecule in which the

open reading frame encoding the *Heliothis* transposase is preferably operably linked to a heterologous promoter active in the invitro transcription system used to prepare the mRNA, for example a T7 promoter.

A *Heliothis* transposon comprises a heterologous polynucleotide flanked by a left transposon end comprising a left ITR with sequence given by SEQ ID NO: 15 and a right transposon end comprising a right ITR with sequence given by SEQ ID NO: 16, and wherein the distal end of each ITR is immediately adjacent to a target sequence. Here and elsewhere when inverted repeats are defined by a sequence including a nucleotide defined by an ambiguity code, the identity of that nucleotide can be selected independently in the two repeats. A preferred target sequence is 5'-TTAA-3', although other useable target sequences may be used; preferably the target sequence on one side of the transposon is a direct repeat of the target sequence on the other side of the transposon. The left transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 13 or SEQ ID NO: 19. The right transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 14, or SEQ ID NO: 20, or SEQ ID NO: 21, or SEQ ID NO: 22. The structure of a representative *Heliothis* transposon is shown in FIG. 1. A *Heliothis* transposon can be transposed by a transposase with a polypeptide sequence given by SEQ ID NO: 39, for example as encoded by a polynucleotide with sequence given by SEQ ID NO: 42 operably linked to a Gall promoter.

Transposon ends, including ITRs and target sequences may be added to the ends of a heterologous polynucleotide sequence to create a synthetic *Heliothis* transposon which may be efficiently transposed into a target eukaryotic genome by an *Heliothis* transposase. For example, SEQ ID NO: 1 and SEQ ID NO: 2 each comprise a left 5'-TTAA-3' target sequence followed by a left transposon ITR followed by additional end sequences that may be added to one side of a heterologous polynucleotide, with the target sequence distal relative to the heterologous polynucleotide, to generate a synthetic *Heliothis* transposon. SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5 and SEQ ID NO: 6 each comprise additional end sequences followed by a right transposon ITR sequence followed by a right 5'-TTAA-3' target sequence that may be added to the other side of a heterologous polynucleotide, with the target sequence distal relative to the heterologous polynucleotide, to generate a synthetic *Heliothis* transposon. The preceding transposon end sequences comprise 5'-TTAA-3' as the target sequence, but SEQ ID NO: 7 and SEQ ID NO: 8 each comprise a left transposon ITR followed by additional end sequences that may be added to one side of a heterologous polynucleotide, with the ITR sequence distal relative to the heterologous polynucleotide, and SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11 and SEQ ID NO: 12 each comprise additional end sequences followed by a right transposon ITR sequence that may be added to the other side of a heterologous polynucleotide, with the target sequence distal relative to the heterologous polynucleotide, to generate a synthetic *Heliothis* transposon which may then be flanked by alternative target sequences.

Heliothis transposases recognize synthetic *Heliothis* transposons. They excise the transposon from a first DNA molecule, by cutting the DNA at the target sequence at the left end of one transposon end and the target sequence at the right end of the second transposon end, re-join the cut ends of the first DNA molecule to leave a single copy of the target

sequence. The excised transposon sequence, including any heterologous DNA that is between the transposon ends, is integrated by the transposase into a target sequence of a second DNA molecule, such as the genome of a target cell. A cell whose genome comprises a synthetic *Heliothis* transposon is an embodiment of the invention.

5.2.5.2 the *Heliothis* Transposase is Active in Mammalian Cells

The looper moth piggyBac transposase has been shown to be active in a very wide variety of eukaryotic cells. In Section 6.1.2 we show that the *Heliothis* transposase can transpose its corresponding transposon into the genome of the yeast *Saccharomyces cerevisiae*. In Section 6.1.4 we show that the *Heliothis* transposase can transpose its corresponding transposon into the genome of a mammalian CHO cell. These results provide evidence that, like the other known active piggy Bac-like transposases, the *Heliothis* transposase is also active in transposing its corresponding transposon into the genomes of most eukaryotic cells. Although the *Heliothis* transposase is active in a wide range of eukaryotic cells, the naturally occurring open reading frame encoding the *Heliothis* transposase (given as SEQ ID NO: 1181) is unlikely to express well in a similarly wide range of cells, as optimal codon usage differs significantly between cell types. It is therefore advantageous to use a sequence other than a naturally occurring sequence to encode the transposase, in other words, to use codon-preferences of the cell type in which expression is to be performed. Likewise, the promoter and other regulatory sequences are selected so as to be active in the cell type in which expression is to be performed. An advantageous polynucleotide for expression of a *Heliothis* transposase comprises at least 2, 5, 10, 20, 30, 40 or 50 synonymous codon differences relative to SEQ ID NO: 1181 at corresponding positions between the polynucleotide and SEQ ID NO: 1181, optionally wherein codons in the polynucleotide at the corresponding positions are selected for mammalian cell expression. The polynucleotide may be DNA or mRNA. Exemplary polynucleotide sequences encoding *Heliothis* transposases with polypeptide sequences given by SEQ ID NOs: 39, 134 and 144, wherein codons have been selected for expression in mammalian cells, are given as SEQ ID NOs: 42, 1182 and 1183 respectively.

5.2.6 Cross-Reacting Piggybac-Like Transposases

5.2.6.1 the *Helicoverpa* Transposase and its Corresponding Transposon

A putative piggyBac-like transposase and its corresponding transposon has been identified from *Helicoverpa armigera* (Sun et al. (2008). Insect Molecular Biology 17, 9-18 “piggyBac like elements in cotton bollworm, *Helicoverpa armigera* (Hübner)”). The authors state that the transposase “is currently being tested for function and the feasibility of its being reconstructed as a transformation vector”. However there have been no further reports in the past 10 years regarding these experiments, nor any filed patents, although positive identification of a novel active piggy Bac-like transposase would be both publishable and patentable. We therefore infer that the authors have failed to identify an active *Helicoverpa* transposase. Nor do the authors suggest that the transposon may be useful in other eukaryotic cells including mammalian cells. Nor do the authors teach how a synthetic *Helicoverpa* transposon may be constructed or used.

Here we show that a synthetic *Helicoverpa* transposon can be constructed by adding *Helicoverpa* transposon ends to either side of a heterologous polynucleotide. The synthetic *Helicoverpa* transposon can be transposed into the

genome of a target eukaryotic cell, including cells other than an insect cell, by a *Helicoverpa* transposase.

A *Helicoverpa* transposase comprises a polypeptide sequence that is at least 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% identical to the sequence given by SEQ ID NO: 41, and is capable of transposing the transposon from transposase reporter construct SEQ ID NO: 68, as described in Section 6.1.3. Although the *Helicoverpa* transposase is active in a wide range of eukaryotic cells, the naturally occurring open reading frame encoding the *Helicoverpa* transposase (given as SEQ ID NO: 1181) is unlikely to express well in a similarly wide range of cells, as optimal codon usage differs significantly between cell types. It is therefore advantageous to use a sequence other than a naturally occurring sequence to encode the transposase, in other words, to use codon-preferences of the cell type in which expression is to be performed. Likewise, the promoter and other regulatory sequences are selected so as to be active in the cell type in which expression is to be performed. An advantageous polynucleotide for expression of a *Helicoverpa* transposase comprises at least 2, 5, 10, 20, 30, 40 or 50 synonymous codon differences relative to SEQ ID NO: 1184 at corresponding positions between the polynucleotide and SEQ ID NO: 1184, optionally wherein codons in the polynucleotide at the corresponding positions are selected for mammalian cell expression. The polynucleotide may be DNA or mRNA. An exemplary polynucleotide sequence encoding a *Helicoverpa* transposase with polypeptide sequence given by SEQ ID NO: 41, wherein codons have been selected for expression in mammalian cells, is given as SEQ ID NO: 44.

A *Helicoverpa* transposon comprises a heterologous polynucleotide flanked by a left transposon end comprising a left ITR with sequence given by SEQ ID NO: 37 and a right transposon end comprising a right ITR with sequence given by SEQ ID NO: 38, and wherein the distal end of each ITR is immediately adjacent to a target sequence. A preferred target sequence is 5'-TTAA-3', although other useable target sequences may be used; preferably the target sequence on one side of the transposon is a direct repeat of the target sequence on the other side of the transposon. The left transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 35. The right transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 36. A *Helicoverpa* transposon can be transposed by a transposase with a polypeptide sequence given by 41, for example as encoded by a polynucleotide with sequence given by SEQ ID NO: 44 operably linked to a Gall promoter.

Transposon ends, including ITRs and target sequences may be added to the ends of a heterologous polynucleotide sequence to create a synthetic *Helicoverpa* transposon which may be efficiently transposed into a target eukaryotic genome by an *Helicoverpa* transposase. For example, SEQ ID NO: 31 comprises a left target sequence followed by a left transposon end sequence that may be added to one side of a heterologous polynucleotide to generate a synthetic *Helicoverpa* transposon. SEQ ID NO: 32 comprises a right transposon end sequence followed by a right target sequence that may be added to the other side of a heterologous polynucleotide to generate a synthetic *Helicoverpa* transposon. Preferably a synthetic *Helicoverpa* transposon comprises a promoter that is active in a eukaryotic cell other than an insect cell. Preferably a synthetic *Helicoverpa* transposon comprises a promoter that is active in a mammalian cell.

Helicoverpa transposases recognize synthetic *Helicoverpa* transposons. They excise the transposon from a first DNA molecule, by cutting the DNA at the target sequence at the left end of one transposon end and the target sequence at the right end of the second transposon end, re-join the cut ends of the first DNA molecule to leave a single copy of the target sequence. The excised transposon sequence, including any heterologous DNA that is between the transposon ends, is integrated by the transposase into a target sequence of a second DNA molecule, such as the genome of a target cell. A cell whose genome comprises a synthetic *Helicoverpa* transposon is an embodiment of the invention.

5.2.6.2 the *Agrotis* Transposase and its Corresponding Transposon

Similar sequences to the putative piggy Bac-like transposase and its corresponding putative transposon from *Helicoverpa armigera* were identified from *Agrotis ypsilon* (Wu et. al. (2011). *Genetica* 139, 149-154. "Cloning and characterization of piggy Bac-like elements in lepidopteran insects") by the same group that identified the putative piggyBac-like transposase and its corresponding putative transposon from *Helicoverpa armigera*. The authors state that "These elements are currently being tested for function, and preliminary results indicate that the transposases support transposition in cell culture, though at varying frequencies. (M. Wu, G. H. Luo, X. F. Wang and Z. J. Han, unpublished data). If HaPLE1.1, Ay-PLE1.1 and AaPLE1.1 are found to be transpositionally active, then comparative and functional assays should allow an assessment of domains critical to their mobility, in addition to the known "DDD" domains. This may allow the development of new vectors for insect transformation, and possibly improvements in the efficiency of IFP2 piggyBac by modifications to critical domains." Although this text suggests initial encouraging results, there have been no further published reports in the past 8 years regarding these experiments, nor any filed patents, although positive identification of a novel active piggy Bac-like transposase would be both publishable and patentable. We therefore infer that the authors have failed to identify an active *Agrotis* transposase. Nor do the authors suggest that the transposon or transposase may be useful in eukaryotic cells other than insect cells. Nor do the authors teach how a synthetic *Agrotis* transposon may be constructed.

Here we show that a synthetic *Agrotis* transposon can be constructed by adding *Agrotis* transposon ends to either side of a heterologous polynucleotide. The synthetic *Agrotis* transposon can be transposed into the genome of a target eukaryotic cell, including cells other than an insect cell, by an *Agrotis* transposase.

An *Agrotis* transposase comprises a polypeptide sequence that is at least 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% identical to a sequence given by SEQ ID NO: 40, and is capable of transposing the transposon from transposase reporter construct SEQ ID NO: 67, as described in Section 6.1.3. Although the *Agrotis* transposase is active in a wide range of eukaryotic cells, the naturally occurring open reading frame encoding the *Agrotis* transposase (given as SEQ ID NO: 1185) is unlikely to express well in a similarly wide range of cells, as optimal codon usage differs significantly between cell types. It is therefore advantageous to use a sequence other than a naturally occurring sequence to encode the transposase, in other words, to use codon-preferences of the cell type in which expression is to be performed. Likewise, the promoter and other regulatory sequences are selected so as to be active in the cell type in which expression is to be performed. An advantageous polynucleotide for expression of an *Agrotis* transposase

comprises at least 2, 5, 10, 20, 30, 40 or 50 synonymous codon differences relative to SEQ ID NO: 1185 at corresponding positions between the polynucleotide and SEQ ID NO:1185, optionally wherein codons in the polynucleotide at the corresponding positions are selected for mammalian cell expression. The polynucleotide may be DNA or mRNA. An exemplary polynucleotide sequence encoding an *Agrotis* transposase with polypeptide sequence given by SEQ ID NO: 40, wherein codons have been selected for expression in mammalian cells, is given as SEQ ID NO: 43.

An *Agrotis* transposon comprises a heterologous polynucleotide flanked by a left transposon end comprising a left ITR with sequence given by SEQ ID NO: 29 and a right transposon end comprising a right ITR with sequence given by SEQ ID NO: 30, and wherein the distal end of each ITR is immediately adjacent to a target sequence. A preferred target sequence is 5'-TTAA-3', although other useable target sequences may be used; preferably the target sequence on one side of the transposon is a direct repeat of the target sequence on the other side of the transposon. The left transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 27. The right transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 28. An *Agrotis* transposon can be transposed by a transposase with a polypeptide sequence given by SEQ ID NO: 40, for example as encoded by a polynucleotide with sequence given by SEQ ID NO: 43 operably linked to a Gall promoter.

Transposon ends, including ITRs and target sequences may be added to the ends of a heterologous polynucleotide sequence to create a synthetic *Agrotis* transposon which may be efficiently transposed into a target eukaryotic genome by an *Agrotis* transposase. For example, SEQ ID NO: 23 comprises a left target sequence followed by a left transposon end sequence that may be added to one side of a heterologous polynucleotide to generate a synthetic *Agrotis* transposon. SEQ ID NO: 24 comprises a right transposon end sequence followed by a right target sequence that may be added to the other side of a heterologous polynucleotide to generate a synthetic *Agrotis* transposon. Preferably a synthetic *Agrotis* transposon comprises a promoter that is active in a eukaryotic cell other than an insect cell. Preferably a synthetic *Agrotis* transposon comprises a promoter that is active in a mammalian cell.

Agrotis transposases recognize synthetic *Agrotis* transposons. They excise the transposon from a first DNA molecule, by cutting the DNA at the target sequence at the left end of one transposon end and the target sequence at the right end of the second transposon end, re-join the cut ends of the first DNA molecule to leave a single copy of the target sequence. The excised transposon sequence, including any heterologous DNA that is between the transposon ends, is integrated by the transposase into a target sequence of a second DNA molecule, such as the genome of a target cell. A cell whose genome comprises a synthetic *Agrotis* transposon is an embodiment of the invention.

5.2.6.3 Cross-Reactivity of the *Agrotis* and *Helicoverpa* Transposases

As described in Section 6.1.3, the *Agrotis* and *Helicoverpa* transposases are more than 92% identical. The ITRs for the corresponding transposons for these two transposases are identical, and the *Agrotis* transposase can transpose a *Helicoverpa* transposon and the *Helicoverpa* transposase can transpose an *Agrotis* transposon. The sequences adjacent to the ITRs differ between the two transposons: the sequences

adjacent to the *Agrotis* left ITR (SEQ ID NO: 27) is about 90% identical with the sequences adjacent to the *Helicoverpa* left ITR (SEQ ID NO: 35), similarly the sequences adjacent to the *Agrotis* right ITR (SEQ ID NO: 28) is about 90% identical with the sequences adjacent to the *Helicoverpa* right ITR (SEQ ID NO: 36). As described in Section 6.1.3, these differences do not significantly affect the relative excision or transposition frequencies of the transposons with the different transposases. The ITRs thus appear to be the critical element for specificity of piggy Bac-like transposases.

5.2.7 Hyperactive *Heliothis* Transposases

Individual favorable mutations may be combined in a variety of different ways, for example by “DNA shuffling” or by methods described in U.S. Pat. No. 8,635,029 B2 and Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”). A transposase with modified activity, either for activity on a new target sequence, or increased activity on an existing target sequence may be obtained by using variations of the selection scheme described herein (for example Section 6.1.6) with an appropriate corresponding transposon.

An alignment of known active piggyBac-like transposases may be used to identify amino acid changes likely to result in enhanced activity. Transposases are often deleterious to their hosts, so tend to accumulate mutations that inactivate them. However the mutations that accumulate in different transposases are different, as each occurs by random chance. A consensus sequence can be obtained from an alignment of sequences, and this can be used to improve activity (Ivics et al, 1997, Cell 91:501-510. “Molecular reconstruction of Sleeping Beauty, a Tc1-like transposon from fish, and its transposition in human cells.”). We aligned known active piggy Bac-like transposases using the CLUSTAL algorithm, and enumerated the amino acids found at each position. This diversity is shown in Table 1 relative to a *Heliothis* transposase (relative to SEQ ID NO: 39), the amino acids shown in column C are found in known active piggyBac-like transposases at the equivalent position in an alignment, and are thus likely to be acceptable changes in a *Heliothis* transposase. Column D shows amino acid changes found in known active piggy Bac-like transposases other than the *Heliothis* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Heliothis* transposase sequence is an outlier. Mutation of the position shown in column A to an amino acid shown in column D is particularly likely to result in enhanced transposase activity. Table 2 shows this diversity relative to a *Helicoverpa* transposase (relative to SEQ ID NO: 41), the amino acids shown in column C are found in known active piggy Bac-like transposases at the equivalent position in an alignment, and are thus likely to be acceptable changes in a *Helicoverpa* transposase. Column D shows amino acid changes found in known active piggy Bac-like transposases other than the *Helicoverpa* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Helicoverpa* transposase sequence is an outlier. Mutation of the position shown in column A to an amino acid shown in column D is particularly likely to result in enhanced transposase activity. Table 3 shows this diversity relative to an *Agrotis* transposase (relative to SEQ ID NO: 40), the amino acids shown in column C are found in known active piggy Bac-like transposases at the equivalent position in an alignment, and are thus likely to be acceptable changes in a *Agrotis* transposase. Column D shows amino acid changes

found in known active piggy Bac-like transposases other than the *Agrotis* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Agrotis* transposase sequence is an outlier. Mutation of the position shown in column A to an amino acid shown in column D is particularly likely to result in enhanced transposase activity.

We selected amino acid substitutions to make in *Heliothis* transposase SEQ ID NO: 39 from column D in Table 1. Genes encoding *Heliothis* transposase variants comprising combinations of these substitutions were synthesized and tested for transposase activity as described in Section 6.1.7. As described in Section 6.1.7, we identified 27 *Heliothis* transposases with transposition activity over 10-fold greater than SEQ ID NO: 39 (sequences given by SEQ ID NOs: 131-157), 211 *Heliothis* transposases with transposition activity between 2 and 10-fold greater than SEQ ID NO: 39 (sequences given by SEQ ID NOs: 158-368), and 33 *Heliothis* transposases with transposition activity less than 2-fold greater than SEQ ID NO: 39 (sequences given by SEQ ID NOs: 369-400). Exemplary sequences of hyperactive *Heliothis* transposases comprises a polypeptide with the sequence of one of these transposases.

Heliothis transposases can thus be created that are not naturally occurring sequences, but that are at least 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 99.5% identical to SEQ ID NO: 39. Such variants can retain partial activity of the transposase of SEQ ID NO: 39 (as determined by either or both of transposition and/or excision activity), can be functionally equivalent of the transposase of SEQ ID NO: 39 in either or both of transposition and excision, or can have enhanced activity relative to the transposase of SEQ ID NO: 39 in transposition, excision activity or both. Such variants can include mutations shown herein to increase transposition and/or excision, mutations shown herein to be neutral as to transposition and/or excision, and mutations detrimental to transposition and/or integration. Preferred variants include mutations shown to be neutral or to enhance transposition/and or excision. Some such variants lack mutations shown to be detrimental to transposition and/or excision. Some such variants include only mutations shown to enhance transposition, only mutations shown to enhance excision, or mutations shown to enhance both transposition and excision.

Enhanced activity means activity (e.g., transposition or excision activity) that is greater beyond experimental error than that of a reference transposase from which a variant was derived. The activity can be greater by a factor of e.g., 1.1, 2, 5, 10, 15, 20, 50 or 100 fold of the reference transposase. The enhanced activity can lie within a range of for example 1.1-100 fold, 2-50 fold, 5-50 fold or 1.1-10 fold of the reference transposase. Here and elsewhere activities can be measured as demonstrated in the examples. Functional equivalence means a variant transposase can mediate transposition and/or excision of the same transposon with a comparable efficiency (within experimental error) to a reference transposase.

Furthermore, variant sequences of SEQ ID NO: 39 can be created by combining two, three, four, or five or more substitutions selected from Table 1 column D. Combining beneficial substitutions, for example those shown in column D of Table 1 can result in hyperactive variants of SEQ ID NO: 39. We identified more than 220 *Heliothis* transposases with transposition or excision frequencies more than about 2-fold greater than that of naturally occurring *Heliothis* transposase SEQ ID NO: 39. These preferred hyperactive *Heliothis* transposases comprised one or more of the following substitutions (relative to SEQ ID NO: 39): S41V,

L43S, V81E, S136V, K152L, L153F, D155T, T162I, I239L, H241N, S264R, T268D, V280P, T297C, L299F, M300R, M305N, L312I, Y421F, V450I, R485K, Y495F, L507I, H512N, Q603E, D83S, V85L, P125S, Q126S, Q131T, Q131R, E140C, E140A, N151Q, K169E, N212S, C316A, N322T, P351G, H357R, K360Q, E379P, K397S, A447N, A447D, A449V, A449S, K476L, V492A, I500M, L585K and T595K. A preferred hyperactive *Heliothis* transposase comprises an amino acid substitution selected from this set. Some hyperactive *Heliothis* transposases may further comprise a heterologous nuclear localization sequence.

We used machine learning methods as described in Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”) to determine the effect of various amino acid substitutions on the excision and transposition activities of a *Heliothis* transposase. Each substitution was empirically tested in a minimum of 5 different sequence contexts (i.e. in the presence of different other amino acid substitutions). As described in Liao et. al., the mean value for the regression weight for a substitution is a measure of the average effect of that substitution within multiple different *Heliothis* transposases. A substitution with a positive mean regression weight is one that on average has a positive effect on the transposition activity of *Heliothis* transposases. Addition of a substitution with a positive mean regression weight to an active *Heliothis* transposase that does not already comprise such a substitution is thus expected to improve the transposition activity of that *Heliothis* transposase. Tables 11 and 14 identify 56 substitutions within a *Heliothis* transposase with positive mean regression weights for transposition: S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K.

Preferred hyperactive *Heliothis* transposases comprise an amino acid sequence, other than a naturally occurring protein (e.g., not a transposase whose amino acid sequence comprises SEQ ID NO: 39), that is at least 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the amino acid sequence of any of SEQ ID NOs: 39 or 131-400 and which comprise a substitution at a position selected from amino acid 41, 43, 81, 83, 85, 125, 126, 131, 136, 140, 149, 151, 152, 153, 155, 162, 169, 212, 238, 239, 241, 264, 268, 280, 297, 299, 300, 305, 312, 316, 322, 357, 360, 396, 397, 421, 430, 447, 449, 450, 476, 485, 492, 495, 507, 512, 585, 589, 595, and 603, relative to SEQ ID NO: 39 and numbered according to SEQ ID NO: 39. Preferably the substitution is one shown in Table 1 columns C or D. Preferably the hyperactive *Heliothis* transposase comprises an amino acid substitution, relative to the sequence of SEQ ID NO: 39, selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, or any combination of substitutions thereof including at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or all of these substitutions. The hyperactive *Heliothis* transposase may

also comprise substitutions relative to SEQ ID NO: 39 at many other positions that are not associated with enhanced transposition, for example conservative substitutions that have a neutral effect on transposition.

Methods of creating transgenic cells using naturally occurring or hyperactive *Heliothis* transposases are an aspect of the invention. A method of creating a transgenic cell comprises (i) introducing into a eukaryotic cell a naturally occurring or hyperactive *Heliothis* transposase (as a protein or as a polynucleotide encoding the transposase) and a corresponding *Heliothis* transposon. Creating the transgenic cell may further comprise (ii) identifying a eukaryotic cell whose genome comprises a *Heliothis* transposon. Identifying the cell in which an *Heliothis* transposon is incorporated into the genome of the eukaryotic cell may comprise selecting the eukaryotic cell for a selectable marker encoded on the *Heliothis* transposon. The selectable marker may be any selectable polypeptide, including any described herein. The eukaryotic cell into which the *Heliothis* transposon is introduced, and whose genome comprises a *Heliothis* transposon may be an animal cell, a fungal cell or a plant cell. The animal cell may be vertebrate or invertebrate. Preferred vertebrate cells include cells from mammals including rodents such as rats, mice, and hamsters; ungulates, such as cows, goats or sheep; swine and cells from primates including humans. Types of cells whose genomes may comprise a *Heliothis* transposon include hepatocytes, neural cells, muscle cells, blood cells, embryonic stem cells, somatic stem cells, hematopoietic cells, and immune cells including lymphocytes such as T cells, B cells and natural killer cells, T-helper cells, antigen-presenting cells, dendritic cells, neutrophils and macrophages. Preferred culture cells are Chinese hamster ovary (CHO) cells or Human embryonic kidney (HEK293) cells. Preferred fungal cells are yeast cells including *Saccharomyces cerevisiae* and *Pichia pastoris*.

Activity of transposases may also be increased by fusion of nuclear localization signal (NLS) at the N-terminus, C-terminus, both at the N- and C-termini or internal regions of the transposase protein, as long as transposase activity is retained. A nuclear localization signal or sequence (NLS) is an amino acid sequence that ‘tags’ or facilitates interaction of a protein, either directly or indirectly with nuclear transport proteins for import into the cell nucleus. Nuclear localization signals (NLS) used can include consensus NLS sequences, viral NLS sequences, cellular NLS sequences, and combinations thereof.

Transposases may also be fused to other protein functional domains. Such protein functional domains can include DNA binding domains, flexible hinge regions that can facilitate one or more domain fusions, and combinations thereof. Fusions can be made either to the N-terminus, C-terminus, or internal regions of the transposase protein so long as transposase activity is retained. Fusions to DNA binding domains can be used to direct the *Heliothis* transposase or the *Agrotis* transposase or the *Helicoverpa* transposase to a specific genomic locus or loci. DNA binding domains may include a helix-turn-helix domain, a zinc-finger domain, a leucine zipper domain, a TALE (transcription activator-like effector) domain, a CRISPR-Cas protein or a helix-loop-helix domain. Specific DNA binding domains used can include a Gal4 DNA binding domain, a LexA DNA binding domain, or a Zif268 DNA binding domain. Flexible hinge regions used can include glycine/serine linkers and variants thereof.

5.3 Kits

The present invention also features kits comprising a *Heliothis* transposase as a protein or encoded by a nucleic

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acid, and/or a *Heliothis* transposon; or a gene transfer system as described herein comprising a *Heliothis* transposase as a protein or encoded by a nucleic acid as described herein, in combination with a *Heliothis* transposon; optionally together with a pharmaceutically acceptable carrier, adjuvant or vehicle, and optionally with instructions for use. Any of the components of the inventive kit may be administered and/or transfected into cells in a subsequent order or in parallel, e.g. a *Heliothis* transposase protein or its encoding nucleic acid may be administered and/or transfected into a cell as defined above prior to, simultaneously with or subsequent to administration and/or transfection of a *Heliothis* transposon. Alternatively, a *Heliothis* transposon may be transfected into a cell as defined above prior to, simultaneously with or subsequent to transfection of a *Heliothis* transposase protein or its encoding nucleic acid. If transfected in parallel, preferably both components are provided in a separated formulation and/or mixed with each other directly prior to administration to avoid transposition prior to transfection. Additionally, administration and/or transfection of at least one component of the kit may occur in a time staggered mode, e.g. by administering multiple doses of this component.

In addition, the present invention also features kits comprising a *Helicoverpa* transposase as a protein or encoded by a nucleic acid, and/or a *Helicoverpa* transposon; or a gene transfer system as described herein comprising a *Helicoverpa* transposase as a protein or encoded by a nucleic acid as described herein, in combination with a *Helicoverpa* transposon; optionally together with a pharmaceutically acceptable carrier, adjuvant or vehicle, and optionally with instructions for use. Any of the components of the inventive kit may be administered and/or transfected into cells in a subsequent order or in parallel, e.g. a *Helicoverpa* transposase protein or its encoding nucleic acid may be administered and/or transfected into a cell as defined above prior to, simultaneously with or subsequent to administration and/or transfection of a *Helicoverpa* transposon. Alternatively, a *Helicoverpa* transposon may be transfected into a cell as defined above prior to, simultaneously with or subsequent to transfection of a *Helicoverpa* transposase protein or its encoding nucleic acid. If transfected in parallel, preferably both components are provided in a separated formulation and/or mixed with each other directly prior to administration the to avoid transposition prior to transfection. Additionally, administration and/or transfection of at least one component of the kit may occur in a time staggered mode, e.g. by administering multiple doses of this component.

In addition, the present invention also features kits comprising a *Agrotis* transposase as a protein or encoded by a nucleic acid, and/or a *Agrotis* transposon; or a gene transfer system as described herein comprising a *Agrotis* transposase as a protein or encoded by a nucleic acid as described herein, in combination with a *Agrotis* transposon; optionally together with a pharmaceutically acceptable carrier, adjuvant or vehicle, and optionally with instructions for use. Any of the components of the inventive kit may be administered and/or transfected into cells in a subsequent order or in parallel, e.g. a *Agrotis* transposase protein or its encoding nucleic acid may be administered and/or transfected into a cell as defined above prior to, simultaneously with or subsequent to administration and/or transfection of a *Agrotis* transposon. Alternatively, a *Agrotis* transposon may be transfected into a cell as defined above prior to, simultaneously with or subsequent to transfection of a *Agrotis* transposase protein or its encoding nucleic acid. If transfected in parallel, preferably both components are provided in a

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separated formulation and/or mixed with each other directly prior to administration the to avoid transposition prior to transfection. Additionally, administration and/or transfection of at least one component of the kit may occur in a time staggered mode, e.g. by administering multiple doses of this component.

6. EXAMPLES

The following examples illustrate the methods, compositions and kits disclosed herein and should not be construed as limiting in any way. Various equivalents will be apparent from the following examples; such equivalents are also contemplated to be part of the invention disclosed herein.

6.1 New Transposases

6.1.1 Measuring Transposase Activity

As described in Section 5.2.5, transposition frequencies for active transposases may be measured using a system in which a transposon interrupts a selectable marker. Transposase reporter polynucleotides were constructed in which the open reading frame of the yeast *Saccharomyces cerevisiae* URA3 open reading frame was interrupted by a yeast TRP 1 open reading frame operably linked to a promoter and terminator such that it was expressible in the yeast *Saccharomyces cerevisiae*. The TRP1 gene was flanked by putative transposon ends with 5'-TTAA-3' target sites, such that excision of the putative transposon would leave a single copy of the 5'-TTAA-3' target site and exactly reconstitute the URA3 open reading frame. A yeast transposase reporter strain was constructed by integrating the transposase reporter polynucleotide into the URA3 gene of a haploid yeast strain auxotrophic for LEU2 and TRP1, such that the strain became LEU2-, URA3- and TRP1+.

Transposases were tested for their ability to transposase the TRP1 gene-containing transposons from within the URA3 open reading frame. Each open reading frame encoding a putative transposase was cloned into a *Saccharomyces cerevisiae* expression vector comprising a 2 micron origin of replication and a LEU2 gene expressible in *Saccharomyces*. Each transposase open reading frame was operably linked to a Gall promoter. Each cloned transposase open reading frame was transformed into a yeast transposase reporter strain and plated on minimal media lacking leucine. After 2 days, all LEU+ colonies were harvested by scraping the plates. The Gal promoter was induced by growing in galactose for 4 hours, and cells were then plated onto 3 different plates: plates lacking only leucine, plates lacking leucine and uracil, and plates lacking leucine, uracil and tryptophan. These plates were incubated for 2-4 days, and the colonies on each plate were counted, measuring the number of live cells, the number of transposon excision events and the number of transposon excision and re-integration (i.e. transposition events) respectively.

6.1.2 Identification of a New Piggybac-Like Transposase

As described in Section 5.2.5, thirteen putative piggyBac-like transposases were identified from Genbank as being at least 20% identical to the piggyBac transposase from *Trichoplusia ni*. The flanking gene sequences were analyzed for the presence of inverted repeat sequences including the 5'-TTAA-3' target sequence characteristic of piggyBac transposition. Putative left and right transposon end sequences were taken from these flanking sequences. These transposon ends were incorporated into transposase reporter constructs configured as described in Section 6.1.1 and integrated into the genome of *Saccharomyces cerevisiae* thereby generating transposase reporter strains. The corresponding transposase sequence for each reporter strain was back-translated,

cloned into a *Saccharomyces cerevisiae* expression vector and transformed into the reporter strain. Transposase activities were measured as described in Section 6.1.1.

The following twenty combinations showed no excision or transposition: reporter construct SEQ ID NO: 71 (comprising putative left transposon end SEQ ID NO: 91, and putative right transposon end SEQ ID NO: 92) with transposase SEQ ID NO: 46, reporter construct SEQ ID NO: 72 (comprising putative left transposon end SEQ ID NO: 93, and putative right transposon end SEQ ID NO: 94) with transposase SEQ ID NO: 47, reporter construct SEQ ID NO: 73 (comprising putative left transposon end SEQ ID NO: 95, and putative right transposon end SEQ ID NO: 96) with transposase SEQ ID NO: 48, reporter construct SEQ ID NO: 74 (comprising putative left transposon end SEQ ID NO: 97, and putative right transposon end SEQ ID NO: 98) with transposase SEQ ID NO: 49, reporter construct SEQ ID NO: 75 (comprising putative left transposon end SEQ ID NO: 99, and putative right transposon end SEQ ID NO: 100) with transposase SEQ ID NO: 50, reporter construct SEQ ID NO: 76 (comprising putative left transposon end SEQ ID NO: 101, and putative right transposon end SEQ ID NO: 102) with transposase SEQ ID NO: 51, reporter construct SEQ ID NO: 77 (comprising putative left transposon end SEQ ID NO: 103, and putative right transposon end SEQ ID NO: 104) with transposase SEQ ID NO: 52, reporter construct SEQ ID NO: 78 (comprising putative left transposon end SEQ ID NO: 105, and putative right transposon end SEQ ID NO: 106) with transposase SEQ ID NO: 53, reporter construct SEQ ID NO: 79 (comprising putative left transposon end SEQ ID NO: 107, and putative right transposon end SEQ ID NO: 108) with transposase SEQ ID NO: 54, reporter construct SEQ ID NO: 80 (comprising putative left transposon end SEQ ID NO: 109, and putative right transposon end SEQ ID NO: 110) with transposase SEQ ID NO: 55, reporter construct SEQ ID NO: 81 (comprising putative left transposon end SEQ ID NO: 111, and putative right transposon end SEQ ID NO: 112) with transposase SEQ ID NO: 56, reporter construct SEQ ID NO: 82 (comprising putative left transposon end SEQ ID NO: 113, and putative right transposon end SEQ ID NO: 114) with transposase SEQ ID NO: 57, reporter construct SEQ ID NO: 83 (comprising putative left transposon end SEQ ID NO: 115, and putative right transposon end SEQ ID NO: 116) with transposase SEQ ID NO: 58, reporter construct SEQ ID NO: 84 (comprising putative left transposon end SEQ ID NO: 117, and putative right transposon end SEQ ID NO: 118) with transposase SEQ ID NO: 59, reporter construct SEQ ID NO: 85 (comprising putative left transposon end SEQ ID NO: 119, and putative right transposon end SEQ ID NO: 120) with transposase SEQ ID NO: 60, reporter construct SEQ ID NO: 86 (comprising putative left transposon end SEQ ID NO: 121, and putative right transposon end SEQ ID NO: 122) with transposase SEQ ID NO: 61, reporter construct SEQ ID NO: 87 (comprising putative left transposon end SEQ ID NO: 123, and putative right transposon end SEQ ID NO: 124) with transposase SEQ ID NO: 62, reporter construct SEQ ID NO: 88 (comprising putative left transposon end SEQ ID NO: 125, and putative right transposon end SEQ ID NO: 126) with transposase SEQ ID NO: 63, reporter construct SEQ ID NO: 89 (comprising putative left transposon end SEQ ID NO: 127, and putative right transposon end SEQ ID NO: 128) with transposase SEQ ID NO: 64, reporter construct SEQ ID NO: 90 (comprising putative left transposon end SEQ ID NO: 129, and putative right transposon end SEQ ID NO: 130) with transposase SEQ ID NO: 65. This is consistent with reports in the literature that while

computational recognition of sequences that are homologous to the piggyBac transposase from *Trichoplusia ni* is straightforward, most of these sequences are transpositionally inactive, even when they appear to have intact terminal repeats and the transposases appear to comprise the DDDE motif found in active piggy Bac-like transposases. It is therefore necessary to measure excision and transposition activity, in order to identify novel active piggyBac-like transposases and transposons.

One transposase that showed good activity in excising its corresponding transposon from the reporter construct (shown by the appearance of URA⁺ colonies) and transposing the TRP gene in the transposon into another genomic location in the *Saccharomyces cerevisiae* reporter strain was transposase SEQ ID NO: 39. Transposase SEQ ID NO: 39 was able to transpose the transposon from reporter construct SEQ ID NO: 66. This is shown in Table 4: the number of excision events, measured by the appearance of URA⁺ colonies, is shown in column G; the number of full transposition events, measured by the appearance of URA⁺ TRP⁺ colonies, is shown in column H.

6.1.3 Identification of a Pair of Cross-Reacting Piggybac-Like Transposases

Two putative piggyBac-like transposases, one from a *Helicoverpa* species and another from an *Agrotis* species were identified from Genbank as being at least 20% identical to the piggy Bac transposase from *Trichoplusia ni*. The flanking gene sequences were analyzed for the presence of inverted repeat sequences including the 5'-TTAA-3' target sequence characteristic of piggyBac transposition. Putative left and right transposon end sequences were taken from these flanking sequences. These transposon ends were incorporated into transposase reporter constructs configured as described in Section 6.1.1 and integrated into the genome of *Saccharomyces cerevisiae* thereby generating transposase reporter strains.

Unlike other known piggy Bac-like transposons, the putative transposons from *Helicoverpa* and *Agrotis* had identical 16 bp ITR sequences given by SEQ ID NO: 37 (5'-CCCTAGAAGCCCAATC-3') immediately adjacent to a 5'-TTAA-3' target sequence at each end of the putative transposon. We hypothesized that if these ITR sequences are what a corresponding transposase recognizes, the *Agrotis* transposase would be able to transpose the *Helicoverpa* transposon, and vice versa. The *Agrotis* and *Helicoverpa* transposases were back-translated, cloned into a *Saccharomyces cerevisiae* expression vector and each was transformed separately into the *Agrotis* and *Helicoverpa* reporter strains. Activities for each transposase in each reporter strain were measured as described in Section 6.1.1.

Table 5 shows the number of excision events (measured by the appearance of URA⁺ colonies, shown in column G) and the number of full transposition events (measured by the appearance of URA⁺ TRP⁺ colonies, shown in column H). The relative excision frequency, determined by dividing the number of live cells (colonies on leu plates) by the number of URA⁺ colonies, is shown in column I. The relative transposition frequency, determined by dividing the number of live cells (colonies on leu plates) by the number of URA⁺ colonies, is shown in column J.

The two transposases showed comparable activity in excising their corresponding transposons from the reporter construct (shown by the appearance of URA⁺ colonies) and transposing the TRP gene in their corresponding transposons into another genomic location in the *Saccharomyces cerevisiae* reporter strain (shown by the appearance of URA⁺ TRP⁺ colonies). These activities were also very comparable

with the activities of the *Agrotis* transposase in excising or transposing the *Helicoverpa* transposon (compare rows 1 and 4 for the activity of the *Agrotis* transposase on the *Helicoverpa* (row 4) and *Agrotis* (row 1) transposon), and of the *Helicoverpa* transposase in excising or transposing the *Agrotis* transposon (compare rows 2 and 3 for the activity of the *Helicoverpa* transposase on the *Helicoverpa* (row 3) and *Agrotis* (row 2) transposon). The relative excision and relative transposition frequencies for the two transposases acting on their corresponding transposons differed by less than 2-fold from their activities on the heterologous transposon.

The *Agrotis* and *Helicoverpa* transposase sequences (SEQ ID NOs 40 and 41 respectively) differ by 45 amino acids over their total length of 599 amino acids, so they are 92.48% identical. The sequences between the transposon ITRs and the open reading frame encoding the transposase in the naturally occurring transposon are slightly shorter in the *Agrotis* transposon than the *Helicoverpa* transposon. Over the length of the 303 bp of the left sequence (i.e. to the 5' of the transposase when considering the direction of translation) but excluding the ITR, there are 24 differences between the *Agrotis* sequence (SEQ ID NO: 27) and the *Helicoverpa* sequence (SEQ ID NO: 35), so these two left end sequences are 92.1% identical. Over the length of the 347 bp of the right sequence but excluding the ITRs, there are 37 differences between the *Agrotis* sequence (SEQ ID NO: 28) and the *Helicoverpa* sequence (SEQ ID NO: 36), so the two right end sequences are 89.3% identical. This suggests the degree of identity between transposons and transposases that may be required for the transposase to act on a heterologous transposon. If a first and second transposon share identical ITR sequences, and the first transposon left transposon end is more than 93% identical with the second transposon left transposon end, and the first transposon right transposon end is more than 93% identical with the second transposon right transposon end, if their corresponding transposases are also at least 93% identical, then the first transposase may be expected to transpose the second transposon and the second transposase may be expected to transpose the first transposon.

6.1.4 the *Heliothis* Transposase is Active in Mammalian Cells

PiggyBac-like transposases can transpose their corresponding transposons into the genomes of eukaryotic cells including yeast cells such as *Pichia pastoris* and *Saccharomyces cerevisiae*, and mammalian cells such as human embryonic kidney (HEK) and Chinese hamster ovary (CHO) cells. To determine the activity of the *Heliothis* transposase in mammalian cells, we constructed a gene transfer polynucleotide comprising transposon ends, and further comprising a selectable marker encoding glutamine synthetase with a polypeptide sequence given by SEQ ID NO: 527, encoded by DNA sequence given by SEQ ID NO: 532 and operably linked to regulatory elements that give weak glutamine synthetase expression, the sequence of the glutamine synthetase and its associated regulatory elements given by SEQ ID NO: 570. The gene transfer polynucleotides further comprised open reading frames encoding the heavy and light chains of an antibody, each operably linked to a promoter and polyadenylation signal sequence. The gene transfer polynucleotide (with SEQ ID NO: 507) comprised a left transposon end comprising a 5'-TTAA-3' target integration sequence immediately followed by a *Heliothis* left transposon end with ITR sequence given by SEQ ID NO: 17, which is an embodiment of SEQ ID NO: 15. The gene transfer polynucleotide further comprised a *Heliothis* right transposon

son end with ITR sequence given by SEQ ID NO: 18, which is an embodiment of SEQ ID NO: 16. The two *Heliothis* transposon ends were placed on either side of the heterologous polynucleotide comprising the glutamine synthetase selectable marker and the open reading frames encoding the heavy and light chains of the antibody. The left transposon end further comprised a sequence given by SEQ ID NO: 13 immediately adjacent to the left ITR. The right transposon end further comprised a sequence given by SEQ ID NO: 14 immediately adjacent to the right ITR.

Gene transfer polynucleotides were transfected into CHO cells which lacked a functional glutamine synthetase gene. Cells were transfected by electroporation: 25 µg of gene transfer polynucleotide DNA with SEQ ID NO: 507 was co-transfected with 3 µg of DNA comprising an open reading frame encoding *Heliothis* transposase SEQ ID NO: 39 operably linked to a human CMV promoter and a polyadenylation signal sequence. The cells were incubated in media containing 4 mM glutamine for 48 hours following electroporation, and subsequently diluted to 300,000 cells per ml in media lacking glutamine. Cells were exchanged into fresh glutamine-free media every 5 days. The viability of the cells from each transfection were measured at various times following transfection using a Beckman-Coulter Vi-Cell. The total number of viable cells were also measured with the same instrument. The results are shown in Table 6.

As shown in Table 6, the viability of cells transfected with the gene transfer polynucleotide but no transposase fell to about 27% by 12 days post-transfection (column B). The total number of live cells fell to fewer than 50,000 per ml within 7 days (column C). At or below this density of live cells, viability measurements become inaccurate. The culture never recovered. In contrast when the gene transfer polynucleotide was co-transfected with the *Heliothis* transposase, cells recovered to greater than 90% viability within 21 days (Table 6 column D), by which time the density of live cells exceeded 1 million per ml (Table 6 column E). This shows that a gene transfer polynucleotide comprising a left and right *Heliothis* transposon end can be efficiently transposed into the genome of a mammalian target cell by a corresponding *Heliothis* transposase.

The recovered pools of CHO cells comprising piggyBac-like transposons integrated into their genomes were grown in a 14 day fed-batch using Sigma Advanced Fed Batch media. Antibody titers were measured in culture supernatant using an Octet. Table 7 shows the titers measured at 7, 10, 12 and 14 days of the fed batch culture. The titer of antibody from cells comprising gene transfer polynucleotide with SEQ ID NO: 507, that had been integrated by co-transfection with the *Heliothis* transposase SEQ ID NO: 39 reached approximately 2 g/L after 14 days. This shows that the *Heliothis* transposon and its corresponding transposase, as described in Section 5.2.5, is a novel, piggyBac-like transposon/transposase system that is active in mammalian cells.

6.1.5 Messenger RNA Encoding the *Heliothis* Transposase is Active in Mammalian Cells

We further tested gene transfer polynucleotide with SEQ ID NO: 507, whose configuration is described in Section 6.1.4, to determine whether the synthetic *Heliothis* transposon could be integrated into the genome of a mammalian cell if the corresponding transposase was provided in the form of mRNA.

mRNA encoding transposases was prepared by in vitro transcription using T7 RNA polymerase. The mRNA comprised a 5' sequence SEQ ID NO: 508 preceding the sequence encoding the open reading frame, and a 3'

sequence SEQ ID NO: 509 following the stop codon at the end of the open reading frame. The mRNA had an anti-reverse cap analog (3'-O-Me-m7G (5')ppp(5') G. DNA molecules comprising a sequence encoding a transposase operably linked to a heterologous promoter that is active in vitro are useful for the preparation of transposase mRNA. Isolated mRNA molecules comprising a sequence encoding a transposase are useful for integration of a corresponding transposon into a target genome.

Gene transfer polynucleotide SEQ ID NO: 507 was transfected into CHO cells which lacked a functional glutamine synthetase gene. Cells were transfected by electroporation: 25 µg of gene transfer polynucleotide DNA was co-transfected with 3 µg of mRNA comprising an open reading frame encoding a corresponding transposase (amino acid sequence SEQ ID NO: 39, nucleotide sequence SEQ ID NO: 42). The cells were incubated in media containing 4 mM glutamine for 48 hours following electroporation, and subsequently diluted to 300,000 cells per ml in media lacking glutamine. Cells were exchanged into fresh glutamine-free media every 5 days. The viability of the cells from each transfection were measured at various times following transfection using a Beckman-Coulter Vi-Cell. The total number of viable cells were also measured with the same instrument. The results are shown in Table 8.

When gene transfer polynucleotide with SEQ ID NO: 507 was co-transfected with mRNA encoding *Heliothis* transposase SEQ ID NO: 39, viability fell to around 33% by 14 days post-transfection (Table 8 column B), by which time the density of live cells was around 30,000 per ml (Table 8 column C). Cell viability and the density of live cells then increased until by 30 days post-transfection viability was above 95% and there were over 2 million live cells per ml. This shows that a gene transfer polynucleotide comprising a left and right *Heliothis* transposon end can be efficiently transposed into the genome of a mammalian target cell when co-transfected with mRNA encoding a corresponding *Heliothis* transposase.

6.1.6 *Heliothis* Transposon End Sequences Active in Mammalian Cells

When we originally tested the *Heliothis* transposon, we used the entire sequence between the 5'-TTAA-3' target sequences and the transposase open reading frame as transposon ends. We have found that for other piggy Bac-like sequences this full sequence is generally not required for transposition activity. We therefore constructed synthetic *Heliothis* transposons with truncated ends to determine whether these were transposable by a *Heliothis* transposase. A heterologous polynucleotide with SEQ ID NO: 70 encoded glutamine synthetase with a polypeptide sequence given by SEQ ID NO: 528, operably linked to regulatory elements that give weak glutamine synthetase expression as a selectable marker. On one side of the heterologous polynucleotide was a left *Heliothis* transposon end comprising a 5'-TTAA-3' integration target sequence immediately followed by a transposon ITR sequence with SEQ ID NO: 17 (which is an embodiment of SEQ ID NO: 15). On the other side of the heterologous polynucleotide was a right *Heliothis* transposon end comprising a transposon ITR sequence with SEQ ID NO: 18 (which is an embodiment of SEQ ID NO: 16) immediately followed by a 5'-TTAA-3' integration target sequence. The transposon further comprised an additional sequence selected from SEQ ID NOs: 13 or 19 immediately adjacent to the left transposon ITR sequence. The transposon further comprised an additional sequence selected from SEQ ID NOs: 14, or 20-22 immediately adjacent to the right transposon ITR sequence. Transposons were transfected into

CHO cells which lacked a functional glutamine synthetase gene. Cells were transfected by electroporation: 25 µg of gene transfer polynucleotide DNA were transfected, optionally the cells were co-transfected with 3 µg of a second plasmid comprising an open reading frame encoding a corresponding transposase (amino acid sequence SEQ ID NO: 144) operably linked to a CMV promoter and expressible in a mammalian cell. The cells were incubated in media containing 4 mM glutamine for 48 hours following electroporation, and subsequently diluted to 300,000 cells per ml in media lacking glutamine. Cells were exchanged into fresh glutamine-free media every 5 days. The viability of the cells from each transfection were measured at various times following transfection using a Beckman-Coulter Vi-Cell. The total number of viable cells were also measured with the same instrument. The results are shown in Table 9.

Table 9 columns A and B show the reduction in cell viability and viable cell density when cells were transfected with a transposon comprising full length transposon ends in the absence of transposase. Cell viability and viable cell density can both be seen to fall throughout the experiment. In contrast when any the same transposon was co-transfected with DNA encoding a *Heliothis* transposase, the cell viability and viable cell density fell initially, but had begun to recover by day 12 and was fully recovered between day 17 and 21 (Table 9 columns C and D). A comparable result was obtained when the left transposon end was truncated from the sequence given by SEQ ID NO: 13, to the sequence given by SEQ ID NO: 19 (compare Table 9 columns E and F with columns G and H respectively). A comparable result was also obtained when the right transposon end was truncated from the sequence given by SEQ ID NO: 14, to the sequence given by SEQ ID NO: 20 (compare Table 9 columns I and J with columns K and L respectively); or to the sequence given by SEQ ID NO: 21 (compare Table 9 columns M and N with columns O and P respectively); or to the sequence given by SEQ ID NO: 22 (compare Table 9 columns Q and R with columns S and T respectively). This shows that in addition to an integration target sequence immediately adjacent to a transposon ITR sequence with SEQ ID NO: 15, a *Heliothis* synthetic transposon left transposon end may further comprise an additional sequence selected from SEQ ID NOs: 13 and 19 immediately adjacent to the left transposon ITR sequence; and a *Heliothis* synthetic transposon right transposon end may comprise an additional sequence selected from SEQ ID NOs: 14 or 20-22, immediately adjacent to an ITR with sequence SEQ ID NO: 16, immediately adjacent to an integration target sequence.

6.1.7 Identifying Hyperactive *Heliothis* Transposases

To identify *Heliothis* transposase mutations that led to either increased transposition activity, or increased excision activity, relative to the naturally occurring *Heliothis* transposase sequence given by SEQ ID NO: 39, we analyzed a CLUSTAL alignment of active piggy Bac-like transposases. Table 1 column C shows the amino acids found in active piggyBac-like transposases relative to each position in the *Heliothis* transposase (position shown in Table 1 column A). The amino acid present in *Heliothis* transposase given by SEQ ID NO: 39 is shown in column B of Table 1. Because transposases are often deleterious to their hosts, they tend to accumulate mutations that inactivate them. The mutations that accumulate in different transposases are different, as each occurs by random chance. A consensus sequence can therefore be used to approximate an ancestral sequence that pre-dates the accumulation of deleterious mutations. It is difficult to accurately calculate an ancestral sequence from a

small number of extant sequences, so we chose to focus on positions where active transposases were more highly conserved, and where the consensus amino acid(s) differed from the one in the *Heliothis* transposase. We considered that mutating these amino acids to the consensus amino acids found in other active transposases would be likely to increase the activity of the *Heliothis* transposase. These candidate beneficial amino acid substitutions are shown in Table 1 column D.

6.1.7.1 First Set of *Heliothis* Transposase Variants

A set of 95 open reading frames encoding variant *Heliothis* transposases comprised one or more substitutions selected from S41V, L43S, Q74S, V81E, V81P, V135G, S136V, S136P, N137K, N137T, I149C, W150F, K152L, L153F, D155T, T162I, L188T, K233R, K233L, V238F, I239L, H241N, H241R, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M303I, M305N, L312I, L314F, L314I, R334K, I378Y, K385L, K396R, Y419F, Y421F, L428V, H430Y, V450I, I459M, R485K, R485N, Y495F, D499N, S501A, L507I, H512N, L552I, Q586D, Q586R, S589T, V600I, Q603K, Q603E and C604H. Each substitution was represented at least 5 times within the set of 96 variants, and the number of different pairwise combinations of substitutions was maximized so that each substitution was tested in as many different sequence contexts as possible. Each variant open reading frame was cloned into a vector comprising a leucine selectable marker; each open reading frame encoding a transposase variant was operably linked to the *Saccharomyces cerevisiae* Gal-1 promoter. Each of these variants was then individually transformed into a *Saccharomyces cerevisiae* strain comprising a chromosomally integrated copy of SEQ ID NO: 66, as described above. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases, then a 100x-diluted aliquot was plated on media lacking leucine, uracil and tryptophan (to count transposition), a 100x-diluted aliquot was plated on media lacking leucine and uracil (to count excision) and a 25,000x-diluted aliquot was plated on media lacking leucine (to count total live cells). Two days later, colonies were counted to determine transposition (=number of cells on -leu -ura -trp media divided by (250x number of cells on -leu media)) and excision (=number of cells on -leu-ura media divided by (250xnumber of cells on -leu media)) frequencies. The results are shown in Table 10. We identified 2 *Heliothis* transposase variants (with sequences given by SEQ ID NO: 158 and 159) with excision or transposition activities that were between 2 and 4-fold higher than the activities measured for the naturally occurring *Heliothis* transposase. We also identified 8 *Heliothis* transposase variants (with sequences given by SEQ ID NO: 369-376) with excision or transposition activities that were improved by less than 2-fold relative to the naturally occurring *Heliothis* transposase. We also identified 45 *Heliothis* transposase variants (with sequences given by SEQ ID NO: 401-445) whose excision or transposition activities ranged from equal to the naturally occurring *Heliothis* transposase to around one fifth as active, and we identified 24 variants that were less than one fifth as active as the naturally occurring *Heliothis* transposase, but whose activity was still measurable (with sequences given by SEQ ID NO: 466-489). Only 16 of the 95 variants (with sequences given by SEQ ID NO: 491-506) possessed activities that were so low that they were essentially inactive.

The effects of sequence changes on excision and transposition activities were modelled as described in U.S. Pat. No. 8,635,029 and Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 "Engineering proteinase K using machine learning and synthetic genes"). Mean values and standard deviations for the regression weights were calculated for each substitution, these are shown in Table 11. The effect of an individual substitution upon transposase activity may vary depending on the context (i.e. the other substitutions present). A positive mean regression weight indicates that on average, considering all of the different sequence contexts in which it has been tested, the substitution has a positive influence on the measured property. Incorporation of substitutions with positive mean regression weights into a sequence generally results in variants with improved activity (Liao et. al., ibid). A measure of this context-dependent variability is the standard deviation of the regression weight. If the mean regression weight plus the standard deviation of the regression weight for a substitution is zero or greater, then there are contexts within which the substitution has a positive effect. Thirty-one of the sixty substitutions we selected by looking for changes toward the consensus in other active piggyBac-like transposases had a mean regression weight plus the standard deviation of the regression weight of zero or greater: S41V, L43S, V81E, V81P, S136V, I149C, K152L, L153F, D155T, T162I, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, K396R, Y421F, H430Y, V450I, Y495F, L507I, H512N, S589T, Q603K. In addition to identifying specific substitutions with a beneficial effect, this also provides an indication of positions at which analogous substitutions may be beneficial. For example, replacement of the nucleophilic threonine at position 268 with either the acidic residue aspartate or the acidic residue glutamate has a positive effect. This provides evidence that multiple different analogous substitutions at a position can be beneficial. Analogous substitutions are those in which properties of the amino acids are conserved. For example: glycine and alanine are in the "small" amino acid group; valine, leucine, isoleucine and methionine are in the "hydrophobic" amino acid group; phenylalanine, tyrosine and tryptophan are in the "aromatic" amino acid group; aspartate and glutamate are in the "acidic" amino acid group; asparagine and glutamine are in the "amide" amino acid group; histidine, lysine and arginine are in the "basic" amino acid group; cysteine, serine and threonine are in the "nucleophilic" amino acid group. If a substitution at an amino acid position within the *Heliothis* transposase is beneficial for excision or transposition activity, other substitutions at the same position drawn from the same amino acid group are likely to be beneficial. For example, since replacing the hydrophobic residue valine at position 81 with the acidic residue glutamate (V81E) is beneficial, replacing with the acidic residue aspartate (i.e. V81D) is likely also to be beneficial. Similarly, since replacing the hydrophobic residue methionine at position 300 with the basic residue arginine (M300R) is beneficial, replacing with the basic residues histidine or lysine (i.e. M300H or M300K) are likely also to be beneficial. An advantageous hyperactive *Heliothis* transposase comprises a substitution at one or more positions selected from amino acids 41, 43, 81, 81, 136, 149, 152, 153, 155, 162, 238, 239, 241, 264, 268, 268, 280, 297, 299, 300, 305, 312, 396, 421, 430, 450, 495, 507, 512, 589 and 603, for example one or more substitutions selected from S41V, L43S, V81E, V81P, S136V, I149C, K152L, L153F, D155T, T162I, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C,

L299F, M300R, M305N, L312I, K396R, Y421F, H430Y, V450I, Y495F, L507I, H512N, S589T and Q603K, or analogous changes at the same positions.

Of the 60 amino acid substitutions selected, only one (R485N) was found exclusively in essentially inactive transposases. This indicates that amino acid changes that make the *Heliothis* transposase sequence closer to a consensus sequence for active piggy Bac-like transposases are substitutions that can be incorporated the *Heliothis* transposase to create variants that retain transposition and excision activity. These are the changes listed in Table 1 column D. The positive results from the initial set of substitutions provide evidence that other changes in column D of Table 1 can also be incorporated into an *Heliothis* transposase to improve its activity.

6.1.7.2 Second Set of *Heliothis* Transposase Variants

As described in Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”) and U.S. Pat. No. 8,635,029, Sections 5.4.2 and 5.4.3, substitutions that have been tested several times in the contexts of different combinations of other substitutions and that have “a positive regression coefficient, weight or other value describing its relative or absolute contribution to one or more activity” of a protein are usefully incorporated into a protein to obtain a protein that is “improved for one or more property, activity or function of interest”. Based on the substitution weights shown in Table 11, we designed a set of open reading frames encoding 95 new variant transposases combining some of the most positive substitutions (S41V, L43S, V81E, S136V, K152L, L153F, D155T, T162I, 1239L, H241N, S264R, T268D, V280P, T297C, L299F, M300R, M305N, L312I, Y421F, V450I, Y495F, L507I and H512N). Each substitution was represented at least 10 times within the set of 95 variants, and the number of pairwise combinations was maximized so that each substitution was tested in as many different sequence contexts as possible. Each variant open reading frame was cloned into a vector comprising a leucine selectable marker; each open reading frame encoding a transposase variant was operably linked to the *Saccharomyces cerevisiae* Gal-1 promoter. Each of these variants was then individually transformed into a *Saccharomyces cerevisiae* strain comprising a chromosomally integrated copy of SEQ ID NO: 66, as described in Section 6.1.7.1. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases, then a 100×-diluted aliquot was plated on media lacking leucine, uracil and tryptophan (to count transposition), a 100×-diluted aliquot was plated on media lacking leucine and uracil (to count excision) and a 25,000×-diluted aliquot was plated on media lacking leucine (to count total live cells). Two days later, colonies were counted to determine transposition (=number of cells on -leu -ura -trp media divided by (250×number of cells on -leu media)) and excision (=number of cells on -leu -ura media divided by (250×number of cells on -leu media)) frequencies. The results are shown in Table 12.

In addition to the activities of the 95 new *Heliothis* transposase variants, Table 12 also shows the activities of 6 variants from the first set that were among the most active variants in that set (transposases with polypeptide sequences given by SEQ ID NOs: 158, 159, 369, 372, 375 and 376). More than 60 of the new set of variants had greater transposition activity than the best variants from the first set. Every variant in the new set of variants, which all comprised

various combinations of amino acid substitutions selected from S41V, L43S, V81E, S136V, K152L, L153F, D155T, T162I, 1239L, H241N, S264R, T268D, V280P, T297C, L299F, M300R, M305N, L312I, Y421F, V450I, Y495F, L507I and H512N (all of which had positive mean regression weights) was increased in either excision or transposition activity relative to the naturally occurring transposase. A preferred hyperactive *Heliothis* transposase comprises one or more amino acid substitution selected from S41V, L43S, V81E, S136V, K152L, L153F, D155T, T162I, 1239L, H241N, S264R, T268D, V280P, T297C, L299F, M300R, M305N, L312I, Y421F, V450I, Y495F, L507I and H512N, or analogous changes at the same positions.

6.1.7.3 Third Set of *Heliothis* Transposase Variants

Using the activities shown in Table 12, we modelled the effects of substitutions on transposition and excision as described in Section 6.1.7.2. We selected the fourteen most positive substitutions (S41V, L43S, V81E, S136V, 1239L, H241N, T268D, T297C, M300R, M305N, L312I, Y421F, V450I, Y495F) and a new set of 29 substitutions identified from Table 1 column D (D83S, V85L, P125S, Q126S, Q131R, Q131T, E140C, E140A, N151Q, K169E, N212S, C316A, C316M, N322T, P351G, H357R, H357D, K360Q, E379P, K397S, A447N, A447D, A449S, A449V, K476L, V492A, I500M, L585K and T595K). We designed a set of open reading frames encoding 96 new variants combining these substitutions. Each substitution was represented at least 5 times within the set of 96 variants, and the number of different pairwise combinations of substitutions was maximized so that each substitution was tested in as many different sequence contexts as possible. Each variant open reading frame was cloned into a vector comprising a leucine selectable marker; each open reading frame encoding a transposase variant was operably linked to the *Saccharomyces cerevisiae* Gal-1 promoter. Each of these variants was then individually transformed into a *Saccharomyces cerevisiae* strain comprising a chromosomally integrated copy of SEQ ID NO: 66, as described in Section 6.1.7.1. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases, then a 1,000×-diluted aliquot was plated on media lacking leucine, uracil and tryptophan (to count transposition), a 1,000×-diluted aliquot was plated on media lacking leucine and uracil (to count excision) and a 25,000×-diluted aliquot was plated on media lacking leucine (to count total live cells). Two days later, colonies were counted to determine transposition (=number of cells on -leu -ura -trp media divided by (25×number of cells on -leu media)) and excision (=number of cells on -leu -ura media divided by (25×number of cells on -leu media)) frequencies. The results are shown in Table 13.

In addition to the activities of the 95 new *Heliothis* transposase variants, Table 13 also shows the activities of 2 variants from previous sets that were among the most active variants in those sets (transposases with polypeptide sequences given by SEQ ID NO: 158 and 174). More than 60 of the new set of variants had greater transposition activity than the best of these. A preferred hyperactive *Heliothis* transposase comprises one or more amino acid substitution selected from S41V, L43S, V81E, S136V, 1239L, H241N, T268D, T297C, M300R, M305N, L312I, Y421F, V450I, Y495F, D83S, V85L, P125S, Q126S, Q131R, Q131T, E140C, E140A, N151Q, K169E, N212S, C316A, C316M, N322T, P351G, H357R, H357D, K360Q,

E379P, K397S, A447N, A447D, A449S, A449V, K476L, V492A, I500M, L585K and T595K, or analogous changes at the same positions.

The effects of sequence changes on excision and transposition frequencies were modelled as described in Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 "Engineering proteinase K using machine learning and synthetic genes") and U.S. Pat. No. 8,635,029. Mean values and standard deviations for the regression weights were calculated for each substitution, these are shown in Table 14. A positive regression weight indicates that the substitution has a positive influence on the measured property, but this is relative to the mean of the set of variants. In the third set of variants, several substitutions that were identified as positive in rounds 1 and 2 were incorporated into every variant in set 3, for example L43S, V81E, T268D, M300R and M305N. These all had positive regression weights when their effects on activity were modelled in round 1 (see Table 11). However in Table 14 it can be seen that all of these substitutions have regression weights of 0 within the third set of variants. This does not mean that these substitutions are no longer have a positive effect on transposase activity, it means that they are now having a neutral effect relative to the activity of the entire set. Since they are present in every variant in the set, they cannot be increasing the activities of some members of the set but not others. Instead they are now increasing the average activity of the entire set of variants. From Table 14, the following substitutions are seen to have positive regression weights for transposition, or were present in every variant: S41V, L43S, V81E, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, N151Q, K169E, N212S, I239L, T268D, T297C, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K397S, Y421F, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L585K and T595K. A preferred hyperactive *Heliothis* transposase comprises one or more of these amino acid substitutions, or analogous changes at the same positions.

6.1.7.4 Fourth Set of *Heliothis* Transposase Variants

Using the activities shown in Table 13, we modelled the effects of substitutions on transposition and excision as described in Section 6.1.7.2 We selected some of the most positive substitutions (S41V, L43S, V81E, D83S, Q126S, Q131R, Q131T, S136V, E140C, E140A, K169E, T268D, T297C, M300R, M305N, L312I, C316A, N322T, H357R, K397S, Y421F, A449S, V450I, V492A, Y495F and L585K). We designed a set of open reading frames encoding 90 new variants combining these substitutions. Each substitution was represented at least 5 times within the set of 90 variants, and the number of different pairwise combinations of substitutions was maximized so that each substitution was tested in as many different sequence contexts as possible. Each variant open reading frame was cloned into a vector comprising a leucine selectable marker; each open reading frame encoding a transposase variant was operably linked to the *Saccharomyces cerevisiae* Gal-1 promoter. Each of these variants was then individually transformed into a *Saccharomyces cerevisiae* strain comprising a chromosomally integrated copy of SEQ ID NO: 66, as described in Section 6.1.7.1. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases, then a 2,000x-diluted aliquot was plated on media lacking leucine, uracil and tryptophan (to count transposition), a 2,000x-diluted aliquot was plated on media lacking leucine and uracil (to count excision) and

a 25,000x-diluted aliquot was plated on media lacking leucine (to count total live cells). Two days later, colonies were counted to determine transposition (=number of cells on -leu -ura -trp media divided by (12.5xnumber of cells on -leu media)) and excision (=number of cells on -leu -ura media divided by (12.5xnumber of cells on -leu media)) frequencies. The results are shown in Table 15.

In addition to the activities of the 90 new *Heliothis* transposase variants, Table 15 also shows the activities of 3 variants from previous sets that were among the most active variants in that set. More than 50 of the new set of variants had greater transposition activity than the best of these. A preferred hyperactive *Heliothis* transposase comprises one or more amino acid substitution selected from S41V, L43S, V81E, D83S, Q126S, Q131R, Q131T, S136V, E140C, E140A, K169E, T268D, T297C, M300R, M305N, L312I, C316A, N322T, H357R, K397S, Y421F, A449S, V450I, V492A, Y495F and L585K.

We also analyzed the variants with transposition activities at least twice as high as the naturally occurring transposase with sequence given by SEQ ID NO: 39. Twenty-seven of these were more than 10-fold more active (sequences given by SEQ ID NO: 131-157), and 211 were between 2 and 10-fold more active (sequences given by SEQ ID NO: 158-368). The following substitutions occurred at least once within this set: S41V, L43S, V81E, S136V, K152L, L153F, D155T, T162I, I239L, H241N, S264R, T268D, V280P, T297C, L299F, M300R, M305N, L312I, Y421F, V450I, R485K, Y495F, L507I, H512N, Q603E, D83S, V85L, P125S, Q126S, Q131T, Q131R, E140C, E140A, N151Q, K169E, N212S, C316A, N322T, P351G, H357R, K360Q, E379P, K397S, A447N, A447D, A449V, A449S, K476L, V492A, I500M, L585K and T595K. A preferred hyperactive *Heliothis* transposase comprises one or more amino acid substitution selected from this set, or analogous changes at the same positions.

We also analyzed the mean regression weights for each set of variants. A preferred hyperactive *Heliothis* transposase comprises an amino acid substitution at one or more of the positions at which we observed positive mean regression weights for transposition, that is at one of the following amino acid positions relative to SEQ ID NO: 39: 41, 43, 81, 81, 83, 85, 125, 126, 131, 131, 136, 140, 140, 149, 151, 152, 153, 155, 162, 169, 212, 238, 239, 241, 264, 268, 268, 280, 297, 299, 300, 305, 312, 316, 322, 357, 360, 396, 397, 421, 421, 430, 447, 447, 449, 449, 450, 476, 485, 492, 495, 495, 507, 512, 585, 589, 595, 603. A preferred hyperactive *Heliothis* transposase comprises one or more amino acid substitution selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, Y495F, L507I, H512N, L585K, S589T, T595K, Q603K, or an analogous substitution at the same position.

Brief Description of Tables

Table 1. Amino Acid Changes Likely to Result in Enhanced Activity of the *Heliothis* Transposase.

Amino acid substitutions with the potential to improve transposase activity were identified as described in Section 5.2.7. Column A shows the position in a *Heliothis* transposase (relative to SEQ ID NO: 39), column B shows the amino

acid in the native protein, column C shows the amino acids found in known active piggy Bac-like transposases at the equivalent position in an alignment, column D shows amino acid changes found in known active piggyBac-like transposases other than the *Heliothis* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Heliothis* transposase sequence is an outlier. Mutation to these amino acids are particularly likely to result in enhanced transposase activity.

Table 2. Amino Acid Changes Likely to Result in Enhanced Activity of the *Helicoverpa* Transposase.

Amino acid substitutions with the potential to improve transposase activity were identified as described in Section 5.2.7. Column A shows the position in a *Helicoverpa* transposase (relative to SEQ ID NO: 41), column B shows the amino acid in the native protein, column C shows the amino acids found in known active piggy Bac-like transposases at the equivalent position in an alignment, column D shows amino acid changes found in known active piggyBac-like transposases other than the *Helicoverpa* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Helicoverpa* transposase sequence is an outlier. Mutation to these amino acids are particularly likely to result in enhanced transposase activity.

Table 3. Amino Acid Changes Likely to Result in Enhanced Activity of the *Agrotis* Transposase.

Amino acid substitutions with the potential to improve transposase activity were identified as described in Section 5.2.7. Column A shows the position in a *Agrotis* transposase (relative to SEQ ID NO: 40), column B shows the amino acid in the native protein, column C shows the amino acids found in known active piggy Bac-like transposases at the equivalent position in an alignment, column D shows amino acid changes found in known active piggyBac-like transposases other than the *Agrotis* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Agrotis* transposase sequence is an outlier. Mutation to these amino acids are particularly likely to result in enhanced transposase activity.

Table 4. Excision and Transposition of Transposons in Yeast.

Transposon and transposase sources are listed in column A. The left sequence with SEQ ID NO shown in column B and the right sequence with SEQ ID NO shown in column C were used to construct reporter plasmids as described in Section 6.1.2. The reporter plasmids have insert sequence given by the SEQ ID NO listed in column D. These reporter plasmids were integrated into the Ura3 gene of a Trp-strain of *Saccharomyces cerevisiae*. The amino acid sequence given by the SEQ ID NO shown in column E was back translated, synthesized and cloned into a plasmid comprising a Leu2 gene expressible in *Saccharomyces cerevisiae* and 2 micron origin of replication. The transposase open reading frame was operably linked to a Gall promoter. The plasmid comprising the transposase was transformed into the reporter strain, expression was induced, and cells were plated as described in Section 6.1.1. Induced culture was diluted 25,000-fold prior to plating 100 μ l on leu dropout plates, and 100-fold prior to plating 100 μ l on leu ura or leu ura trp dropout plates. Column F shows the number of colonies on the leu dropout plates; column G shows the number of colonies on the leu ura dropout plates (indicating excision of the transposon from the middle of the ura open reading frame in the reporter); column H shows the number of colonies on the leu ura trp dropout plates (indicating

excision of the transposon from the middle of the ura open reading frame in the reporter and transposition to another site in the genome).

Table 5. Transposition of Transposons by Related Transposases.

Putative transposons were identified from the sources listed in column A. The ends from these transposons were used to construct reporter plasmids as described in Section 6.1.3. The reporter plasmids have sequence given by the SEQ ID NO listed in column B. These reporter plasmids were integrated into the Ura3 gene of a Trp-strain of *Saccharomyces cerevisiae*. Putative transposases were identified from the sources listed in column C. The amino acid sequence given by the SEQ ID NO shown in column D was back translated to the DNA sequence given by the SEQ ID NO shown in column E. This DNA sequence was cloned into a plasmid comprising a Leu2 gene expressible in *Saccharomyces cerevisiae* and 2 micron origin of replication. The transposase open reading frame was operably linked to a Gall promoter. The plasmid comprising the transposase was transformed into the reporter strain, expression was induced, and cells were plated as described in Section 6.1.1. Induced culture was diluted 25,000-fold prior to plating 100 μ l on leu dropout plates, and 100 μ l of culture was directly plated on leu ura or leu ura trp dropout plates. Column F shows the number of colonies on the leu dropout plates; column G shows the number of colonies on the leu ura dropout plates (indicating excision of the transposon from the middle of the ura open reading frame in the reporter); column H shows the number of colonies on the leu ura trp dropout plates (indicating excision of the transposon from the middle of the ura open reading frame in the reporter and transposition to another site in the genome). The relative excision frequency, determined by dividing the number of live cells (colonies on leu plates) by the number of URA+ colonies, is shown in column I. The relative transposition frequency, determined by dividing the number of live cells (colonies on leu plates) by the number of URA+ colonies, is shown in column J.

Table 6. Transposition of Transposons into the Genome of CHO Target Cells.

Cells were transfected with transposon SEQ ID NO: 507 and transposases as described in Section 6.1.4. The transposase SEQ ID NO is shown in row 1. For each transfection, viability (the percentage of cells that are viable) and the total viable cell density (in millions of cells per ml) are shown in adjacent columns, as indicated in row 2. Rows 3-15 show these measurements at various times post-transfection, the days elapsed are shown in column A.

Table 7. Antibody Production from Transposons Integrated into the Genome of CHO Target Cells.

Cells were transfected with transposon SEQ ID NO: 507 and DNA encoding a transposase with polypeptide sequence given by SEQ ID NO: 39 as described in Section 6.1.4. The transposon SEQ ID NO is shown in column A, the co-transfected transposase SEQ ID NO is shown in column B. Recovery is shown in Table 6. During a 14 day fed batch antibody production run, the culture supernatant contained the concentration of antibody shown: column C shows the titer on Day 7; column D shows the titer on Day 10; column E shows the titer on Day 12; column F shows the titer on Day 14.

Table 8. Transposition of Transposons into the Genome of CHO Target Cells by mRNA-Encoded Transposase.

Cells were transfected with a transposon and mRNA-encoded transposase as described in Section 6.1.5. The transposon with sequence given by SEQ ID NO: 507 was co-transfected with mRNA encoding a transposase with

polypeptide sequence given by SEQ ID NO: 39. The viability (the percentage of cells that are viable) and the total viable cell density (in millions of cells per ml) are shown in adjacent columns, as indicated in row 1. Rows 4-14 show these measurements at various times post-transfection, the days elapsed since transfection are shown in column A. Table 9. Transposition of Transposons with Truncated End Sequences into the Genome of CHO Target Cells.

Cells were transfected with a transposon as described in Section 6.1.6. The transposon comprised a left transposon end comprising a 5'TTAA-3' integration target sequence immediately followed by a transposon ITR sequence with SEQ ID NO: 17 which was immediately followed by a left end sequence with SEQ ID NO shown in row 2. The transposon further comprised a heterologous polynucleotide with sequence SEQ ID NO: 70 which comprised a glutamine synthetase open reading frame operably linked to regulatory sequences to make it expressible in a mammalian cell. The transposon further comprised a right transposon end comprising a right end sequence with SEQ ID NO shown in row 3 immediately followed by a transposon ITR sequence with SEQ ID NO: 18 immediately followed by a 5'TTAA-3' integration target sequence. Row 1 shows transfections in which a plasmid encoding a transposase with amino acid sequence given by SEQ ID NO: 144 was co-transfected with the transposon. The viability (the percentage of cells that are viable) is indicated in columns labelled "V" in row 5 and the total viable cell density (in millions of cells per ml) is indicated in columns labelled "VCD" in row 4. Rows 5-14 show these measurements at various times post-transfection, the days elapsed since transfection are shown in column U. Table 10. Transposition and Excision Activities of *Heliothis* Transposase Variants (First Set).

Open reading frames encoding *Heliothis* transposase variants were designed, synthesized and cloned as described in Section 6.1.7.1. SEQ ID NOs of each variant are given in column A. Genes were transformed into a *Saccharomyces cerevisiae* strain whose genome comprised a single copy of transposase reporter transposon with SEQ ID NO: 66, and plated on media lacking leucine. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases. Cultures were diluted 100-fold into minimal media lacking leucine; one 100 µl aliquot was plated onto minimal media agar plates lacking leucine and uracil (to measure transposon excision) another 100 µl aliquot was plated onto minimal media agar plates lacking leucine, tryptophan and uracil (to measure transposon transposition). Each culture was also diluted 25,000-fold and a 100 µl aliquot was plated onto minimal media agar plates lacking leucine (to measure live cells). After 48 hours colonies on each plate were counted, the number of colonies on plates lacking leucine are shown in column B, the number of colonies on plates lacking leucine, uracil and tryptophan are shown in column C, the number of colonies on plates lacking leucine and uracil are shown in column D. Column E shows the transposition frequency (calculated as the number in column C, divided by the number in column B, and further divided by 250) Column F shows the excision frequency (calculated as the number in column D, divided by the number in column B, and further divided by 250).

Table 11. Model Weights for Amino Acid Substitutions in *Heliothis* Transposase Variants.

The effects of sequence changes on *Heliothis* transposase excision and transposition activities for the first set of

variants were modelled as described in Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 "Engineering proteinase K using machine learning and synthetic genes") and U.S. Pat. No. 8,635,029, see also Section 6.1.7.1. The mean values and standard deviations for the regression weights were calculated for each substitution. The position (relative to SEQ ID NO: 39) is shown in column A, the amino acid found at this position in SEQ ID NO: 39 is shown in column B. The tested amino acid substitution is shown in column C. The mean regression weight for the substitution on transposition activity is shown in column D, the standard deviation for this regression weight is shown in column E. The mean regression weight for the substitution on excision activity is shown in column F, the standard deviation for this regression weight is shown in column G.

Table 12. Transposition and Excision Activities of *Heliothis* Transposase Variants (Second Set).

Open reading frames encoding *Heliothis* transposase variants were designed, synthesized and cloned as described in Section 6.1.7.2. SEQ ID NOs of each variant are given in column A. Genes were transformed into a *Saccharomyces cerevisiae* strain whose genome comprised a single copy of transposase reporter SEQ ID NO: 66, and plated on media lacking leucine. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases. Cultures were diluted 100-fold into minimal media lacking leucine; one 100 µl aliquot was plated onto minimal media agar plates lacking leucine and uracil (to measure transposon excision) another 100 µl aliquot was plated onto minimal media agar plates lacking leucine, tryptophan and uracil (to measure transposon transposition). Each culture was also diluted 25,000-fold and a 100 µl aliquot was plated onto minimal media agar plates lacking leucine (to measure live cells). After 48 hours colonies on each plate were counted, the number of colonies on plates lacking leucine are shown in column B, the number of colonies on plates lacking leucine, uracil and tryptophan are shown in column C, the number of colonies on plates lacking leucine and uracil are shown in column D. Column E shows the transposition frequency (calculated as the number in column C, divided by the number in column B, and further divided by 250) Column F shows the excision frequency (calculated as the number in column D, divided by the number in column B, and further divided by 250).

Table 13. Transposition and Excision Activities of *Heliothis* Transposase Variants (Third Set).

Open reading frames encoding *Heliothis* transposase variants were designed, synthesized and cloned as described in Section 6.1.7.3. SEQ ID NOs of each variant are given in column A. Genes were transformed into a *Saccharomyces cerevisiae* strain whose genome comprised a single copy of transposase reporter SEQ ID NO: 66, and plated on media lacking leucine. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases. Cultures were diluted 1,000-fold into minimal media lacking leucine; one 100 µl aliquot was plated onto minimal media agar plates lacking leucine and uracil (to measure transposon excision) another 100 µl aliquot was plated onto minimal media agar plates lacking leucine, tryptophan and uracil (to measure transposon transposition). Each culture was also diluted 25,000-fold and a 100 µl aliquot was plated onto minimal media agar plates

lacking leucine (to measure live cells). After 48 hours colonies on each plate were counted, the number of colonies on plates lacking leucine are shown in column B, the number of colonies on plates lacking leucine, uracil and tryptophan are shown in column C, the number of colonies on plates lacking leucine and uracil are shown in column D. Column E shows the transposition frequency (calculated as the number in column C, divided by the number in column B, and further divided by 25) Column F shows the excision frequency (calculated as the number in column D, divided by the number in column B, and further divided by 25).

Table 14. Model Weights for Amino Acid Substitutions in *Heliothis* Transposase Variants.

The effects of sequence changes on *Heliothis* transposase excision and transposition activities were modelled for the third set of variants as described in Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”) and U.S. Pat. No. 8,635,029 and in Section 6.1.7.3. The mean values and standard deviations for the regression weights were calculated for each substitution. The position (relative to SEQ ID NO: 39) is shown in column A, the amino acid found at this position in SEQ ID NO: 39 is shown in column B. The tested amino acid substitution is shown in column C. The mean regression weight for the substitution on transposition activity is shown in column D, the standard deviation for this regression weight is shown in column E. The mean regression weight for the substitution on excision activity is shown in column F, the standard deviation for this regression weight is shown in column G. The number of times a substitution occurred within the set of 96 variants is shown in column H.

Table 15. Transposition and Excision Activities of *Heliothis* Transposase Variants.

Open reading frames encoding *Heliothis* transposase variants were designed, synthesized and cloned as described in Section 6.1.7.4. SEQ ID NOs of each variant are given in column A. Genes were transformed into a *Saccharomyces cerevisiae* strain whose genome comprised a single copy of transposase reporter SEQ ID NO: 66, and plated on media lacking leucine. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases. Cultures were diluted 2,000-fold into minimal media lacking leucine; one 100 µl aliquot was plated onto minimal media agar plates lacking leucine and uracil (to measure transposon excision) another 100 µl aliquot was plated onto minimal media agar plates lacking leucine, tryptophan and uracil (to measure transposon transposition). Each culture was also diluted 25,000-fold and a 100 µl aliquot was plated onto minimal media agar plates lacking leucine (to measure live cells). After 48 hours colonies on each plate were counted, the number of colonies on plates lacking leucine are shown in column B, the number of colonies on plates lacking leucine, uracil and tryptophan are shown in column C, the number of colonies on plates lacking leucine and uracil are shown in column D. Column E shows the transposition frequency (calculated as the number in column C, divided by the number in column B, and further divided by 12.5) Column F shows the excision frequency (calculated as the number in column D, divided by the number in column B, and further divided by 12.5).

TABLE 1

	A heliothis_position	B heliothis	C Acceptable	D Beneficial
5	1	M	M	
	2	E	EM	
	3	P	PSEAG	
	4	S	SMK	
	5	T	TARDN	
	6	S	SRQFI	
10	7	S	SGFRIYE	
	8	G	GLTDSRA	
	9	R	RTANDQ	
	10	K	KDEQH	
	11	R	RLEDHN	
	12	S	SEAIR	
	13	I	ILA	
15	14	G	GNLASR	
	15	N	NQLTAH	
	16	V	VIFLCM	
	17	H	HLFM	LFM
	18	N	NEDQA	
20	19	Q	QLSNE	
	20	R	REDSV	
	21	A	ADSLE	
	22	A	AVEDST	
	23	K	KEYLD	
	24	N	NESVfy	
25	25	R	RDIPGS	
	26	R	RVSLEGDY	
	27	A	AIEDS	
	28	V	VFISD	
	29	V	VDES	
	30	P	PGSVE	
	31	G	GDEP	
30	32	T	TEAKVP	
	33	R	RSEAT	
	34	D	DS	
	35	F	FHNRCE	
	36	G	GVDSC	
	37	T	TSIVD	
35	38	T	TIDES	
	39	L	LRDSH	
	40	T	TPDEN	
	41	S	SVDE	VDE
	42	W	WEFQN	
	43	L	LSQY	SQY
40	44	D	DFW	
	45	N	NTSC	
	46	E	EDS	
	47	D	DESQ	
	48	S	SVEA	
	49	S	SEMIFTRA	
45	50	G	GIDV	
	51	S	SPD	
	52	E	ETVYFASP	
	53	V	VLD	
	54	E	EHD	
	55	D	DPSEVL	
	56	I	ITVPL	
50	57	G	GSQ	
	58	D	DRAPN	
	59	N	NQTS	
	60	F	FESRG	
	61	T	TDSQ	
	62	P	PGE	
55	63	E	EDS	
	64	R	RA	
	65	H	HN	
	66	E	EAT	
	67	I	IV	
	68	E	EIV	
	69	S	SA	
60	70	D	DQVNE	
	71	T	TQVNIME	
	72	I	IGSLET	
	73	S	SPADVY	
	74	Q	QSED	SED
	75	S	SDQRE	
65	76	E	ESNDP	
	77	S	SENVd	

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TABLE 1-continued

A heliiothis_position	B heliiothis	C Acceptable	D Beneficial
78	E	ENLAPID	PE
79	E	EDGN	
80	Q	QMLV	
81	V	VPE	
82	A	ALQVGD	
83	D	DSTQ	10
84	H	HNSELA	
85	V	VLSA	
86	T	TAGR	
87	E	ERSDQ	
88	E	ERGHN	15
89	H	HSTRAM	
90	N	NWIDSF	
91	M	MAILC	
92	S	STA	
93	S	SKLAR	20
94	D	DPGQ	
95	D	D	
96	D	D	
97	A	A	
98	P	P	25
99	L	L	
100	S	SV	
101	T	TSG	
102	R	RQG	
103	R	RPS	30
104	S	SFTY	
105	F	FYI	
106	Y	YKCRTS	
107	G	GS	
108	K	K	35
109	N	NDG	
110	R	RNKEG	
111	Y	YTIHP	
112	K	KIVCA	
113	W	W	40
114	A	ANSKY	
115	C	CRTPK	
116	Q	QASTPN	
117	P	PKC	
118	L	LPHGSNQ	45
119	S	SNQRTF	
120	R	RPST	
121	A	ARTN	
122	V	VGSI	
123	R	R	50
124	V	VTL	
125	P	PRS	
126	Q	QSAE	
127	H	HESIL	
128	N	NP	55
129	I	IP	
130	I	IVF	
131	Q	QTKR	
132	R	RGMTSE	
133	T	TNQVR	60
134	N	NPRA	
135	V	VGQL	
136	S	SVP	
137	N	NKT	
138	L	LRVNT	65
139	T	TQFMDIG	
140	E	EACT	
141	D	DKVRS	
142	D	DNT	
143	P	PAVI	CAF
144	K	KLDVYFS	
145	D	DLET	
146	P	PEI	
147	F	FLQIYS	
148	S	SDLNEK	LI
149	I	ICAF	
150	W	WF	
151	N	NHOK	
152	K	KLI	
153	L	LF	F
154	M	MVIF	

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TABLE 1-continued

A heliiothis_position	B heliiothis	C Acceptable	D Beneficial
155	D	DNST	NST
156	D	DEQS	
157	E	ESPAD	
158	I	IM	
159	L	LEI	
160	Q	QSRHD	10
161	E	EVID	
162	T	TIM	
163	L	LV	
164	K	KEDLT	
165	W	WHMY	15
166	T	T	
167	N	N	
168	E	EHLVAS	
169	K	KEYS	
170	I	IGMA	20
171	I	IRSE	
172	Q	QSRVLH	
173	Y	YEVKRS	
174	R	RFLQ	
175	S	SQRTV	25
176	K	KGNQ	
177	F	FKLENTS	
178	S	SNKPA	
179	D	DTMLA	
180	K	KSITPE	30
181	D	DNSR	
182	R	RYS	
183	P	PYEAHFV	
184	E	EAKSTYH	
185	L	LYWFMK	35
186	R	RSKHQ	
187	N	NESDP	
188	L	LTI	
189	D	DTN	
190	M	MLQTEI	40
191	V	VTMDACS	
192	E	ED	
193	L	LMI	
194	H	HRNWKYK	
195	A	AR	45
196	F	FVYL	
197	I	IVF	
198	G	GA	
199	L	LI	
200	L	LT	50
201	L	LYITV	
202	F	FLMAI	
203	T	TAM	
204	A	AG	
205	V	VL	55
206	F	FYMRIT	
207	K	KR	
208	S	SDA	
209	N	NGK	
210	H	HRG	60
211	E	EQLMS	
212	N	NASL	
213	V	VLTE	
214	N	NQSDK	
215	Y	YDSE	65
216	L	LW	
217	F	FWD	
218	A	ANDTR	
219	T	TSAR	
220	D	DETS	CAF
221	G	GEFLV	
222	T	TNSL	
223	G	GS	
224	R	RIV	
225	E	EPTMD	LI
226	I	IRV	
227	F	FY	
228	R	RPVS	
229	C	CMAST	
230	V	VT	F
231	M	M	

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TABLE 1-continued

A heliiothis_position	B heliiothis	C Acceptable	D Beneficial
232	S	S	
233	K	KLR	LR
234	N	NREDQ	
235	R	RT	
236	F	FY	
237	L	LAHEDQY	
238	V	VFML	FML
239	I	IL	
240	L	LVSQ	
241	H	HNR	RN
242	C	CVFNS	
243	L	LIM	
244	R	RH	
245	F	FM	
246	D	DN	
247	N	ND	
248	P	PSRKT	
249	D	DTSA	
250	D	DTLIV	
251	R	RP	
252	E	EVPD	
253	E	EGTD	
254	R	RLQ	
255	R	RAPK	
256	E	EASGQK	
257	S	SIDNHT	
258	D	D	
259	K	KRAVN	
260	I	ILFM	
261	A	AILTH	
262	A	APK	
263	I	IVLF	
264	S	SR	R
265	Y	YQDKPS	
266	I	IVLM	
267	F	FYWI	
268	T	TED	ED
269	K	KEILSQ	
270	F	FWL	
271	V	VIS	
272	G	GKENHQ	
273	N	NIQRC	
274	C	CLF	
275	Q	QKPRIA	
276	K	KDLQAN	
277	I	IVLNA	
278	Y	YH	
279	N	NTVS	
280	V	VP	P
281	C	CYGS	
282	E	EPGSAQ	
283	Y	YNHF	
284	A	ALVI	LVI
285	T	TC	
286	V	VI	
287	D	D	
288	E	E	
289	M	MERQS	
290	L	L	
291	V	VL	
292	P	PAGLS	
293	F	F	
294	R	RK	
295	G	G	
296	R	R	
297	T	TCL	CL
298	H	HKPQL	
299	L	LF	F
300	M	MR	R
301	I	IQMV	
302	Y	Y	
303	M	MLI	
304	P	P	
305	M	MNS	NS
306	K	K	
307	P	PR	
308	A	ADS	

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TABLE 1-continued

A heliiothis_position	B heliiothis	C Acceptable	D Beneficial
309	K	KR	
310	Y	Y	
311	G	G	
312	L	LI	
313	K	KR	
314	L	LIF	IF
315	M	MIWPLYF	
316	C	CAMK	
317	L	LAM	
318	C	CV	
319	D	DAE	
320	A	AS	
321	N	NYKAGS	
322	N	NTS	
323	G	GYSKF	
324	Y	Y	
325	F	FSAMTV	
326	Y	YLWISV	
327	N	NKDY	
328	C	CMAGFL	
329	Y	YEQIML	
330	I	IVP	
331	Y	Y	
332	T	TALE	
333	G	G	
334	R	RDK	
335	G	GQSD	
336	S	SPT	
337	D	DGKQSL	
338	G	GTL	
339	A	APND	
340	G	GYP	
341	L	LKEVPA	
342	T	TVKPG	
343	E	ESNC	
344	E	ENQP	
345	E	EP	
346	K	K	
347	K	KD	
348	F	FL	
349	M	MT	
350	V	VKLRT	
351	P	PGSA	
352	T	THMEGF	
353	Q	QDRFYKE	
354	S	SVYI	
355	V	V	
356	I	IDLKWE	
357	H	HRED	
358	L	LIM	
359	A	AVSTI	
360	K	KQES	
361	P	PGT	
362	L	LIV	
363	F	FSQHILA	
364	G	GQR	
365	S	STQ	
366	N	NGCH	
367	R	RHF	
368	N	NH	
369	I	IVL	
370	T	TY	
371	C	CMVF	
372	D	D	
373	N	N	
374	W	WF	
375	F	FY	
376	T	TS	
377	S	SG	
378	I	IY	
379	E	EPRT	
380	L	LT	
381	I	IYGAFM	
382	E	EAKTL	
383	Y	YHENA	
384	L	LM	
385	K	KLQY	LQY

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TABLE 1-continued

A heliiothis_position	B heliiothis	C Acceptable	D Beneficial
386	K	KQCN	
387	K	KNRAEL	
388	G	GKNDR	
389	L	LT	
390	T	TP	
391	C	CAMIS	
392	V	VLCT	
393	G	G	
394	T	T	
395	M	MVI	
396	K	KRN	
397	K	KSR	
398	N	N	
399	K	KR	
400	R	RTPK	
401	E	ECGQ	
402	I	ILM	
403	P	P	
404	K	KPSERD	
405	E	EKVAS	
406	F	FIL	
407	L	LRKIT	
408	P	PEKNDR	
409	S	SRIKT	
410	K	KQRDG	
411	Q	QGS	
412	R	RN	
413	D	DEPRQ	
414	V	VIMGP	
415	G	GNHEA	
416	S	ST	
417	S	SY	
418	L	LIMAV	
419	Y	YFL	FL
420	G	GACR	
421	Y	YFK	FK
422	A	AQTDN	
423	G	GDEK	
424	Q	QDKPL	
425	N	NFALI	
426	T	TA	
427	I	IVL	
428	L	LVK	VK
429	S	SF	
430	H	HYF	YF
431	V	VICKDA	
432	P	P	
433	K	K	
434	K	KRP	
435	N	NSAK	
436	K	KR	
437	A	ANMV	
438	V	V	
439	I	IFLYV	
440	L	LMVA	
441	L	LM	
442	S	ST	
443	S	ST	
444	M	MLCI	
445	H	HD	
446	H	HTED	
447	A	ADNE	
448	E	ESAN	
449	A	AESV	
450	V	VIL	IL
451	D	DSNR	
452	E	ESTQ	
453	T	TERSQ	
454	T	TDNR	
455	G	GDV	
456	K	K	
457	P	P	
458	E	ESQDL	
459	I	IMC	
460	I	IVS	
461	G	GTLMK	
462	F	FDYE	

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TABLE 1-continued

A heliiothis_position	B heliiothis	C Acceptable	D Beneficial
463	Y	Y	
464	N	NS	
465	K	KSQ	
466	T	TY	
467	K	KM	
468	G	GSA	
469	G	G	
470	V	V	
471	D	D	
472	E	ENSTRV	
473	I	IVLFT	
474	D	D	
475	K	KQE	
476	K	KLVM	
477	C	CITQS	
478	A	ARKSH	
479	I	ITSVYN	
480	Y	YM	
481	T	TDSN	
482	S	SVCA	
483	S	SQNT	
484	R	R	
485	R	RNK	NK
486	T	TS	
487	R	RANK	
488	R	RA	
489	W	W	
490	P	PY	
491	M	MLK	
492	V	VTAK	
493	V	VIL	
494	F	FLG	
495	Y	YFI	
496	R	RWNGY	
497	M	MILV	
498	L	LVI	
499	D	DNQ	
500	I	ITVM	
501	S	SA	
502	T	TGAFCLS	
503	V	VIYR	
504	N	N	
505	S	SA	
506	H	HKYFC	
507	L	LIV	IV
508	I	IVL	
509	Y	YQW	
510	D	DMSCRKQ	
511	I	ILEHAT	
512	H	HNIA	NIA
513	H	HSNKV	
514	D	DGKE	
515	K	KNVA	
516	T	TLPV	
517	T	TPYQSV	
518	E	ETRNSYK	
519	R	RY	
520	G	GRKT	
521	M	MALEKY	
522	F	FQ	
523	L	LIM	
524	K	KERQ	
525	Q	QKENIS	
526	L	L	
527	A	AGSYP	
528	R	RMKTIAL	
529	T	TSADQL	
530	L	LM	
531	V	VITF	
532	L	LATSGY	
533	P	PSGE	
534	Q	QHKWV	
535	M	MQIEL	
536	K	KAQREH	
537	R	REKQS	
538	R	RT	
539	A	AKLVN	

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TABLE 1-continued

A heliothis_position	B heliothis	C Acceptable	D Beneficial
540	L	LTRKQEP	
541	N	NPAEK	
542	E	ESAPMK	
543	R	RKTPN	
544	L	LISP	
545	P	PKS	
546	R	RVADTF	
547	E	ESATYNH	
548	L	LAVI	
549	R	RA	
550	L	LKVDRQ	
551	S	SRINL	
552	L	LIE	IE
553	A	AGKTSE	
554	R	RSINK	
555	V	VHKIQ	
556	L	LF	
557	G	GRKPI	
558	P	PETNKID	
559	D	DSVETP	
560	M	MSNVLT	
561	P	PAQ	
562	V	VSATR	
563	P	PMSH	
564	D	DAEGV	
565	P	PKND	
566	Q	QIVNSM	
567	E	EPDSTR	
568	V	VNTE	
569	D	DE	
570	E	EP	
571	T	TGEV	
572	F	FVPM	
573	K	KG	
574	T	TVKPR	
575	R	RKQY	
576	R	RKSTV	
577	R	RYG	
578	C	C	
579	H	HYQGTR	
580	T	TIVFYDE	
581	C	C	
582	P	PSR	
583	L	LVSYKN	
584	K	KR	
585	L	LKDI	
586	Q	QDR	DR
587	R	RS	
588	K	KMD	
589	S	STA	
590	T	TKSNR	
591	H	HYTAR	
592	T	TISQY	
593	C	CF	
594	Y	YIVCKPN	
595	T	TSKA	
596	C	C	
597	K	KTPA	
598	K	KSNR	
599	H	HFAVNP	
600	V	VIL	LI
601	C	C	
602	L	LRGFM	
603	Q	QEK	KE
604	C	CHP	HP
605	A	ATNC	
606	K	KNVIF	
607	Q	QFTDE	
608	V	VFMIL	
609	C	CY	
610	A	AEPQH	
611	D	DNST	
612	C	CQ	
613	V	VRGFLA	

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TABLE 2

A helicoverpa_position	B helicoverpa	C Acceptable	D Beneficial_Ha
1	M	M	
2	A	PSEAG	
3	S	SMK	
4	R	TARDN	
5	Q	SRQFI	
6	R	SGFRLYE	
7	L	GLTDSRA	
8	N	RTANDQ	
9	H	KDEQH	DE
10	D	RLEDHN	
11	E	SEAIR	
12	I	ILA	
13	A	GNLASR	
14	T	NQLTAH	
15	I	VIFLCM	
16	L	HLFM	
17	E	NEDQA	
18	N	QLSNE	
19	D	REDSV	
20	D	ADSLE	
21	D	AVEDST	
22	Y	KEYLD	
23	S	NESVfy	
24	P	RDIPGS	
25	L	RVSLEGDY	
26	D	AIEDS	
27	S	VFISD	
28	E	VDES	
29	S	PGSVE	
30	E	GDEP	
31	K	TEAKVP	
32	E	RSEAT	
33	D	DS	
34	C	FHNRCE	
35	V	GVDSC	
36	V	TSIVD	
37	E	TIDES	
38	D	LRDSH	
39	D	TPDEN	
40	V	SVDE	
41	W	WEFQN	
42	S	LSQY	
43	D	DFW	
44	N	NTSC	
45	E	EDS	
46	D	DESQ	
47	A	SVEA	
48	I	SEMIFTRA	
49	V	GIDV	
50	D	SPD	
51	F	ETYFASP	
52	V	VLD	
53	E	EHD	
54	D	DPSEVL	
55	T	ITVPL	
56	S	GSQ	
57	A	DRAPN	
58	Q	NQTS	
59	E	FESRG	
60	D	TDSQ	
61	P	PGE	
62	D	EDS	
63	N	DQVNE	
64	N	TQVNIME	
65	I	IGSLET	
66	A	SPADVY	
67	S	QSED	
68	R	SDQRE	
69	E	ESNDP	
70	S	SENVD	
71	P	ENLAPID	
72	N	EDGN	
73	L	QMLV	
74	E	VPE	
75	V	ALQVGD	
76	T	DSTQ	
77	S	HNSELA	

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TABLE 2-continued

A helicopterpa_position	B helicopterpa	C Acceptable	D Beneficial_Ha
78	L	VLSA	5
79	T	TAGR	
80	S	ERSDQ	
81	H	ERGHN	
82	R	HSTRAM	
83	I	NWIDSF	10
84	I	MAILC	
85	T	STA	
86	L	SKLAR	
87	P	DPGQ	
88	Q	RQG	15
89	R	RPS	
90	S	SFTY	
91	I	FYI	
92	R	YKCRTS	
93	G	GS	20
94	K	K	
95	N	NDG	
96	N	RNKEG	
97	H	YTIHP	
98	V	KIVCA	25
99	W	W	
100	S	ANSGY	
101	T	CRTPK	
102	T	QASTPN	
103	K	PKC	30
104	G	LPHGSNQ	
105	R	SNQRTF	
106	T	RPST	
107	T	ARTN	
108	G	VGSI	35
109	R	R	
110	T	VTL	
111	S	PRS	
112	A	QSAE	
113	I	HESIL	40
114	N	NP	
115	I	IP	
116	I	IVF	
117	R	QTKR	
118	T	RGMTSE	45
119	N	TNQVR	
120	R	NPRA	
121	G	VGQL	
122	P	SVP	
123	T	NKT	50
124	R	LRVNT	
125	M	TQFMDIG	
126	C	EACT	
127	R	DKVRS	
128	N	DNT	55
129	I	PAVI	
130	V	KLDVYFS	
131	D	DLET	
132	P	PEI	
133	L	FLQIYS	60
134	L	SDLNEK	
135	C	ICAF	
136	F	WF	
137	Q	NHQB	
138	L	KLI	65
139	F	LF	
140	I	MVIF	
141	T	DNST	
142	D	DEQS	
143	E	ESPAD	60
144	I	IM	
145	I	LEI	
146	H	QSRHD	
147	E	EVID	
148	I	TIM	65
149	V	LV	
150	K	KEDLT	
151	W	WHMY	
152	T	T	
153	N	N	
154	V	EHLVAS	

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TABLE 2-continued

A helicopterpa_position	B helicopterpa	C Acceptable	D Beneficial_Ha
155	E	KEYS	GA
156	I	IGMA	
157	I	IRSE	
158	V	QSRVLH	
159	K	YEVKRS	
160	R	RFLQ	10
161	Q	SQRTV	
162	N	KGNQ	
163	L	FKLENTS	
164	K	SNIKPA	
165	D	DTMLA	15
166	I	KSITPE	
167	S	DNSR	
168	A	PYEAHFV	
169	S	EAKSTYH	
170	Y	LYWFMK	20
171	R	RSKHQ	
172	D	NESDP	
173	T	LTI	
174	N	DTN	
175	T	MLQTEI	25
176	M	VTMDACS	
177	E	ED	
178	I	LMI	
179	W	HRNWKYK	
180	A	AR	30
181	L	FVYL	
182	V	IVF	
183	G	GA	
184	I	LI	
185	L	LT	35
186	T	LYITV	
187	L	FLMAI	
188	T	TAM	
189	A	AG	
190	V	VL	40
191	M	FYMRIT	
192	K	KR	
193	D	SDA	
194	N	NGK	
195	H	HRG	45
196	L	EQLMS	
197	S	NASL	
198	T	VLTE	
199	D	NQSDK	
200	E	YDSE	50
201	L	LW	
202	F	FWD	
203	D	ANDTR	
204	A	TSAR	
205	T	DETS	55
206	F	GEFLV	
207	S	TNSL	
208	G	GS	
209	T	EPTMD	
210	R	IRV	60
211	Y	FY	
212	V	RPVS	
213	S	CMAS	
214	V	VT	
215	M	M	65
216	S	S	
217	R	KLR	
218	E	NREDQ	
219	R	RT	
220	F	FY	60
221	E	LAHEDQY	
222	F	VFML	
223	L	IL	
224	I	LVSQI	
225	R	HNR	65
226	C	CVFNS	
227	I	LIM	
228	R	RH	
229	M	FM	
230	D	DN	LA
231	D	ND	

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TABLE 2-continued

A helicoverpa_position	B helicoverpa	C Acceptable	D Beneficial_Ha
232	K	PSRKT	
233	T	DTSA	
234	L	DTLIV	
235	R	RP	
236	P	EVPD	
237	T	EGTD	
238	L	RLQ	
239	R	RAPK	
240	S	EASGQK	
241	D	SIDNHT	
242	D	D	
243	A	KRAVN	
244	F	ILFM	
245	L	AILTH	
246	P	APK	
247	V	IVLF	
248	R	SR	
249	K	YQDKPS	
250	I	IVLM	
251	W	FYWI	
252	E	TED	
253	I	KEILSQ	
254	F	FWI	
255	I	VIS	
256	N	GKENHQ	
257	Q	NIQRC	
258	C	CLF	
259	R	QKPRIA	
260	Q	KDLQAN	
261	N	IVLNA	
262	H	YH	Y
263	V	NTVS	
264	P	VP	
265	G	CYGS	
266	S	EPGSAQ	
267	N	YNHF	
268	L	ALVI	
269	T	TC	
270	V	VI	
271	D	D	
272	E	E	
273	Q	MERQS	
274	L	L	
275	L	VL	
276	G	PAGLS	
277	F	F	
278	R	RK	
279	G	G	
280	R	R	
281	C	TCL	
282	P	HKPQL	
283	F	LF	
284	R	MR	
285	M	IQMV	
286	Y	Y	
287	I	MLI	
288	P	P	
289	N	MNS	
290	K	K	
291	P	PR	
292	D	ADS	AS
293	K	KR	
294	Y	Y	
295	G	G	
296	I	LI	
297	K	KR	
298	F	LIF	LI
299	P	MIWPLYF	
300	M	CAMK	
301	M	LAM	LA
302	C	CV	
303	A	DAE	DE
304	A	AS	
305	A	NYKAGS	
306	T	NTS	
307	K	GYSKF	
308	Y	Y	

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TABLE 2-continued

A helicoverpa_position	B helicoverpa	C Acceptable	D Beneficial_Ha
309	M	FSAMTV	
310	I	YLWISV	
311	D	NKDY	
312	A	CMAGFL	
313	I	YEQIML	
314	P	IVP	
315	Y	Y	
316	L	TALE	
317	G	G	
318	K	RDK	
319	S	GQSD	
320	T	SPT	
321	K	DGKQSL	
322	T	GTL	
323	N	APND	
324	G	GYP	
325	L	LKEVPA	
326	P	TVKPG	
327	L	VKLRT	
328	G	PGSA	
329	E	THMEGF	
330	F	QDRFYKE	
331	Y	SVYI	
332	V	V	
333	K	IDLKWE	
334	D	HRED	E
335	L	LIM	
336	T	AVSTI	
337	K	KQES	
338	T	PGT	PG
339	V	LIV	
340	H	FSQHLLA	
341	G	GQR	
342	T	STQ	
343	N	NGCH	
344	R	RHF	
345	N	NH	
346	I	IVL	
347	T	TY	
348	C	CMVF	
349	D	D	
350	N	N	
351	W	WF	
352	F	FY	
353	T	TS	
354	S	SG	
355	I	IY	
356	P	EPRT	
357	L	LT	
358	A	IYGAEM	
359	K	EAKTL	
360	N	YHENA	
361	M	LM	L
362	L	KLQY	
363	Q	KQCN	
364	A	KNRAEL	
365	P	P	
366	Y	HY	
367	N	GKNDR	
368	L	LT	
369	T	TP	
370	I	CAMIS	
371	V	VLCT	
372	G	G	
373	T	T	
374	I	MVI	
375	R	KRN	
376	S	KSR	
377	N	N	
378	K	KR	
379	R	RTPK	
380	E	ECGQ	
381	M	ILM	IL
382	P	P	
383	E	KPSERD	
384	E	EKVAS	
385	I	FIL	

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TABLE 2-continued

A helicoverpa_position	B helicoverpa	C Acceptable	D Beneficial_Ha
386	K	LRKIT	
387	N	PEKNDR	
388	S	SRIKT	
389	R	KQRDG	
390	S	QGS�	
391	R	RN	
392	P	DEPRQ	
393	V	VIMGP	
394	G	GNHEA	
395	S	ST	
396	S	SY	
397	M	LIMAV	
398	F	YFL	
399	C	GACR	
400	F	YFK	
401	D	AQTDN	
402	G	GDEK	
403	P	QDKPL	
404	L	NFALI	
405	T	TA	
406	L	IVL	
407	V	LVK	
408	S	SF	
409	Y	HYF	
410	K	VICKDA	
411	P	P	
412	K	K	
413	P	KRP	
414	S	NSAK	
415	K	KR	
416	M	ANMV	
417	V	V	
418	F	IFLYV	
419	L	LMVA	
420	L	LM	
421	S	ST	
422	S	ST	
423	C	MLCI	
424	D	HD	H
425	E	HTED	
426	N	ADNE	
427	A	ESAN	
428	V	AESV	
429	I	VIL	
430	N	DSNR	
431	E	ESTQ	
432	S	TERSQ	
433	N	TDNR	
434	G	GDV	
435	K	K	
436	P	P	
437	D	ESQDL	
438	M	IMC	
439	I	IVS	
440	L	GTLMK	
441	F	FDYE	
442	Y	Y	
443	N	NS	
444	Q	KSQ	
445	T	TY	
446	K	KM	
447	G	GSA	
448	G	G	
449	V	V	
450	D	D	
451	S	ENSTRV	
452	F	IVLFT	
453	D	D	
454	Q	KQE	
455	M	KLVM	
456	C	CITQS	
457	K	ARKSH	
458	S	ITSVYN	
459	M	YM	Y
460	S	TDSN	
461	A	SVCA	
462	N	SQNT	

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TABLE 2-continued

A helicoverpa_position	B helicoverpa	C Acceptable	D Beneficial_Ha
463	R	R	
464	K	RNK	
465	T	TS	
466	N	RANK	
467	R	RA	
468	W	W	
469	P	PY	
470	M	MLK	
471	A	VTAK	
472	V	VIL	
473	F	FLG	
474	Y	YFI	
475	G	RWNGY	
476	M	MILV	
477	L	LVI	
478	N	DNQ	
479	M	ITVM	
480	A	SA	
481	F	TGAFCLS	
482	V	VIYR	
483	N	N	
484	S	SA	
485	Y	HKYFC	
486	I	LIV	
487	I	IVL	
488	Y	YQW	
489	C	DMSCKRQ	
490	H	ILEHAT	
491	N	HNIA	
492	K	HSNKV	
493	I	SQIPN	
494	N	DENSG	
495	K	WKP	
496	Q	AQG	
497	E	DGKE	
498	K	KNVA	
499	P	TLPV	
500	I	TPYIQSV	
501	S	ETRNSYK	
502	R	RY	
503	K	GRKT	
504	E	MALEKY	
505	F	FQ	
506	M	LIM	
507	K	KERQ	
508	K	QKENIS	
509	L	L	
510	S	AGSYP	
511	I	RMKTIAL	
512	Q	TSADQL	
513	L	LM	
514	T	VITF	
515	T	LATSGY	
516	P	PSGE	
517	W	QHKWFV	
518	M	MQIEL	
519	Q	KAQREH	
520	E	REKQS	
521	R	RT	
522	L	AKLVN	
523	Q	LTRKQEP	
524	A	NPAEK	
525	P	ESAPMK	
526	T	RKTPN	
527	L	LISP	
528	K	PKS	
529	R	RVADTF	
530	T	ESATYNH	
531	L	LAVI	
532	R	RA	
533	D	LKVDRQ	
534	N	SRINL	
535	I	LIE	
536	T	AGKTSE	
537	N	RSINK	
538	V	VHKIQ	
539	L	LF	

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TABLE 2-continued

A helicoverpa_position	B helicoverpa	C Acceptable	D Beneficial_Ha
540	K	GRKPI	
541	N	PETNKD	
542	V	DSVETP	
543	V	MSNVLT	
544	P	PAQ	
545	A	FPAG	
546	S	VSATR	
547	S	PMSH	
548	E	DAEGV	
549	N	PKND	
550	I	QIVNSM	
551	S	EPDSTR	
552	N	VNTE	
553	E	DE	
554	P	EP	
555	E	TGEV	
556	P	FVPM	
557	K	KG	
558	K	TVKPR	
559	R	RKQY	
560	R	RKSTV	
561	Y	RYG	
562	C	C	
563	G	HYQGTR	
564	V	TIVFYDE	
565	C	C	
566	S	PSR	PR
567	Y	LVSYKN	
568	K	KR	
569	K	LKDI	
570	R	QDR	
571	R	RS	
572	M	KMD	KD
573	T	STA	
574	K	TKSNR	
575	A	HYTAR	
576	Q	TISQY	
577	C	CF	
578	C	YIVCKPN	
579	K	TSKA	
580	C	C	
581	K	KTPA	
582	K	KSNR	
583	A	HFAVNP	
584	I	VIL	
585	C	C	
586	G	LRGFM	
587	E	QEK	
588	H	CHP	
589	N	ATNC	
590	I	KNVIF	
591	D	QFTDE	
592	V	VFMIL	
593	C	CY	
594	Q	AEPQH	
595	D	DNST	
596	C	CQ	
597	I	VRGFLA	
598	G	REGHD	

TABLE 3

A AGROTIS_position	B agrotis	C Acceptable	D Beneficial_Ag
1	M	M	
2	E	PSEAG	
3	S	SMK	
4	R	TARDN	
5	Q	SRQFI	
6	R	SGFRLYE	
7	L	GLTDSRA	
8	N	RTANDQ	
9	Q	KDEQH	DE

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TABLE 3-continued

A AGROTIS_position	B agrotis	C Acceptable	D Beneficial_Ag
5 10	D	RLEDHN	
11	E	SEAIR	
12	I	ILA	
13	A	GNLASR	
14	T	NQLTAH	
15	I	VIFLCM	
10 16	L	HLFM	
17	E	NEDQA	
18	N	QLSNE	
19	D	REDSV	
20	D	ADSLE	
21	D	AVEDST	
15 22	Y	KEYLD	
23	S	NESVfy	
24	P	RDIPGS	
25	L	RVSLEGDY	
26	D	AIEDS	
27	S	VFISD	
28	D	VDES	
20 29	S	PGSVE	
30	E	GDEP	
31	A	TEAKVP	
32	E	RSEAT	
33	D	DS	
34	R	FHNRCE	
25 35	V	GVDSC	
36	V	TSIVD	
37	E	TIDES	
38	D	LRDSH	
39	D	TPDEN	
40	V	SVDE	
30 41	W	WEFQN	
42	S	LSQY	
43	D	DFW	
44	N	NTSC	
45	E	EDS	
46	D	DESQ	
35 47	A	SVEA	
48	M	SEMIFTRA	
49	I	GIDV	
50	D	SPD	
51	Y	ETYFASP	
52	V	VLD	
53	E	EHD	
40 54	D	DPSEVL	
55	T	ITVPL	
56	S	GSQ	
57	R	DRAPN	
58	Q	NQTS	
59	E	FESRG	
45 60	D	TDSQ	
61	P	PGE	
62	D	EDS	
63	N	DQVNE	
64	N	TQVNIME	
65	I	IGSLET	
50 66	A	SPADVY	
67	S	QSED	
68	Q	SDQRE	
69	E	ESNDP	
70	S	SENV	
71	A	ENLAPIID	
55 72	N	EDGN	
73	L	QMLV	
74	E	VPE	
75	V	ALQVGD	
76	T	DSTQ	
77	S	HNSELA	
60 78	L	VLSA	
79	T	TAGR	
80	S	ERSDQ	
81	H	ERGHN	
82	R	HSTRAM	
83	I	NWIDSF	
84	I	MAILC	
65 85	S	STA	
86	L	SKLAR	

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TABLE 3-continued

A	B	C	D
AGROTIS_position	agrotis	Acceptable	Beneficial_Ag
87	P	DPGQ	
88	Q	RQG	
89	R	RPS	
90	S	SFTY	
91	I	FYI	
92	C	YKCRTS	
93	G	GS	
94	K	K	
95	N	NDG	
96	N	RNKEG	
97	H	YTIHP	
98	V	KIVCA	
99	W	W	
100	S	ANSGY	
101	T	CRTPK	
102	T	QASTPN	
103	K	PKC	
104	G	LPHGSNQ	
105	R	SNQRTF	
106	T	RPST	
107	T	ARTN	
108	G	VGSI	
109	R	R	
110	T	VTL	
111	S	PRS	
112	A	QSAE	
113	I	HESIL	
114	N	NP	
115	I	IP	
116	I	IVF	
117	R	QTKR	
118	T	RGMTSE	
119	N	TNQVR	
120	R	NPRA	
121	G	VGQL	
122	P	SVP	
123	T	NKT	
124	R	LRVNT	
125	M	TQFMDIG	
126	C	EACT	
127	R	DKVRS	
128	N	DNT	
129	I	PAVI	
130	V	KLDVYFS	
131	D	DLET	
132	P	PEI	
133	L	FLQIYS	
134	L	SDLNEK	
135	C	ICAF	
136	F	WF	
137	Q	NHQB	
138	L	KLI	
139	F	LF	
140	I	MVIF	
141	T	DNST	
142	D	DEQS	
143	E	ESPAD	
144	I	IM	
145	I	LEI	
146	H	QSRHD	
147	E	EVID	
148	I	TIM	
149	V	LV	
150	K	KEDLT	
151	W	WHMY	
152	T	T	
153	N	N	
154	V	EHLVAS	
155	E	KEYS	
156	M	IGMA	IGA
157	I	IRSE	
158	V	QSRVLH	
159	K	YEVKRS	
160	R	RFLQ	
161	Q	SQRTV	
162	N	KGNO	
163	L	FKLENTS	

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TABLE 3-continued

A	B	C	D
AGROTIS_position	agrotis	Acceptable	Beneficial_Ag
5 164	I	SNIKPA	
165	D	DTMLA	
166	I	KSITPE	
167	S	DNSR	
168	A	PYEAHFV	
169	S	EAKSTYH	
10 170	Y	LYWFMK	
171	R	RSKHQ	
172	D	NESDP	
173	T	LTI	
174	N	DTN	
175	T	MLQTEI	
15 176	M	VTMDACS	
177	E	ED	
178	M	LMI	LI
179	W	HRNWYK	
180	A	AR	
181	L	FVYL	
182	V	IVF	
20 183	G	GA	
184	I	LI	
185	L	LT	
186	T	LYITV	
187	L	FLMAI	
188	T	TAM	
25 189	A	AG	
190	V	VL	
191	M	FYMRIT	
192	K	KR	
193	D	SDA	SA
194	N	NGK	
30 195	H	HRG	
196	L	EQLMS	
197	S	NASL	
198	T	VLTE	
199	D	NQSDK	
200	E	YDSE	
35 201	L	LW	
202	F	FWD	
203	D	ANDTR	
204	A	TSAR	
205	T	DETS	
206	F	GEFLV	
40 207	S	TNSL	
208	G	GS	
209	T	EPTMD	
210	R	IRV	
211	Y	FY	F
212	V	RPVS	
213	S	CMAS	
45 214	V	VT	
215	M	M	
216	S	S	
217	R	KLR	
218	E	NREDQ	
219	R	RT	
50 220	F	FY	
221	E	LAHEDQY	
222	F	VFML	
223	L	IL	
224	I	LVSQ	
225	R	HNR	
226	C	CVFNS	
55 227	M	LIM	LI
228	R	RH	
229	M	FM	
230	D	DN	
231	D	ND	
232	K	PSRKT	
60 233	T	DTSA	
234	L	DTLIV	
235	R	RP	
236	P	EVPI	
237	T	EGTD	
238	L	RLQ	
65 239	R	RAPK	
240	S	EASGQK	

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TABLE 3-continued

A	B	C	D
AGROTIS_position	agrotis	Acceptable	Beneficial_Ag
241	D	SIDNHT	
242	D	D	
243	A	KRAVN	
244	F	ILFM	
245	I	AILTH	
246	P	APK	
247	V	IVLF	
248	R	SR	
249	K	YQDKPS	
250	L	IVLM	
251	W	FYWI	
252	E	TED	
253	I	KEILSQ	
254	F	FWL	
255	I	VIS	
256	N	GKENHQ	
257	Q	NIQRC	
258	C	CLF	
259	R	QKPRIA	
260	L	KDLQAN	
261	N	IVLNA	
262	Y	YH	
263	V	NTVS	
264	P	VP	
265	G	CYGS	
266	G	EPGSAQ	
267	N	YNHF	
268	L	ALVI	
269	T	TC	
270	V	VI	
271	D	D	
272	E	E	
273	Q	MERQS	
274	L	L	
275	L	VL	
276	G	PAGLS	
277	F	F	
278	R	RK	
279	G	G	
280	R	R	
281	C	TCL	
282	P	HKPQL	
283	F	LF	
284	R	MR	
285	M	IQMV	
286	Y	Y	
287	I	MLI	
288	P	P	
289	N	MNS	
290	K	K	
291	P	PR	
292	D	ADS	AS
293	K	KR	
294	Y	Y	
295	G	G	
296	I	LI	
297	R	KR	K
298	F	LIF	LI
299	P	MIWPLYF	
300	M	CAMK	
301	M	LAM	LA
302	C	CV	
303	D	DAE	E
304	A	AS	
305	A	NYKAGS	
306	T	NTS	
307	K	GYSKF	
308	Y	Y	
309	M	FSAMTV	
310	I	YLWISV	
311	D	NKDY	
312	A	CMAGFL	
313	I	YEQIML	
314	P	IVP	
315	Y	Y	
316	L	TALE	
317	G	G	

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TABLE 3-continued

A	B	C	D
AGROTIS_position	agrotis	Acceptable	Beneficial_Ag
5 318	K	RDK	
319	S	GQSD	
320	T	SPT	
321	K	DGKQSL	
322	T	GTL	
323	N	APND	
10 324	G	GYP	
325	L	LKEVPA	
326	P	TVKPG	
327	L	VKLRT	
328	G	PGSA	
329	E	THMEGF	
15 330	F	QDRFYKE	
331	Y	SVYI	
332	V	V	
333	K	IDLKWE	
334	E	HRED	
335	L	LIM	
336	T	AVSTI	
20 337	K	KQES	
338	T	PGT	PG
339	V	LIV	
340	H	FSQHLLA	
341	G	GQR	
342	T	STQ	
25 343	N	NGCH	
344	R	RHF	
345	N	NH	
346	V	IVL	
347	T	TY	
348	C	CMVF	
30 349	D	D	
350	N	N	
351	W	WF	
352	F	FY	
353	T	TS	
354	S	SG	
355	I	IY	
35 356	P	EPRT	
357	L	LT	
358	A	IYGAFM	
359	K	EAKTL	
360	N	YHENA	
361	M	LM	L
40 362	L	KLQY	
363	Q	KQCN	
364	A	KNRAEL	
365	P	P	
366	Y	HY	
367	N	GKNDR	
45 368	L	LT	
369	T	TP	
370	I	CAMIS	
371	V	VLCT	
372	G	G	
373	T	T	
50 374	I	MVI	
375	R	KRN	
376	S	KSR	
377	N	N	
378	K	KR	
379	R	RTPK	
55 380	E	ECGQ	
381	I	ILM	LM
382	P	P	
383	E	KPSERD	
384	E	EKVAS	
385	I	FIL	
386	K	LRKIT	
60 387	N	PEKNDR	
388	S	SRIKT	
389	R	KQRDG	
390	S	QGS	
391	R	RN	
392	P	DEPRQ	
65 393	V	VIMGP	
394	G	GNHEA	

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TABLE 3-continued

A	B	C	D
AGROTIS_position	agrotis	Acceptable	Beneficial_Ag
395	S	ST	
396	S	SY	
397	M	LIMAV	
398	F	YFL	
399	C	GACR	
400	F	YFK	
401	D	AQTDN	
402	G	GDEK	
403	P	QDKPL	
404	L	NFALI	
405	T	TA	
406	L	IVL	
407	V	LVK	
408	S	SF	
409	Y	HYF	
410	K	VICKDA	
411	P	P	
412	K	K	
413	P	KRP	
414	S	NSAK	
415	R	KR	K
416	M	ANMV	
417	V	V	
418	F	IFLYV	
419	L	LMVA	
420	L	LM	
421	S	ST	
422	S	ST	
423	C	MLCI	
424	D	HD	H
425	E	HTED	
426	N	ADNE	
427	A	ESAN	
428	V	AESV	
429	I	VIL	
430	N	DSNR	
431	E	ESTQ	
432	S	TERSQ	
433	N	TDNR	
434	G	GDV	
435	K	K	
436	P	P	
437	D	ESQDL	
438	M	IMC	
439	I	IVS	
440	L	GTLMK	
441	F	FDYE	
442	Y	Y	
443	N	NS	
444	Q	KSQ	
445	T	TY	
446	K	KM	
447	G	GSA	
448	G	G	
449	V	V	
450	D	D	
451	S	ENSTRV	
452	F	IVLFT	
453	D	D	
454	Q	KQE	
455	M	KLVM	
456	C	CITOS	
457	K	ARKSH	
458	S	ITSVYN	
459	M	YM	Y
460	S	TDSN	
461	A	SVCA	
462	N	SQNT	
463	R	R	
464	K	RNK	
465	T	TS	
466	N	RANK	
467	R	RA	
468	W	W	
469	P	PY	
470	M	MLK	
471	A	VTAK	

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TABLE 3-continued

A	B	C	D
AGROTIS_position	agrotis	Acceptable	Beneficial_Ag
5 472	V	VIL	
473	F	FLG	
474	Y	YFI	
475	G	RWNGY	
476	M	MILV	
477	L	LVI	
10 478	N	DNQ	
479	M	ITVM	
480	A	SA	
481	F	TGAFCLS	
482	V	VTYR	
483	N	N	
15 484	S	SA	
485	Y	HKYFC	
486	I	LIV	
487	I	IVL	
488	Y	YQW	
489	C	DMSCKRQ	
490	H	ILEHAT	
20 491	N	HNIA	
492	K	HSNKV	
493	I	SQIPN	
494	N	DENSG	
495	K	WKP	
496	0	AQG	
25 497	K	DGKE	
498	K	KNVA	
499	P	TLPV	
500	I	TPYIQSV	
501	N	ETRNSYK	
502	R	RY	
30 503	K	GRKT	
504	E	MALEKY	
505	F	FQ	
506	M	LIM	
507	K	KERQ	
508	N	QKENIS	
35 509	L	L	
510	S	AGSYP	
511	T	RMKTIAL	
512	D	TSADQL	
513	L	LM	
514	T	VITF	
515	T	LATSGY	
40 516	P	PSGE	
517	W	QHKWFV	
518	M	MQIEL	
519	Q	KAQREH	
520	E	REKQS	
521	R	RT	
45 522	L	AKLVN	
523	K	LTRKQEP	
524	A	NPAEK	
525	P	ESAPMK	
526	T	RKTPN	
527	L	LISP	
50 528	K	PKS	
529	R	RVADTF	
530	T	ESATYNH	
531	L	LAVI	
532	R	RA	
533	D	LKVDRQ	
534	N	SRINL	
55 535	I	LIE	
536	T	AGKTSE	
537	N	RSINK	
538	V	VHKIQ	
539	L	LF	
60 540	K	GRKPI	
541	N	PETNKD	
542	V	DSVETP	
543	V	MSNVLT	
544	P	PAQ	
545	P	FPAG	
546	S	VSATR	
65 547	P	PMSH	
548	A	DAEGV	

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TABLE 3-continued

A	B	C	D
AGROTIS_position	agrotis	Acceptable	Beneficial_Ag
549	N	PKND	
550	N	QIVNSM	
551	S	EPDSTR	
552	E	VNTE	
553	E	DE	
554	P	EP	
555	G	TGEV	
556	P	FVPM	
557	K	KG	
558	K	TVKPR	
559	R	RKQY	
560	S	RKSTV	
561	Y	RYG	
562	C	C	
563	G	HYQGTR	
564	F	TIVFYDE	
565	C	C	
566	S	PSR	PR
567	Y	LV'SYKN	
568	K	KR	
569	K	LKDI	
570	R	QDR	
571	R	RS	
572	M	KMD	KD
573	T	STA	
574	K	TKSNR	

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TABLE 3-continued

A	B	C	D
AGROTIS_position	agrotis	Acceptable	Beneficial_Ag
575	T	HYTAR	
576	Q	TISQY	
577	F	CF	C
578	Y	YIVCKPN	
579	K	TSKA	
580	C	C	
581	K	KTPA	
582	K	KSNR	
583	A	HFAVNP	
584	I	VIL	
585	C	C	
586	G	LRGFM	
587	E	QEK	
588	H	CHP	
589	N	ATNC	
590	I	KNVIF	
591	D	QFTDE	
592	V	VFML	
593	C	CY	
594	Q	AEPQH	
595	D	DNST	
596	C	CQ	
597	V	VRGIFLA	
598	G	REGHD	

TABLE 4

A	B	C	D	E	F	G	H
Source	Tposon left end	Tposon right end	Tposon SEQ ID NO	Tpase SEQ ID	leu	leu ura	leu ura trp
Heliothis	1	3	66	39	427	215	130
Spodoptera litura	91	92	71	46	>250	0	0
Pieris rapae	93	94	72	47	>250	0	0
Myzus persicae	95	96	73	48	>250	0	0
Onthophagus taurus	97	98	74	49	>250	0	0
Temnothorax curvispinosus	99	100	75	50	>250	0	0
Agrius planipenn	101	102	76	51	>250	0	0
Parasteatoda tepidariorum	103	104	77	52	>250	0	0
Pectinophora gossypiella	105	106	78	53	>250	0	0
Ctenopusia agnata	107	108	79	54	>250	0	0
Macrostromum lignano	109	110	80	55	>250	0	0
Orussus abietinus	111	112	81	56	>250	0	0
Eufriesea mexicana	113	114	82	57	323	0	0
Spodoptera litura	115	116	83	58	400	0	0
Vanessa tameamea	117	118	84	59	389	0	0
Blattella germanica	119	120	85	60	248	0	0
Onthophagus taurus	121	122	86	61	>250	0	0
Onthophagus taurus	123	124	87	62	>250	0	0
Onthophagus taurus	125	126	88	63	>250	0	0
Megachile rotundata	127	128	89	64	>250	0	0
Xiphophorus maculatus	129	130	90	65	>250	0	0

TABLE 5

A	B	C	D	E	F	G	H	I	J
Transposon	Tposon	Transposase	D	E	F	G	H	I	J
Source	SEQ ID NO	source	Tpase SEQ ID	Tpase DNA seq ID	leu	leu ura	leu ura trp	rel ex	rel trans
1 Agrotis	67	Agrotis	40	43	273	471	120	1.73	0.44
2 Agrotis	67	Helicoverpa	41	44	302	563	144	1.86	0.48
3 Helicoverpa	68	Helicoverpa	41	44	412	547	211	1.33	0.51
4 Helicoverpa	68	Agrotis	40	43	330	551	245	1.67	0.74

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TABLE 6

	A	B	C	D	E
1	Transposase SEQ ID NO	none	none	39	39
2	Day	viability	viable cells	viability	viable cells
3	1	94.12	1.03	93.10	0.87
4	3	92.15	0.55	91.01	0.27
5	5	80.66	0.22	73.53	0.20
6	7	57.58	0.05	54.03	0.06
7	10	27.18	0.03	34.38	0.02
8	12	27.05	0.04	32.48	0.04
9	14	31.88	0.04	21.48	0.03
10	17	41.46	0.04	50.98	0.06
11	19	no live cells	no live cells	73.71	0.23
12	21	no live cells	no live cells	94.96	1.03
13	24	no live cells	no live cells	98.38	4.14
14	26	no live cells	no live cells	99.02	>4
15	27	no live cells	no live cells	99.50	>4

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TABLE 7

A Day 7	B Day 10	C Day 12	D Day 14
1,064	1,524	1,771	2,053

TABLE 8

	A	B	C
1	Days post-transfection	viability	viable cells
2	1	94.82	1.07
3	2	93.91	0.10
4	5	84.29	0.11
5	7	48.39	0.03
6	9	37.89	0.04
7	14	33.00	0.03
8	16	34.26	0.04
9	19	54.69	0.11
10	21	49.44	0.09
11	23	53.14	0.13
12	26	75.00	0.10
13	28	90.56	0.70
14	30	95.69	2.16

TABLE 9

[illegible]

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TABLE 10

A seq id no	B live	C int	D ex	E int freq	F ex freq
39	363	56	92	0.0006	0.0010
158	123	76	92	0.0025	0.0030
159	201	63	81	0.0013	0.0016
369	103	19	31	0.0007	0.0012
370	145	40	41	0.0011	0.0011
371	107	17	29	0.0006	0.0011
372	180	45	80	0.0010	0.0018
373	140	25	31	0.0007	0.0009
374	160	26	16	0.0007	0.0004
375	165	46	42	0.0011	0.0010
376	188	57	62	0.0012	0.0013
401	275	31	57	0.0005	0.0008
402	294	23	18	0.0003	0.0002
403	149	5	7	0.0001	0.0002
404	198	7	11	0.0001	0.0002
405	145	7	16	0.0002	0.0004
406	137	10	14	0.0003	0.0004
407	121	8	7	0.0003	0.0002
408	131	17	30	0.0005	0.0009
409	154	5	12	0.0001	0.0003
410	195	21	25	0.0004	0.0005
411	193	20	52	0.0004	0.0011
412	150	12	14	0.0003	0.0004
413	193	7	19	0.0001	0.0004
414	172	13	12	0.0003	0.0003
415	135	8	7	0.0002	0.0002
416	234	12	19	0.0002	0.0003
417	151	23	34	0.0006	0.0009
418	131	20	34	0.0006	0.0010
419	161	19	29	0.0005	0.0007
420	249	9	17	0.0001	0.0003
421	211	8	13	0.0002	0.0002
422	240	35	42	0.0006	0.0007
423	240	33	49	0.0006	0.0008
424	196	7	7	0.0001	0.0001
425	172	11	12	0.0003	0.0003
426	292	12	31	0.0002	0.0004
427	185	22	27	0.0005	0.0006
428	145	11	23	0.0003	0.0006
429	310	38	54	0.0005	0.0007
430	163	12	20	0.0003	0.0005
431	94	7	12	0.0003	0.0005
432	212	19	31	0.0004	0.0006
433	213	15	14	0.0003	0.0003
434	189	8	4	0.0002	0.0001
435	165	13	16	0.0003	0.0004
436	311	22	50	0.0003	0.0006
437	118	8	7	0.0003	0.0002
438	124	13	15	0.0004	0.0005
439	169	8	5	0.0002	0.0001
440	101	8	16	0.0003	0.0006
441	111	6	3	0.0002	0.0001
442	163	5	5	0.0001	0.0001
443	160	7	7	0.0002	0.0002
444	87	3	10	0.0001	0.0005
445	63	4	7	0.0003	0.0004
466	280	2	7	0.0000	0.0001
467	129	2	4	0.0001	0.0001
468	132	2	10	0.0001	0.0003
469	178	4	12	0.0001	0.0003
470	206	2	5	0.0000	0.0001
471	245	1	2	0.0000	0.0000
472	165	2	1	0.0000	0.0000
473	266	6	9	0.0001	0.0001
474	195	5	7	0.0001	0.0001
475	239	2	6	0.0000	0.0001
476	182	3	10	0.0001	0.0002
477	117	2	10	0.0001	0.0003
478	170	3	11	0.0001	0.0003
479	138	2	9	0.0001	0.0003
480	148	1	2	0.0000	0.0001
481	152	2	3	0.0001	0.0001
482	211	3	8	0.0001	0.0002
483	202	2	4	0.0000	0.0001
484	102	1	3	0.0000	0.0001
485	95	2	5	0.0001	0.0002
486	95	2	1	0.0001	0.0000

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TABLE 10-continued

A seq id no	B live	C int	D ex	E int freq	F ex freq
5 487	94	2	5	0.0001	0.0002
488	68	1	7	0.0001	0.0004
489	111	3	3	0.0001	0.0001
491	209	0	0	0.0000	0.0000
492	252	0	2	0.0000	0.0000
493	217	0	1	0.0000	0.0000
10 494	183	0	2	0.0000	0.0000
495	127	0	0	0.0000	0.0000
496	210	1	0	0.0000	0.0000
497	101	0	0	0.0000	0.0000
498	293	0	0	0.0000	0.0000
499	216	0	6	0.0000	0.0001
15 500	300	0	1	0.0000	0.0000
501	55	0	2	0.0000	0.0001
502	127	0	0	0.0000	0.0000
503	79	0	0	0.0000	0.0000
504	333	0	0	0.0000	0.0000
505	147	0	0	0.0000	0.0000
20 506	208	0	0	0.0000	0.0000

TABLE 11

	A Position	B From	C To	D Int Weight	E Int Weight Std	F Ex Weight	G Ex Weight Std
25	300	M	R	0.620	0.085	0.746	0.064
	43	L	S	0.374	0.045	0.207	0.047
30	162	T	I	0.306	0.066	0.018	0.061
	239	I	L	0.278	0.078	0.281	0.040
	507	L	I	0.268	0.055	0.193	0.063
	421	Y	F	0.234	0.063	0.278	0.064
	603	Q	K	0.222	0.051	0.471	0.070
	305	M	N	0.213	0.048	0.459	0.079
35	81	V	E	0.200	0.093	0.068	0.063
	241	H	N	0.192	0.052	0.036	0.048
	450	V	I	0.191	0.077	0.170	0.052
	136	S	V	0.177	0.043	0.293	0.049
	155	D	T	0.166	0.067	0.069	0.048
	153	L	F	0.157	0.060	-0.280	0.068
40	264	S	R	0.149	0.046	0.256	0.067
	152	K	L	0.143	0.060	-0.061	0.065
	312	L	I	0.133	0.041	0.136	0.043
	268	T	D	0.122	0.080	0.116	0.066
	280	V	P	0.111	0.065	0.185	0.046
	299	L	F	0.106	0.060	0.016	0.064
45	430	H	Y	0.081	0.072	-0.070	0.059
	81	V	P	0.069	0.057	0.027	0.042
	238	V	F	0.063	0.066	0.091	0.050
	512	H	N	0.062	0.073	0.136	0.066
	495	Y	F	0.060	0.068	0.208	0.090
	297	T	C	0.055	0.059	0.213	0.070
	396	K	R	0.050	0.071	-0.067	0.075
50	149	I	C	0.023	0.078	0.055	0.069
	268	T	E	0.014	0.049	0.069	0.049
	41	S	V	0.014	0.057	0.112	0.055
	589	S	T	0.003	0.069	0.067	0.040
	600	V	I	-0.021	0.065	-0.106	0.046
55	150	W	F	-0.025	0.051	0.080	0.054
	604	C	H	-0.032	0.058	-0.207	0.085
	428	L	V	-0.045	0.058	-0.119	0.058
	233	K	R	-0.045	0.062	-0.122	0.054
	459	I	M	-0.052	0.053	-0.126	0.062
	188	L	T	-0.080	0.078	-0.320	0.076
	419	Y	F	-0.084	0.041	-0.653	0.072
60	552	L	I	-0.084	0.055	-0.079	0.063
	603	Q	E	-0.087	0.094	-0.162	0.082
	586	Q	D	-0.097	0.052	-0.128	0.049
	137	N	K	-0.125	0.058	-0.022	0.066
	137	N	T	-0.137	0.044	-0.022	0.050
	136	S	P	-0.149	0.054	-0.143	0.063
	501	S	A	-0.155	0.091	-0.123	0.075
65	378	I	Y	-0.219	0.106	-0.268	0.083
	314	L	F	-0.221	0.068	-0.009	0.067

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TABLE 11-continued

A Position	B From	C To	D Int Weight	E Int Weight Std	F Ex Weight	G Ex Weight Std
74	Q	S	-0.225	0.080	0.074	0.043
499	D	N	-0.258	0.059	-0.656	0.090
303	M	I	-0.278	0.082	-0.251	0.065
241	H	R	-0.293	0.058	-0.311	0.100
485	R	K	-0.312	0.061	-0.343	0.055
586	Q	R	-0.312	0.071	-0.032	0.053
135	V	G	-0.336	0.079	-0.493	0.055
485	R	N	-0.359	0.121	-0.675	0.214
334	R	K	-0.396	0.073	-0.233	0.045
314	L	I	-0.424	0.070	-0.306	0.076
233	K	L	-0.426	0.087	-0.399	0.047
385	K	L	-0.583	0.049	-0.554	0.062

TABLE 12

A SEQ ID NO.	B live	C int	D ex	E int freq	F ex freq
39	604	360	308	0.0024	0.0020
369	612	260	424	0.0017	0.0028
158	512	668	1000	0.0052	0.0078
372	668	444	604	0.0027	0.0036
159	736	1004	840	0.0055	0.0046
375	568	428	512	0.0030	0.0036
376	616	264	416	0.0017	0.0027
377	888	844	1056	0.0038	0.0048
160	616	1892	1920	0.0123	0.0125
161	384	744	1048	0.0078	0.0109
162	404	832	828	0.0082	0.0082
163	576	1432	1292	0.0099	0.0090
164	648	880	936	0.0054	0.0058
165	576	848	1360	0.0059	0.0094
166	568	900	1200	0.0063	0.0085
378	1460	1100	1468	0.0030	0.0040
167	592	1400	1756	0.0095	0.0119
168	184	364	488	0.0079	0.0106
379	340	332	352	0.0039	0.0041
169	436	1056	968	0.0097	0.0089
170	560	1072	1188	0.0077	0.0085
171	220	1256	1256	0.0228	0.0228
172	328	688	1836	0.0084	0.0224
173	648	1016	1216	0.0063	0.0075
174	452	2144	2584	0.0190	0.0229
175	460	664	1280	0.0058	0.0111
176	320	588	656	0.0074	0.0082
177	380	808	1456	0.0085	0.0153
178	476	940	1160	0.0079	0.0097
380	648	528	1120	0.0033	0.0069
179	360	772	1232	0.0086	0.0137
180	664	948	1300	0.0057	0.0078
181	236	820	1084	0.0139	0.0184
182	396	916	960	0.0093	0.0097
381	440	320	496	0.0029	0.0045
382	376	392	560	0.0042	0.0060
183	384	572	1260	0.0060	0.0131
446	528	308	592	0.0023	0.0045
184	444	648	1076	0.0058	0.0097
185	316	860	1012	0.0109	0.0128
186	356	992	1184	0.0111	0.0133
187	312	500	740	0.0064	0.0095
188	424	636	1016	0.0060	0.0096
189	252	632	1080	0.0100	0.0171
190	296	748	1104	0.0101	0.0149
191	320	416	964	0.0052	0.0121
192	356	1444	1504	0.0162	0.0169
193	904	1500	2244	0.0066	0.0099
194	332	1068	2152	0.0129	0.0259
195	360	632	924	0.0070	0.0103
383	488	420	312	0.0034	0.0026
196	528	992	676	0.0075	0.0051
197	440	692	656	0.0063	0.0060
198	612	960	1688	0.0063	0.0110
199	600	1680	2044	0.0112	0.0136

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TABLE 12-continued

A SEQ ID NO.	B live	C int	D ex	E int freq	F ex freq
447	440	213	288	0.0019	0.0026
200	580	1064	864	0.0073	0.0060
201	544	780	696	0.0057	0.0051
202	216	295	380	0.0055	0.0070
203	312	473	508	0.0061	0.0065
204	308	477	416	0.0062	0.0054
448	484	283	348	0.0023	0.0029
205	396	702	696	0.0071	0.0070
206	572	1704	1104	0.0119	0.0077
207	516	1780	1140	0.0138	0.0088
208	508	868	968	0.0068	0.0076
209	300	712	720	0.0095	0.0096
210	492	800	996	0.0065	0.0081
211	328	676	804	0.0082	0.0098
384	564	360	564	0.0026	0.0040
212	424	720	796	0.0068	0.0075
213	544	664	656	0.0049	0.0048
214	540	1056	940	0.0078	0.0070
215	520	960	876	0.0074	0.0067
385	464	360	324	0.0031	0.0028
386	628	724	1196	0.0046	0.0076
216	472	728	572	0.0062	0.0048
387	576	600	700	0.0042	0.0049
217	428	1176	612	0.0110	0.0057
218	552	1080	744	0.0078	0.0054
388	492	496	696	0.0040	0.0057
219	276	532	640	0.0077	0.0093
389	508	600	500	0.0047	0.0039
220	404	736	612	0.0073	0.0061
390	576	384	512	0.0027	0.0036
391	388	432	796	0.0045	0.0082
221	480	636	820	0.0053	0.0068
222	388	560	560	0.0058	0.0058
223	452	1244	1180	0.0110	0.0104
224	384	848	880	0.0088	0.0092
225	236	340	516	0.0058	0.0087
226	548	1040	1060	0.0076	0.0077
227	260	816	852	0.0126	0.0131
228	492	672	848	0.0055	0.0069
229	288	1032	884	0.0143	0.0123
230	580	820	760	0.0057	0.0052
392	584	568	996	0.0039	0.0068
393	780	884	1380	0.0045	0.0071
231	340	628	796	0.0074	0.0094
449	788	380	600	0.0019	0.0030
394	532	400	724	0.0030	0.0054
232	512	756	900	0.0059	0.0070

TABLE 13

A SEQ ID NO.	B live	C int	D ex	E int freq	F ex freq
465	224	11	15	0.002	0.003
298	215	108	129	0.020	0.024
297	161	86	94	0.021	0.023
296	228	119	169	0.021	0.030
295	264	150	177	0.023	0.027
294	194	67	105	0.014	0.022
400	179	26	53	0.006	0.012
293	199	141	152	0.028	0.031
399	200	21	21	0.004	0.004
292	265	122	138	0.018	0.021
291	209	49	84	0.009	0.016
290	205	109	111	0.021	0.022
289	234	69	56	0.012	0.010
288	200	100	134	0.020	0.027
287	292	124	223	0.017	0.031
464	248	19	11	0.003	0.002
463	239	8	15	0.001	0.003
286	210	154	133	0.029	0.025
137	229	243	363	0.042	0.063
285	236	103	117	0.017	0.020
284	293	145	150	0.020	0.020

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TABLE 13-continued

A SEQ ID NO.	B live	C int	D ex	E int freq	F ex freq
283	253	157	156	0.025	0.025
136	171	164	191	0.038	0.045
282	227	108	90	0.019	0.016
281	233	92	96	0.016	0.016
280	314	167	225	0.021	0.029
279	337	176	189	0.021	0.022
278	258	124	129	0.019	0.020
277	275	122	134	0.018	0.019
276	311	150	149	0.019	0.019
275	239	119	176	0.020	0.029
274	249	79	103	0.013	0.017
462	160	12	29	0.003	0.007
273	294	129	138	0.018	0.019
272	224	140	138	0.025	0.025
271	254	75	102	0.012	0.016
135	195	164	233	0.034	0.048
461	270	19	34	0.003	0.005
270	199	155	184	0.031	0.037
269	286	147	105	0.021	0.015
268	253	118	139	0.019	0.022
267	264	101	78	0.015	0.012
398	254	30	30	0.005	0.005
266	245	143	269	0.023	0.044
265	326	212	374	0.026	0.046
397	239	32	40	0.005	0.007
264	251	61	78	0.010	0.012
263	297	113	160	0.015	0.022
134	159	163	162	0.041	0.041
262	256	126	121	0.020	0.019
261	295	164	215	0.022	0.029
260	274	136	183	0.020	0.027
259	335	195	247	0.023	0.029
396	390	35	46	0.004	0.005
460	297	21	27	0.003	0.004
459	244	8	11	0.001	0.002
458	249	5	7	0.001	0.001
457	232	10	16	0.002	0.003
258	264	167	180	0.025	0.027
257	209	142	180	0.027	0.034
256	295	191	283	0.026	0.038

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TABLE 13-continued

A SEQ ID NO.	B live	C int	D ex	E int freq	F ex freq
5 255	265	79	92	0.012	0.014
254	219	155	198	0.028	0.036
253	256	137	176	0.021	0.028
456	207	6	8	0.001	0.002
252	308	123	144	0.016	0.019
455	320	23	25	0.003	0.003
10 251	260	130	109	0.020	0.017
490	357	4	8	0.000	0.001
250	234	149	142	0.025	0.024
454	240	14	19	0.002	0.003
249	180	105	116	0.023	0.026
248	244	120	77	0.020	0.013
15 453	257	7	4	0.001	0.001
452	326	9	12	0.001	0.001
133	235	263	526	0.045	0.090
247	314	72	129	0.009	0.016
246	247	143	193	0.023	0.031
245	243	181	201	0.030	0.033
20 244	276	128	143	0.019	0.021
243	224	148	120	0.026	0.021
242	184	79	84	0.017	0.018
241	246	92	140	0.015	0.023
132	135	110	169	0.033	0.050
131	209	176	227	0.034	0.043
240	284	208	263	0.029	0.037
25 451	269	14	15	0.002	0.002
239	278	140	209	0.020	0.030
238	283	120	139	0.017	0.020
450	246	8	7	0.001	0.001
237	239	93	114	0.016	0.019
30 236	302	137	175	0.018	0.023
395	256	21	40	0.003	0.006
235	340	176	210	0.021	0.025
234	214	132	150	0.025	0.028
233	285	172	232	0.024	0.033
174	380	134	209	0.014	0.022
35 158	351	61	89	0.007	0.010
39	381	31	51	0.003	0.005

TABLE 14

A Position	B From	C To	int D Weight	E Weight Std	ex F Weight	G Weight Std	H Number Of Occurrences
41	S	V	0.21	0.03	0.17	0.03	19
43	L	S	0.00	0.00	0.00	0.00	96
81	V	E	0.00	0.00	0.00	0.00	96
83	D	S	0.28	0.05	0.22	0.05	5
85	V	L	0.10	0.05	0.07	0.04	5
125	P	S	0.01	0.05	0.08	0.04	5
126	Q	S	0.23	0.06	0.21	0.07	5
131	Q	R	0.38	0.05	0.43	0.04	5
131	Q	T	0.37	0.06	0.22	0.05	5
136	S	V	0.24	0.02	0.16	0.02	19
140	E	C	0.40	0.05	0.23	0.05	5
140	E	A	0.24	0.05	0.20	0.04	5
151	N	Q	0.00	0.06	-0.11	0.05	5
169	K	E	0.43	0.06	0.48	0.05	5
212	N	S	0.13	0.04	0.02	0.03	5
239	I	L	0.17	0.03	0.12	0.02	19
268	T	D	0.00	0.00	0.00	0.00	96
297	T	C	0.21	0.03	0.20	0.04	9
300	M	R	0.00	0.00	0.00	0.00	96
305	M	N	0.00	0.00	0.00	0.00	96
312	L	I	0.28	0.03	0.23	0.03	10
316	C	A	0.22	0.04	0.23	0.03	5
322	N	T	0.31	0.07	0.29	0.06	5
357	H	R	0.32	0.06	0.21	0.05	5
360	K	Q	0.14	0.05	0.08	0.04	5
397	K	S	0.32	0.05	0.32	0.04	5
421	Y	F	0.22	0.03	0.16	0.02	19
447	A	N	0.18	0.06	0.11	0.06	5

TABLE 14-continued

A	B	C	int	E	ex	G	H
Position	From	To	D Weight	Weight Std	F Weight	Weight Std	Number Of Occurrences
447	A	D	0.07	0.05	0.08	0.05	5
449	A	S	0.22	0.03	0.07	0.03	5
449	A	V	0.10	0.06	-0.08	0.06	5
450	V	I	0.26	0.05	0.23	0.04	9
476	K	L	0.06	0.03	-0.22	0.04	5
485	R	K	0.11	0.05	0.06	0.04	5
492	V	A	0.22	0.06	0.17	0.05	5
495	Y	F	0.42	0.03	0.31	0.02	19
585	L	K	0.46	0.08	0.27	0.06	5
595	T	K	0.00	0.08	-0.08	0.05	5
241	H	N	-0.07	0.04	0.03	0.03	10
351	P	G	-0.19	0.08	-0.22	0.05	5
500	I	M	-0.52	0.07	-0.36	0.06	5
379	E	P	-1.13	0.10	-0.67	0.06	5
316	C	M	-1.74	0.10	-1.17	0.07	5
485	R	N	-1.77	0.08	-1.33	0.05	5
357	H	A	-1.90	0.05	-1.41	0.07	5

TABLE 15

A	B	C	D	E	F
SEQ ID NO.	live	int	ex	int freq	ex freq
368	288	117	220	0.016	0.031
157	268	280	452	0.042	0.067
367	160	98	146	0.025	0.037
366	296	127	232	0.017	0.031
365	67	15	24	0.009	0.014
364	248	151	224	0.024	0.036
156	300	212	316	0.028	0.042
363	528	320	464	0.024	0.035
155	236	172	292	0.029	0.049
154	268	220	348	0.033	0.052
362	288	169	292	0.023	0.041
361	268	132	252	0.020	0.038
360	180	49	97	0.011	0.022
359	376	162	296	0.017	0.031
358	420	170	404	0.016	0.038
357	652	324	576	0.020	0.035
153	312	218	36	0.028	0.005
356	404	187	276	0.019	0.027
355	360	119	207	0.013	0.023
354	340	150	316	0.018	0.037
353	220	60	90	0.011	0.016
352	216	106	173	0.020	0.032
351	348	172	356	0.020	0.041
350	400	209	400	0.021	0.040
349	284	107	276	0.015	0.039
152	292	218	376	0.030	0.052
348	211	104	188	0.020	0.036
151	220	266	476	0.048	0.087
347	222	83	133	0.015	0.024
150	313	214	324	0.027	0.041
149	262	223	500	0.034	0.076
346	452	193	328	0.017	0.029
345	300	187	336	0.025	0.045
344	360	142	332	0.016	0.037
148	283	248	296	0.035	0.042
343	195	68	113	0.014	0.023
342	201	76	133	0.015	0.026
341	266	138	248	0.021	0.037
340	338	204	280	0.024	0.033
339	355	241	364	0.027	0.041
338	276	181	280	0.026	0.041
147	305	385	824	0.050	0.108
337	156	83	192	0.021	0.049
336	206	113	124	0.022	0.024
335	224	132	220	0.024	0.039
334	252	93	184	0.015	0.029
333	319	166	320	0.021	0.040
332	710	469	1336	0.026	0.075

TABLE 15-continued

A	B	C	D	E	F
SEQ ID NO.	live	int	ex	int freq	ex freq
331	275	173	404	0.025	0.059
330	300	135	324	0.018	0.043
329	211	127	320	0.024	0.061
328	173	79	168	0.018	0.039
327	169	98	140	0.023	0.033
326	242	73	128	0.012	0.021
146	290	204	360	0.028	0.050
325	330	192	384	0.023	0.047
145	290	415	748	0.057	0.103
324	313	204	348	0.026	0.044
144	211	441	936	0.084	0.177
323	226	115	212	0.020	0.038
322	212	104	164	0.020	0.031
143	243	188	548	0.031	0.090
321	324	176	324	0.022	0.040
320	475	201	264	0.017	0.022
319	339	170	352	0.020	0.042
318	290	163	276	0.022	0.038
317	299	176	316	0.024	0.042
316	352	178	348	0.020	0.040
315	176	64	180	0.015	0.041
314	300	101	260	0.013	0.035
313	274	158	316	0.023	0.046
312	264	136	320	0.021	0.048
311	270	142	260	0.021	0.039
310	346	185	408	0.021	0.047
309	284	163	180	0.023	0.025
142	326	373	868	0.046	0.107
141	220	150	320	0.027	0.058
308	360	151	300	0.017	0.033
307	304	171	308	0.023	0.041
306	461	184	388	0.016	0.034
140	328	288	664	0.035	0.081
305	447	181	384	0.016	0.034
304	268	140	266	0.021	0.040
303	362	129	267	0.014	0.030
302	262	161	290	0.025	0.044
301	343	101	183	0.012	0.021
300	24	142	247	0.023	0.040
139	452	344	564	0.030	0.050
299	326	105	190	0.013	0.023
138	301	450	896	0.060	0.119
134	568	284	432	0.020	0.030
174	236	56	133	0.009	0.023
158	272	37	109	0.005	0.016
39	55	0	0	0.000	0.000

7. References

All references cited herein are incorporated herein by reference in their entirety and for all purposes to the same extent as if each individual publication or patent or patent application was specifically and individually indicated to be incorporated by reference in its entirety for all purposes. To the extent the information associated with a citation may change with time, the version in effect at the effective filing date of this application is meant, the effective filing date being the filing date of the application or priority application in which the citation was first mentioned.

Many modifications and variations of this invention can be made without departing from its spirit and scope, as will be apparent to those skilled in the art. The specific embodiments described herein are offered by way of example only, and the invention is to be limited only by the terms of the appended claims, along with the full scope of equivalents to which such claims are entitled. Unless otherwise apparent from the context, any embodiment, aspect, element, feature or step can be used in combination with any other.

8. The polynucleotide of claim 1, wherein the open reading frame further encodes a nuclear localization sequence fused to the transposase.

9. The polynucleotide of claim 1, wherein the open reading frame further encodes a heterologous DNA binding domain fused to the transposase.

10. The polynucleotide of claim 9, wherein the heterologous DNA binding domain is derived from a Cas system, or a zinc finger protein, or a TALE nuclease.

11. An isolated mRNA encoding a transposase, the amino acid sequence of which is at least 90% identical with SEQ ID NO: 39, and wherein the mRNA sequence comprises at least 10 synonymous codon differences relative to SEQ ID NO: 1181, optionally selected for mammalian expression, wherein the amino acid sequence comprises at least 2 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F,

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12404504B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

What is claimed is:

1. A polynucleotide comprising an open reading frame encoding a transposase, the amino acid sequence of which is at least 90% identical to SEQ ID NO: 39, operably linked to a heterologous promoter, wherein the amino acid sequence comprises at least 2 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.

2. A polynucleotide comprising an open reading frame encoding a transposase operably linked to a heterologous promoter, wherein the amino acid sequence of the transposase is selected from SEQ ID NO: 131-490.

3. The polynucleotide of claim 1, wherein the transposase can excise or transpose the transposon from SEQ ID NO: 66.

4. The polynucleotide of claim 3, wherein the excision activity or transposition activity of the transposase is at least 2-fold higher than the activity of the wild type transposase set forth by SEQ ID NO: 39.

5. The polynucleotide of claim 1, wherein the heterologous promoter is active in an in vitro transcription reaction.

6. The polynucleotide of claim 1, wherein the heterologous promoter is active in a eukaryotic cell.

7. The polynucleotide of claim 6, wherein the eukaryotic cell is a mammalian cell, optionally, wherein codons of the open reading frame are selected for expression in the mammalian cell.

8. H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.

12. A non-naturally occurring polynucleotide encoding a transposase comprising an amino acid sequence at least 90% identical to SEQ ID NO: 39 wherein the amino acid sequence comprises at least 2 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.

13. A non-naturally occurring transposase encoded by the polynucleotide of claim 1.

14. The polynucleotide of claim 4, wherein the excision activity or transposition activity of the transposase is 2 to 10 fold higher than the activity of the transposase of SEQ ID NO:39.

15. The polynucleotide of claim 1, wherein the amino acid sequence comprises at least 4 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I,

H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.

16. The isolated mRNA of claim **11**, wherein the amino acid sequence comprises at least 4 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, 5
Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, 10
A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.

17. The non-naturally occurring polynucleotide of claim **12**, wherein the amino acid sequence comprises at least 4 15
mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, 20
C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39. 25

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